Pure-MSE gamma short | Test IC 04

T_0=1317.5 K P_0=1.244 atm ϕ_0=1.984 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] · Y_{H_2} · Y_H · Y_O · Y_{O_2} · Y_{OH} · Y_{H_2O} · Y_{HO_2} · $Y_{H_2O_2}$ · Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 04

Repaired | Cantera vs NN NRMSE | T0=1317.5K, P0=1.244atm, phi0=1.984

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0323	0.0465	0.642	0.499	0.0957	0.644	0.0217	0.234	0.327	0	Complete
1 ANN10	0.1	0.0108	0.0583	0.593	2.33	0.0953	0.831	0.0354	0.149	0.418	0	Complete
1 ANN10	0.2	0.04	0.0755	0.995	0.628	0.0488	0.877	0.0298	2.86	104	0	Complete
1 ANN10	0.3	7.94e-03	0.0266	0.298	0.441	0.0382	0.572	0.0176	12.7	370	0	Complete
1 ANN10	0.5	0.104	0.142	0.111	1.12	1.9	2.99	0.181	2.9	75.8	0	Complete
1 ANN10	1	0.19	0.0563	1.14	3.77	0.819	5.73	0.188	49.7	349	0	Complete
2 P-only DeepONet10	0.05	0.424	0.362	0.43	0.826	0.184	22	0.671	0.649	7.37	0	Complete
2 P-only DeepONet10	0.1	8.74e-03	0.0113	0.165	0.217	0.0395	0.0155	2.22e-03	0.206	0.266	0	Complete
2 P-only DeepONet10	0.2	7.86e-03	7.31e-03	0.0897	0.219	0.11	0.107	6.39e-03	0.422	0.242	0	Complete
2 P-only DeepONet10	0.3	0.0102	5.05e-03	0.106	0.403	0.0721	0.122	6.60e-03	0.324	0.41	0	Complete
2 P-only DeepONet10	0.5	8.51e-03	0.0152	0.0994	0.867	0.129	0.236	0.0144	0.994	0.875	0	Complete
2 P-only DeepONet10	1	0.698	0.176	7.04	25.9	1.66	6.51	0.342	0.988	25.8	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.595	0.547	0.979	0.801	0.232	31.5	0.965	13	6.96	0	Complete
3 ANN10 + P-only DeepONet10	0.1	9.87e-03	0.0137	0.196	0.258	0.0434	0.031	2.92e-03	0.155	0.2	0	Complete
3 ANN10 + P-only DeepONet10	0.2	3.50e-03	3.52e-03	0.0364	0.121	0.0555	0.0576	3.34e-03	0.207	0.175	0	Complete
3 ANN10 + P-only DeepONet10	0.3	5.57e-03	1.41e-03	0.0545	0.16	0.0184	0.0829	3.54e-03	0.128	0.232	0	Complete
3 ANN10 + P-only DeepONet10	0.5	1.36e-03	9.76e-03	0.0719	0.326	0.0506	0.176	8.05e-03	0.311	1.23	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0186	0.0391	0.25	0.685	0.686	0.277	0.0284	2.7	325	0	Complete
4 T/P DeepONet10	0.05	0.512	0.224	5.82	2.23	0.107	7.34	0.232	4.25	23.4	0	Complete
4 T/P DeepONet10	0.1	0.0189	0.0104	0.227	0.392	0.0858	0.218	9.30e-03	0.336	0.28	0	Complete
4 T/P DeepONet10	0.2	1.54e-03	6.87e-03	0.0403	0.0963	0.0979	0.0359	4.59e-03	0.324	0.148	0	Complete
4 T/P DeepONet10	0.3	0.752	0.228	4.02	78.9	2.48	8.46	0.588	50.8	2.95	0	Complete
4 T/P DeepONet10	0.5	0.616	0.323	2.02	91.9	1.44	7.92	0.556	114	353	0	Complete
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.048 ms
5 ANN10 + T/P DeepONet10	0.05	0.0779	0.0406	0.961	0.838	0.125	0.987	0.0326	0.335	9.27	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.016	8.47e-03	0.186	0.267	0.0804	0.21	8.43e-03	0.233	0.181	0	Complete
5 ANN10 + T/P DeepONet10	0.2	2.45e-03	3.75e-03	0.0339	0.0983	0.0518	0.0297	2.67e-03	0.176	0.0921	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.481	0.345	0.585	65.5	0.475	11.4	0.545	71.8	43.1	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.445	0.215	1.38	47.9	0.0747	10.6	0.453	63	152	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0351	0.0298	0.455	0.978	0.359	0.315	0.02	0.899	7.15	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0789	0.0409	0.976	0.875	0.0613	0.995	0.0323	0.324	0.471	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0714	0.0485	0.943	0.811	0.196	0.973	0.0223	2.13	23.7	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0641	8.46e-03	0.473	2.82	0.154	1.28	0.052	1.11	2.99	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0148	0.051	0.539	0.745	0.244	0.963	0.0341	0.684	0.755	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.692	0.337	8.87	0.533	3.21	2.57	0.231	0.886	5.71	0	Complete
6 T/P/ϕ DeepONet10	1	0.515	1.09	0.998	1	21.2	0.999	0.999	0.989	1.23	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0753	0.0463	0.967	0.846	0.167	0.991	0.0271	0.331	7.62	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	4.70e-03	3.41e-03	0.059	0.0529	0.0354	0.0625	2.65e-03	0.16	0.106	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	9.45e-03	6.52e-03	0.0528	0.294	0.0431	0.339	0.0119	0.187	0.335	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0275	0.0638	0.79	0.646	0.0566	0.944	0.0315	0.385	1.08	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0349	0.0111	0.339	0.731	0.148	0.627	0.0234	0.352	0.538	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	6.46e-03	0.0183	0.107	0.24	0.294	0.0553	0.0131	0.664	4.84	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 04

T_0=1317.5 K P_0=1.244 atm ϕ_0=1.984 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 04

Repaired | Cantera vs NN NRMSE | T0=1317.5K, P0=1.244atm, phi0=1.984

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0441	0.0697	0.966	0.681	0.0278	0.758	0.025	0.101	0.407	0	Complete
1 ANN10	0.1	2.12e-03	0.0205	0.193	0.103	0.0155	0.543	0.0172	0.0581	0.191	0	Complete
1 ANN10	0.2	0.0495	0.107	0.599	0.304	0.205	4.03	0.133	0.0865	0.563	0	Complete
1 ANN10	0.3	0.0343	0.0685	0.276	0.137	0.534	1.95	0.081	0.0528	0.459	0	Complete
1 ANN10	0.5	0.0124	3.19e-03	0.0912	0.142	0.0544	0.341	0.0119	0.196	3.1	0	Complete
1 ANN10	1	0.0186	0.0435	0.0911	0.251	0.806	0.0336	0.0388	0.743	2.17	0	Complete
2 P-only DeepONet10	0.05	0.26	0.184	3.66	0.93	0.0489	1.88	0.0616	0.409	1.8	0	Complete
2 P-only DeepONet10	0.1	0.0344	0.0178	0.422	0.771	0.0508	0.366	0.0153	0.15	1.29	0	Complete
2 P-only DeepONet10	0.2	0.0176	3.88e-03	0.163	0.407	0.0428	0.333	0.0119	0.143	0.548	0	Complete
2 P-only DeepONet10	0.3	5.47e-03	4.36e-03	0.0611	0.131	0.0645	0.0741	4.10e-03	0.177	0.168	0	Complete
2 P-only DeepONet10	0.5	0.483	0.255	1.57	52.9	1.17	9.44	0.486	76.8	1.05e+03	0	Complete
2 P-only DeepONet10	1	0.853	0.691	12.9	1.34	1.94	0.894	0.0714	6.06	2.83e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.245	0.177	3.47	0.183	0.0295	1.86	0.0576	0.374	2.95	0	Complete
3 ANN10 + P-only DeepONet10	0.1	6.98e-03	4.49e-03	0.088	0.175	0.0173	0.0927	3.80e-03	0.0457	0.32	0	Complete
3 ANN10 + P-only DeepONet10	0.2	4.00e-03	2.02e-03	0.0231	0.116	0.0281	0.12	4.25e-03	0.0804	0.406	0	Complete
3 ANN10 + P-only DeepONet10	0.3	2.38e-03	1.26e-03	0.0221	0.0642	0.0201	0.0416	1.86e-03	0.0626	0.0894	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.11	0.2	0.794	9.51	0.951	5.23	0.23	3.73	280	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0719	0.029	0.563	4.21	0.428	1.36	0.0588	4.1	1.33e+03	0	Complete
4 T/P DeepONet10	0.05	0.0948	0.0586	1.24	0.886	0.04	1.02	0.0347	0.184	72.6	0	Complete
4 T/P DeepONet10	0.1	4.80e-03	1.36e-03	0.0479	0.0893	7.71e-03	0.0873	3.02e-03	0.138	0.108	0	Complete
4 T/P DeepONet10	0.2	8.30e-03	4.22e-03	0.0995	0.189	0.0351	0.0859	3.86e-03	0.0966	0.165	0	Complete
4 T/P DeepONet10	0.3	0.0185	8.46e-03	0.216	0.363	0.0632	0.221	8.96e-03	0.171	0.249	0	Complete
4 T/P DeepONet10	0.5	0.485	0.06	3.15	38.4	0.251	8.33	0.367	29.7	326	0	Complete
4 T/P DeepONet10	1	0.121	0.0599	1.52	3.38	0.663	0.436	0.0491	1.12	58.2	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.416	0.145	4.4	1.82	0.0288	7.19	0.225	0.51	5.41	0	Complete
5 ANN10 + T/P DeepONet10	0.1	2.60e-03	1.18e-03	0.0226	0.0407	0.0129	0.0671	2.21e-03	0.0615	0.101	0	Complete
5 ANN10 + T/P DeepONet10	0.2	3.25e-03	1.70e-03	0.0376	0.0855	0.0173	0.0381	1.75e-03	0.0551	0.101	0	Complete
5 ANN10 + T/P DeepONet10	0.3	5.71e-03	3.97e-03	0.0629	0.149	0.0573	0.0784	3.98e-03	0.143	0.123	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.156	0.145	2.53	3.11	0.0822	0.145	5.02e-03	4.57	13.2	0	Complete
5 ANN10 + T/P DeepONet10	1	0.104	0.0487	1.3	2.82	0.667	0.29	0.045	1.04	69.9	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.317	0.127	3.64	11	0.32	2.83	0.131	1.05	1.75	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.829	0.783	13.1	0.832	0.0384	0.654	0.0203	0.321	11	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0944	0.0606	1.3	2.23	0.122	0.395	0.023	0.294	1.13	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.111	0.0713	1.52	2.86	0.295	0.451	0.0297	17.7	14.3	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.189	0.121	2.55	4.62	0.316	1.23	0.0646	7.81	44.5	0	Complete
6 T/P/ϕ DeepONet10	1	0.12	0.0302	1.25	2.46	1.26	0.398	0.0776	1.41	82.4	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.418	0.143	4.42	4.37	0.0652	6.79	0.221	0.743	4.31	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.349	0.148	3.93	0.626	0.0357	5.61	0.171	0.148	2.31	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.082	0.0615	1.21	1.65	0.078	0.181	0.0132	0.16	0.432	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.117	0.0882	1.73	2.79	0.134	0.167	0.0175	0.504	0.534	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.111	0.103	1.79	2.23	0.125	0.342	0.016	8.26	26.4	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.141	0.0339	1.36	2.18	1.95	0.119	0.103	1.22	162	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 05

T_0=1049.0 K P_0=4.666 atm ϕ_0=0.674 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 05

Repaired | Cantera vs NN NRMSE | T0=1049.0K, P0=4.666atm, phi0=0.674

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.289	0.856	1.6	1.11	0.643	8.46	0.515	1.03	1.01	0	Complete
1 ANN10	0.1	0.299	0.774	1.07	1.02	0.614	11.4	0.509	1.03	1.01	0	Complete
1 ANN10	0.2	0.528	1.32	1	1	0.732	1	1	32	180	0	Complete
1 ANN10	0.3	0.3	0.791	0.944	0.547	0.501	0.797	0.532	1.04	1.04	0	Complete
1 ANN10	0.5	0.364	0.93	0.945	0.606	0.576	0.723	0.64	1.04	1	0	Complete
1 ANN10	1	0.467	1.24	1	0.994	0.786	0.989	0.838	1	0.953	0	Complete
2 P-only DeepONet10	0.05	0.291	0.824	0.998	6.6	0.566	2.07	0.538	0.976	0.999	0	Complete
2 P-only DeepONet10	0.1	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
2 P-only DeepONet10	0.2	0.565	1.48	1	1	0.929	1	1	0.991	1	0	Complete
2 P-only DeepONet10	0.3	0.565	1.48	1	1	0.929	1	1	0.976	0.999	0	Complete
2 P-only DeepONet10	0.5	0.522	1.16	20.7	3.33	0.725	0.759	0.832	1	0.989	0	Complete
2 P-only DeepONet10	1	0.565	1.47	1	1	0.925	1	1	9.29	0.955	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.292	0.828	6.77	6.66	0.574	2.56	0.538	2.9	1.02	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.216	0.553	8.86	0.677	0.349	1.87	0.376	0.912	0.995	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.565	1.48	1	1	0.929	1	1	0.96	0.997	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.37	0.951	1.06	0.686	0.594	0.559	0.651	0.897	1	0	Complete
3 ANN10 + P-only DeepONet10	1	0.548	1.43	0.879	0.967	0.903	0.999	0.971	0.999	0.986	0	Complete
4 T/P DeepONet10	0.05	0.562	1.47	0.999	1	0.924	1	0.996	2.76	3.97	0	Complete
4 T/P DeepONet10	0.1	0.35	0.942	2.92	0.887	0.604	1.32	0.626	0.989	0.997	0	Complete
4 T/P DeepONet10	0.2	0.565	1.48	1	1	0.929	1	1	0.996	1	0	Complete
4 T/P DeepONet10	0.3	0.565	1.48	1	1	0.929	1	1	0.99	1	0	Complete
4 T/P DeepONet10	0.5	0.568	1.48	2.1	1.12	0.928	1.03	1.01	15.8	2.05	0	Complete
4 T/P DeepONet10	1	0.43	1.06	0.955	6.82	0.636	0.639	0.725	1	1.08	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.359	0.889	35	0.801	0.719	14.2	0.536	2.44	1.36	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.34	0.913	2.09	0.728	0.586	1.34	0.606	0.921	0.996	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.314	0.82	1.31	0.595	0.515	0.548	0.557	0.903	0.998	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.565	1.48	1	1	0.929	1	1	0.983	0.999	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.358	0.955	6.27	1.35	0.603	0.879	0.641	0.89	0.997	0	Complete
5 ANN10 + T/P DeepONet10	1	0.474	1.16	0.878	7.42	0.687	0.61	0.795	1	1.67	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.292	0.885	2.65	2.57	0.658	7.16	0.531	1.77	0.999	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.565	1.48	1	1	0.929	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.565	1.48	1	1	0.929	1	1	0.998	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.565	1.48	1	1	0.929	1	1	0.99	1	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.36	0.964	7.81	2.48	0.618	1.41	0.643	4.87	1.02	0	Complete
6 T/P/ϕ DeepONet10	1	0.33	0.847	0.764	5.47	0.524	0.583	0.574	1	0.957	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.283	0.858	4.01	1.64	0.639	7.49	0.511	1.65	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.565	1.48	1	1	0.929	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.565	1.48	1	1	0.929	1	1	0.997	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.565	1.48	1	1	0.929	1	1	0.986	0.999	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.359	0.957	7.1	1.56	0.604	0.874	0.643	0.896	0.998	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.471	1.1	1.18	9.7	0.632	0.807	0.768	1	1.71	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 05

T_0=1049.0 K P_0=4.666 atm ϕ_0=0.674 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 05

Repaired | Cantera vs NN NRMSE | T0=1049.0K, P0=4.666atm, phi0=0.674

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.275	0.693	1.06	1.15	0.509	8.49	0.473	1.07	1.05	0	Complete
1 ANN10	0.1	0.287	0.585	1.06	1.1	0.432	10.1	0.472	1.09	1.17	0	Complete
1 ANN10	0.2	0.262	0.67	0.878	1.11	0.423	2.74	0.462	1.46	2.8	0	Complete
1 ANN10	0.3	0.184	0.43	1.72	1.02	0.251	2.45	0.325	4.76	9.46	0	Complete
1 ANN10	0.5	0.201	0.53	0.925	0.466	0.334	0.488	0.358	0.825	0.929	0	Complete
1 ANN10	1	0.453	1.21	0.91	1	0.764	0.988	0.814	1	1.07	0	Complete
2 P-only DeepONet10	0.05	0.565	1.48	1	1	0.929	1	1	0.999	1	0	Complete
2 P-only DeepONet10	0.1	0.617	1.45	1.74	11.9	0.829	0.814	0.988	3.57	0.96	0	Complete
2 P-only DeepONet10	0.2	0.565	1.48	1	1	0.929	1	1	0.979	0.999	0	Complete
2 P-only DeepONet10	0.3	0.565	1.48	1	1	0.929	1	1	0.999	1	0	Complete
2 P-only DeepONet10	0.5	0.392	0.943	15.3	8.44	0.644	7.57	0.641	1.4	1.93	0	Complete
2 P-only DeepONet10	1	0.607	0.91	65.9	56.3	0.899	9.09	0.564	0.976	8.32	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.316	0.658	8.07	0.999	0.604	15.6	0.497	1.09	1.19	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.653	1.45	2.09	17.3	0.785	1.11	0.999	1.82	0.959	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.222	0.581	0.872	0.445	0.365	0.407	0.394	0.986	1.14	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.565	1.48	1	1	0.929	1	1	0.998	1	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.306	0.942	49.9	0.913	0.544	0.588	0.59	1	0.996	0	Complete
3 ANN10 + P-only DeepONet10	1	0.51	0.968	14.8	15.4	0.534	3.73	0.74	0.852	2.82	0	Complete
4 T/P DeepONet10	0.05	0.385	0.869	41.5	8.23	0.707	12.5	0.532	2.69	0.998	0	Complete
4 T/P DeepONet10	0.1	0.307	0.928	9.34	0.594	0.644	5.55	0.563	1.33	0.99	0	Complete
4 T/P DeepONet10	0.2	0.565	1.48	1	1	0.929	1	1	1	1	0	Complete
4 T/P DeepONet10	0.3	0.565	1.48	1	1	0.929	1	1	1	1	0	Complete
4 T/P DeepONet10	0.5	0.565	1.48	1	1	0.929	1	1	1	1	0	Complete
4 T/P DeepONet10	1	0.363	0.923	22.2	6.89	0.582	1.37	0.626	2.19	1.01	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.418	0.866	39.2	0.704	0.764	18	0.565	1.61	1.01	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.32	0.909	15.4	0.827	0.737	13.2	0.534	1.77	0.982	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.565	1.48	1	1	0.929	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.565	1.48	1	1	0.929	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.565	1.48	1	1	0.929	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.414	1.05	23.2	4.46	0.658	1.64	0.717	3.74	1.09	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.428	0.853	50.2	2.8	0.704	15.4	0.558	15.8	1.24	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.553	1.44	6.09	0.992	0.905	1.79	0.976	1.1	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.565	1.48	1	1	0.929	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.565	1.48	1	1	0.929	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.565	1.48	0.999	1	0.929	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	1	0.326	0.934	10.3	3.33	0.603	1.41	0.609	1.09	0.999	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.344	0.852	39.4	0.853	0.661	12.7	0.511	1.7	1.01	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.376	0.874	58.4	0.851	0.632	11.5	0.528	3.23	0.985	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.565	1.48	1	1	0.929	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.565	1.48	1	1	0.929	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.565	1.48	0.999	1	0.929	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.639	0.829	35.5	51.2	0.622	6.17	0.651	0.95	9.85	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 07

T_0=930.6 K P_0=3.722 atm ϕ_0=0.754 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 07

Repaired | Cantera vs NN NRMSE | T0=930.6K, P0=3.722atm, phi0=0.754

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	2.85e-06	2.61e-06	0.999	181	4.29e-06	1	1	0.987	0.998	0	Complete
1 ANN10	0.1	2.26e-06	1.72e-06	1	360	2.48e-06	1	1	0.531	0.464	0	Complete
1 ANN10	0.2	1.73e-04	8.59e-05	0.986	1	6.25e-05	0.998	102	0.992	1	0	Complete
1 ANN10	0.3	0.49	0.276	1.19e+07	1.28e+09	0.226	1.53e+09	3.06e+05	50.6	2.73	0	Complete
1 ANN10	0.5	1.31	0.716	8.83e+06	7.61e+07	0.546	4.01e+08	8.27e+05	94.8	253	0	Complete
1 ANN10	1	0.0504	0.0268	2.20e+05	3.17e+07	0.0212	2.61e+05	3.11e+04	0.944	6.79	0	Complete
2 P-only DeepONet10	0.05	2.79e-06	2.57e-06	1	3.53	4.23e-06	0.676	0.978	0.964	0.993	0	Complete
2 P-only DeepONet10	0.1	2.84e-06	2.61e-06	1	0.68	4.31e-06	0.986	0.999	0.994	1	0	Complete
2 P-only DeepONet10	0.2	2.80e-06	2.59e-06	0.606	0.586	4.27e-06	0.764	0.98	0.983	0.999	0	Complete
2 P-only DeepONet10	0.3	2.75e-06	2.56e-06	14	50.4	4.27e-06	18.7	0.942	0.995	1	0	Complete
2 P-only DeepONet10	0.5	0.0327	0.0438	1.05e+06	4.75e+08	0.0518	3.67e+08	4.54e+04	16.8	9.54	0	Complete
2 P-only DeepONet10	1	0.368	0.251	1.26e+06	9.93e+08	0.232	1.15e+09	2.75e+05	7.02	104	0	Complete
3 ANN10 + P-only DeepONet10	0.05	2.80e-06	2.58e-06	1	4.05	4.24e-06	0.753	0.983	0.97	0.995	0	Complete
3 ANN10 + P-only DeepONet10	0.1	2.84e-06	2.61e-06	1	0.623	4.31e-06	0.987	0.999	0.992	1	0	Complete
3 ANN10 + P-only DeepONet10	0.2	2.80e-06	2.58e-06	3.3	0.874	4.27e-06	0.679	0.974	0.985	0.999	0	Complete
3 ANN10 + P-only DeepONet10	0.3	2.52e-06	2.43e-06	24.8	15.8	4.17e-06	17.4	0.823	0.994	1	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0637	0.0638	6.44e+06	5.37e+08	0.0681	3.57e+08	6.79e+04	23.2	6.07	0	Complete
3 ANN10 + P-only DeepONet10	1	0.389	0.266	3.70e+07	8.23e+08	0.231	8.67e+08	2.90e+05	1.05	337	0	Complete
4 T/P DeepONet10	0.05	1.11	0.885	1.52e+08	6.98e+08	0.788	1.25e+10	8.21e+05	200	464	0	Complete
4 T/P DeepONet10	0.1	2.85e-06	2.61e-06	29.5	0.994	4.31e-06	1.56	1	0.993	1	0	Complete
4 T/P DeepONet10	0.2	2.84e-06	2.62e-06	1	1	4.32e-06	1	1	1	1	0	Complete
4 T/P DeepONet10	0.3	2.24e-06	2.14e-06	133	1.54e+03	3.87e-06	4.89e+03	0.733	1	1	0	Complete
4 T/P DeepONet10	0.5	0.744	0.472	1.66e+08	4.28e+08	0.369	1.74e+09	5.08e+05	4.56e+03	2e+03	0	Complete
4 T/P DeepONet10	1	0.636	0.399	3.60e+07	9.54e+08	0.335	7.49e+08	4.46e+05	1	243	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.747	0.85	3.52e+08	3.11e+08	0.774	1.79e+10	6.65e+05	280	2.38e+03	0	Complete
5 ANN10 + T/P DeepONet10	0.1	2.86e-06	2.49e-06	105	0.972	4.01e-06	14.1	0.99	0.861	0.984	0	Complete
5 ANN10 + T/P DeepONet10	0.2	2.84e-06	2.62e-06	0.933	0.951	4.32e-06	0.983	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	2.54e-06	2.06e-06	1.23e+03	82.6	3.92e-06	4.67e+03	0.721	1	1	0	Complete
5 ANN10 + T/P DeepONet10	0.5	1.64	0.957	8.75e+07	2.91e+08	0.745	2.60e+09	1.06e+06	63.7	5.3	0	Complete
5 ANN10 + T/P DeepONet10	1	0.408	0.289	3.66e+07	1.09e+09	0.258	9.72e+08	3.15e+05	1	554	0	Complete
6 T/P/ϕ DeepONet10	0.05	2.84e-06	2.60e-06	0.593	0.988	4.29e-06	0.853	0.999	0.985	1	0	Complete
6 T/P/ϕ DeepONet10	0.1	1.48	0.931	5.41e+07	3.02e+08	0.768	6.67e+09	9.77e+05	107	599	0	Complete
6 T/P/ϕ DeepONet10	0.2	2.84e-06	2.62e-06	1	1	4.32e-06	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	2.33e-06	1.90e-06	295	3.57e+03	3.42e-06	3.43e+04	0.757	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	0.5	1.22	0.762	2.48e+08	8.22e+08	0.607	3.78e+09	8.20e+05	937	1.24e+03	0	Complete
6 T/P/ϕ DeepONet10	1	1.02	0.586	1.43e+07	6.65e+08	0.47	8.27e+08	6.67e+05	1	72.2	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.86	0.813	2.77e+08	2.53e+08	0.717	1.39e+10	6.92e+05	219	1.32e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	1.38	0.888	5.22e+07	2.11e+08	0.74	7.44e+09	9.16e+05	119	900	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	2.84e-06	2.62e-06	0.939	0.956	4.32e-06	0.984	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	2.84e-06	2.61e-06	3.22	4.11	4.32e-06	2.1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	1.62	0.949	1.05e+08	3.50e+08	0.739	2.50e+09	1.06e+06	70.1	4.01	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.418	0.329	4.88e+07	1.34e+09	0.306	2.10e+09	3.43e+05	1.01	552	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 07

T_0=930.6 K P_0=3.722 atm ϕ_0=0.754 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

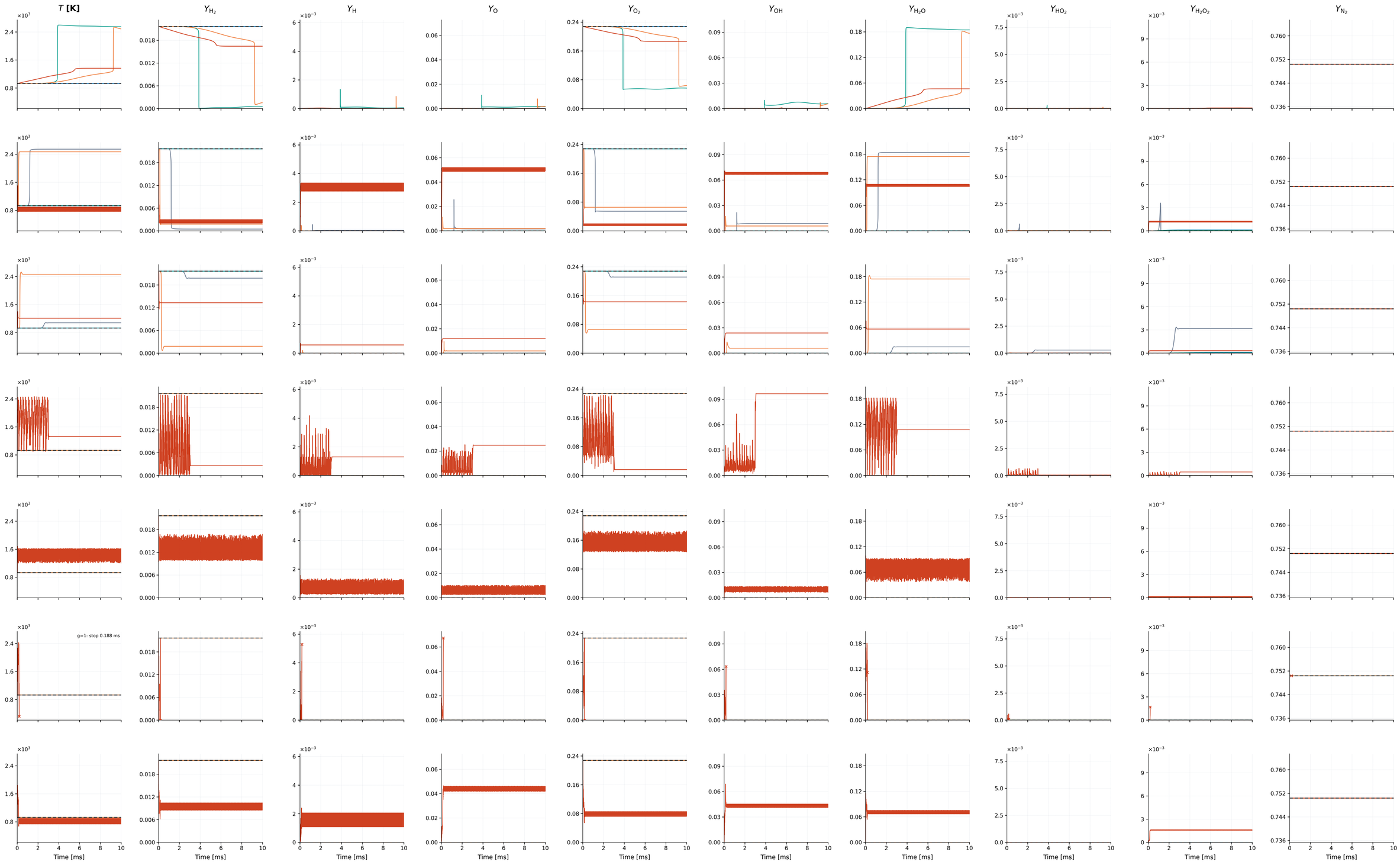
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 07

Repaired | Cantera vs NN NRMSE | T0=930.6K, P0=3.722atm, phi0=0.754

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	2.88e-06	2.60e-06	0.999	525	4.26e-06	0.999	1	0.979	0.995	0	Complete
1 ANN10	0.1	6.48e-07	1.31e-06	1	1.67e+03	3.09e-06	1	1	0.501	1.6	0	Complete
1 ANN10	0.2	2.72e-06	2.35e-06	1	4.94	3.71e-06	1	1	0.79	0.883	0	Complete
1 ANN10	0.3	1.38	0.771	1.83e+07	1.54e+08	0.593	1.03e+09	8.79e+05	33.9	4.65	0	Complete
1 ANN10	0.5	0.498	0.273	5.93e+06	6.32e+07	0.21	2.77e+08	3.14e+05	33.8	104	0	Complete
1 ANN10	1	0.351	0.181	4.68e+06	8.66e+05	0.137	3.29e+07	2.10e+05	0.975	146	0	Complete
2 P-only DeepONet10	0.05	2.84e-06	2.61e-06	1	0.898	4.32e-06	0.976	1	0.996	1	0	Complete
2 P-only DeepONet10	0.1	1.63	0.917	1.03e+07	2.20e+08	0.714	1.82e+09	1.04e+06	69.7	771	0	Complete
2 P-only DeepONet10	0.2	2.07e-03	1.21e-03	543	443	1.23e-03	1.14e+03	1.01e+03	50	288	0	Complete
2 P-only DeepONet10	0.3	1.25e-03	7.36e-04	33.3	1.11e+03	7.55e-04	4.05e+03	604	35.5	178	0	Complete
2 P-only DeepONet10	0.5	1.64	0.912	1.95e+06	2.52e+08	0.708	1.28e+09	1.04e+06	38.4	1.94	0	Complete
2 P-only DeepONet10	1	0.129	0.887	8.74e+08	6.81e+09	0.925	1.51e+10	6.45e+05	1.45	3.01e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	2.84e-06	2.61e-06	1	0.854	4.31e-06	0.967	0.999	0.995	1	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.139	0.0741	36.3	3.23e+05	0.0623	6.50e+04	7.70e+04	507	6.97e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.2	1.61e-03	9.47e-04	465	304	9.64e-04	754	781	42.3	228	0	Complete
3 ANN10 + P-only DeepONet10	0.3	1.22e-03	7.08e-04	29.1	666	7.09e-04	2.45e+03	604	32.4	154	0	Complete
3 ANN10 + P-only DeepONet10	0.5	1.63	0.905	1.29e+06	2.51e+08	0.703	1.31e+09	1.03e+06	39.4	2.7	0	Complete
3 ANN10 + P-only DeepONet10	1	0.301	0.385	1.61e+08	1.63e+09	0.373	5.27e+09	3.40e+05	39.9	806	0	Complete
4 T/P DeepONet10	0.05	2.84e-06	2.61e-06	0.921	0.913	4.30e-06	0.965	0.999	0.987	1	0	Complete
4 T/P DeepONet10	0.1	7.80e-06	1.77e-06	1.07e+04	4.56	3.96e-06	1.12e+03	0.749	0.937	0.999	0	Complete
4 T/P DeepONet10	0.2	6.50e-05	4.87e-05	5.23	56.5	6.77e-05	8.32	26.3	11	17.2	0	Complete
4 T/P DeepONet10	0.3	6.49e-04	3.69e-04	6.06e+04	8.91e+03	3.01e-04	7.63e+04	402	18.5	0.703	0	Complete
4 T/P DeepONet10	0.5	4.49e-04	2.44e-04	3.43e+04	1.51e+04	1.82e-04	1.26e+05	276	0.82	0.997	0	Complete
4 T/P DeepONet10	1	0.723	0.842	3.23e+08	2.84e+09	0.842	1.80e+10	6.94e+05	133	966	0	Complete
5 ANN10 + T/P DeepONet10	0.05	2.82e-06	2.56e-06	0.903	0.889	4.19e-06	0.956	0.991	0.942	0.991	0	Complete
5 ANN10 + T/P DeepONet10	0.1	2.97e-06	2.51e-06	462	0.848	4.27e-06	35.3	0.961	0.987	1	0	Complete
5 ANN10 + T/P DeepONet10	0.2	5.36e-05	4.03e-05	2.12	37.4	5.63e-05	4.89	21.6	9.21	14.3	0	Complete
5 ANN10 + T/P DeepONet10	0.3	8.95e-04	4.84e-04	5.15e+04	4.66e+03	3.79e-04	5.27e+04	542	12.9	0.711	0	Complete
5 ANN10 + T/P DeepONet10	0.5	7.22e-04	3.81e-04	3.46e+04	1.30e+04	2.84e-04	8.44e+04	435	0.703	0.997	0	Complete
5 ANN10 + T/P DeepONet10	1	0.557	0.438	1.90e+08	7.56e+08	0.354	2.32e+09	4.41e+05	17	292	0	Complete
6 T/P/ϕ DeepONet10	0.05	2.84e-06	2.61e-06	0.752	0.939	4.31e-06	0.926	0.999	0.995	1	0	Complete
6 T/P/ϕ DeepONet10	0.1	2.84e-06	2.61e-06	3.66	0.925	4.31e-06	0.752	0.998	0.996	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	1.98e-04	1.36e-04	1.66e+03	89.5	1.72e-04	459	78.3	17	53.2	0	Complete
6 T/P/ϕ DeepONet10	0.3	6.22e-04	3.32e-04	1.1e+03	3.59e+04	2.85e-04	1.44e+05	368	21.3	0.826	0	Complete
6 T/P/ϕ DeepONet10	0.5	1.07e-03	5.69e-04	9.99e+03	1.01e+05	4.45e-04	1.30e+06	642	1.18	0.998	0	Complete
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.188 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	2.83e-06	2.60e-06	0.88	0.92	4.29e-06	0.955	0.997	0.984	0.999	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	2.83e-06	2.61e-06	1.27	0.93	4.30e-06	0.79	0.997	0.989	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	1.90e-04	1.31e-04	751	71.5	1.67e-04	159	74.8	16.7	51.5	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	6.53e-04	3.41e-04	969	2.22e+04	2.81e-04	6.85e+04	384	15.1	0.891	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	1.19e-03	6.22e-04	8.01e+03	8.09e+04	4.81e-04	1.05e+06	707	1.04	0.999	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.148	0.561	4.49e+08	5.86e+09	0.651	9.60e+09	4.26e+05	13.7	4.03e+03	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 18

T_0=1069.1 K P_0=2.170 atm ϕ_0=1.544 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] (×10³) | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

g=0.1: stop 0.250 ms

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 18

Repaired | Cantera vs NN NRMSE | T0=1069.1K, P0=2.170atm, phi0=1.544

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0467	0.0709	0.627	0.863	0.33	0.928	0.0473	1.01	1.22	0	Complete
1 ANN10	0.1	0.0255	0.115	0.793	0.959	0.336	1.94	0.0671	1.02	1.19	0	Complete
1 ANN10	0.2	0.0534	0.175	0.657	2.97	0.573	1.66	0.103	1.21	1.17	0	Complete
1 ANN10	0.3	0.0336	0.108	0.254	1.53	0.502	0.937	0.0602	2.51	30.7	0	Complete
1 ANN10	0.5	0.0448	0.108	0.581	1.5	0.511	0.858	0.0602	1.21	1.16	0	Complete
1 ANN10	1	0.546	1.63	0.994	0.702	7.97	0.857	0.906	0.999	10.8	0	Complete
2 P-only DeepONet10	0.05	0.219	0.209	2.11	1.1	0.977	6.74	0.221	1.55	2.72	0	Complete
2 P-only DeepONet10	0.1	0.067	0.161	0.806	1.05	0.768	0.82	0.0892	1.54	2.36	0	Complete
2 P-only DeepONet10	0.2	0.608	1.8	1	1	8.83	1	1	2.98	13.7	0	Complete
2 P-only DeepONet10	0.3	0.495	1.05	0.917	62.4	3.13	4.95	0.654	7.27	40.4	0	Complete
2 P-only DeepONet10	0.5	0.37	0.659	0.647	41.1	1.34	8.6	0.486	0.995	19.6	0	Complete
2 P-only DeepONet10	1	0.592	1.76	0.973	0.99	8.63	0.987	0.976	0.999	7.26	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.143	0.16	2.7	1.03	0.61	2.32	0.0918	1.47	2.74	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.029	0.0992	0.314	0.756	0.462	0.297	0.051	1.31	1.86	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0262	0.099	0.244	0.611	0.464	0.121	0.0501	1.37	2.81	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0198	0.0745	0.205	0.698	0.348	0.133	0.0377	1.15	1.59	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.132	0.303	0.888	15.3	0.611	4.34	0.209	1.01	22.3	0	Complete
3 ANN10 + P-only DeepONet10	1	0.587	1.73	0.933	0.874	8.51	0.866	0.966	0.999	16.6	0	Complete
4 T/P DeepONet10	0.05	0.128	0.357	1	1.16	1.74	1	0.197	1.97	3.82	0	Complete
4 T/P DeepONet10	0.1	0.0652	0.169	1.06	1.06	0.778	1.03	0.0828	1.98	5.67	0	Complete
4 T/P DeepONet10	0.2	0.0338	0.115	0.305	0.579	0.549	0.147	0.0598	1.56	2.37	0	Complete
4 T/P DeepONet10	0.3	0.617	1.64	0.961	30.5	6.47	0.985	0.987	2.45e+03	2.15e+04	0	Complete
4 T/P DeepONet10	0.5	0.407	0.617	0.544	74	0.821	7.95	0.458	25.8	45.4	0	Complete
4 T/P DeepONet10	1	0.102	0.207	1.37	4.74	1.01	0.589	0.119	1.7	4.48	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0715	0.233	1	1.08	1.09	0.998	0.114	1.62	3.44	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0681	0.149	1.03	1.02	0.693	1.02	0.0796	1.62	2.2	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.04	0.136	0.298	0.595	0.649	0.147	0.0711	1.52	1.99	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0208	0.0804	0.249	0.665	0.371	0.125	0.0399	1.58	3.44	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.211	0.448	1.46	14.9	1.77	4.4	0.293	6.01	12.1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0651	0.17	0.653	2.9	0.822	0.322	0.094	1.56	4.35	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.626	0.357	10.4	1.12	0.98	11.4	0.348	15.6	163	0	Complete
6 T/P/ϕ DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 0.250 ms
6 T/P/ϕ DeepONet10	0.2	0.036	0.0883	0.936	0.573	0.339	0.124	0.0362	1.43	4.04	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.283	0.214	1.92	10.4	0.588	8.08	0.288	4.26	7.4	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.811	0.901	18.9	1.98	0.803	4.35	0.161	70.8	1.08e+03	0	Complete
6 T/P/ϕ DeepONet10	1	0.143	0.191	2.72	6.96	0.784	1.15	0.0965	1.8	1.64	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.6	0.565	1.66	1.06	0.915	24.9	0.73	1.82	4.78	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.304	0.175	3.44	0.629	0.833	8.52	0.265	2.04	5.77	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0352	0.119	0.452	0.614	0.558	0.144	0.0608	1.6	2.52	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0699	0.18	2.53	0.689	0.358	0.895	0.0479	1.3	2.23	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.179	0.354	1.52	6.54	1.5	3.89	0.235	3.82	11.3	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.139	0.402	0.296	1.08	1.99	0.228	0.228	1.61	16.2	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 18

T_0=1069.1 K P_0=2.170 atm ϕ_0=1.544 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] (×10³) | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 18

Repaired | Cantera vs NN NRMSE | T0=1069.1K, P0=2.170atm, phi0=1.544

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0493	0.0808	0.27	0.613	0.119	2.83	0.0803	0.305	0.382	0	Complete
1 ANN10	0.1	0.017	0.0653	0.369	6	0.255	0.799	0.0286	0.906	5.14	0	Complete
1 ANN10	0.2	0.0547	0.151	0.64	0.668	0.334	3.05	0.11	0.868	1.03	0	Complete
1 ANN10	0.3	0.0177	0.0788	0.367	0.521	0.303	0.881	0.0428	0.84	2.64	0	Complete
1 ANN10	0.5	0.0122	0.0411	0.143	0.493	0.159	0.462	0.0236	0.405	0.603	0	Complete
1 ANN10	1	0.169	0.123	2.07	3.57	0.585	3.97	0.14	12.9	125	0	Complete
2 P-only DeepONet10	0.05	0.0558	0.241	0.95	11.3	0.743	0.295	0.111	0.507	0.672	0	Complete
2 P-only DeepONet10	0.1	0.0202	0.0779	0.212	0.545	0.364	0.0941	0.0391	1.02	1.16	0	Complete
2 P-only DeepONet10	0.2	0.0314	0.113	0.246	0.619	0.533	0.131	0.058	1.27	1.24	0	Complete
2 P-only DeepONet10	0.3	0.0119	0.055	0.176	0.449	0.248	0.0814	0.0263	0.739	0.913	0	Complete
2 P-only DeepONet10	0.5	0.0546	0.192	1.55	1.08	0.74	0.984	0.0881	1.04	1	0	Complete
2 P-only DeepONet10	1	0.737	0.745	17.1	17.5	1.05	0.544	0.15	0.984	723	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.175	0.273	4.96	0.791	0.201	0.207	0.0215	0.545	0.842	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.019	0.0754	0.21	0.539	0.349	0.0927	0.0376	0.989	1.13	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0264	0.0983	0.235	0.585	0.461	0.115	0.0501	1.22	1.24	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0178	0.073	0.205	0.522	0.337	0.0931	0.0361	0.994	1.23	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.042	0.162	1.3	1.05	0.605	0.967	0.0736	1.11	1.01	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0841	0.164	0.475	5.79	0.786	1.86	0.107	0.998	159	0	Complete
4 T/P DeepONet10	0.05	0.0782	0.176	1	1.07	0.842	0.999	0.0988	1.75	4.63	0	Complete
4 T/P DeepONet10	0.1	0.185	0.223	3.06	3.19	1.02	2.43	0.144	1.73	3.59	0	Complete
4 T/P DeepONet10	0.2	0.425	1.25	0.712	8.55	6.08	1.56	0.695	9.23	119	0	Complete
4 T/P DeepONet10	0.3	0.0877	0.271	0.312	0.912	1.32	0.227	0.148	2.95	9.69	0	Complete
4 T/P DeepONet10	0.5	0.608	1.8	1	1	8.84	1	1	1	1	0	Complete
4 T/P DeepONet10	1	0.607	1.8	1	0.994	8.83	0.995	1	1	0.983	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0578	0.109	0.918	1.04	0.495	0.996	0.0595	1.34	2.44	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.376	0.331	7.73	0.683	0.66	4.77	0.154	1.47	2.07	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.261	0.766	0.49	6.46	3.72	1.34	0.428	6.97	114	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0671	0.213	0.296	0.775	1.03	0.195	0.115	1.63	2.2	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.608	1.8	1	1	8.84	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.797	0.857	18.7	23.2	0.895	0.944	0.134	1.81	2.35e+03	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.641	0.2	9.35	1.24	0.547	13.9	0.405	1.45	10.5	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.645	0.951	17.2	1.31	1.22	1.24	0.141	1.62	3.93	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.139	0.295	3.19	1.76	1.2	0.258	0.134	1.8	3.96	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.221	0.435	5.8	2.05	1.45	0.588	0.164	10.1	6.01	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.753	0.877	18.1	1.7	0.711	3.2	0.135	19.3	455	0	Complete
6 T/P/ϕ DeepONet10	1	0.85	0.987	20.3	29.9	0.914	0.936	0.125	4.38	6.93e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.326	0.107	4.78	1.01	0.364	7.6	0.221	1.09	1.86	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.374	0.265	6.61	1.26	0.885	6.15	0.209	1.58	2.59	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.113	0.241	2.98	1.02	0.852	0.387	0.0946	1.62	2.53	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.192	0.366	5.49	1.27	0.952	0.645	0.107	1.62	1.79	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.143	0.239	4.25	3	0.201	0.685	0.0253	8.36	6.35	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.803	0.834	18.2	2.52	0.894	4.44	0.189	1.1	877	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 32

T_0=1140.7 K P_0=3.354 atm ϕ_0=1.463 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

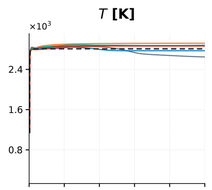
3 ANN10 + P-only DeepONet10

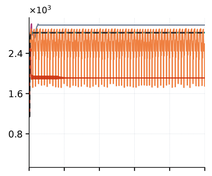
4 T/P DeepONet10

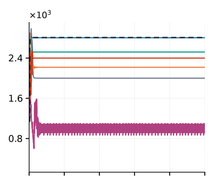
5 ANN10 + T/P DeepONet10

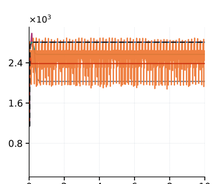
6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10









Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 32

Repaired | Cantera vs NN NRMSE | T0=1140.7K, P0=3.354atm, phi0=1.463

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0202	0.0641	0.331	0.873	0.151	0.963	0.039	0.354	0.474	0	Complete
1 ANN10	0.1	0.0312	0.126	1.48	0.511	0.0533	0.83	0.0317	0.118	1.03	0	Complete
1 ANN10	0.2	0.012	0.108	1.02	0.603	0.165	0.916	0.0382	0.387	0.67	0	Complete
1 ANN10	0.3	0.0236	0.0452	0.135	1.33	0.219	0.951	0.0363	2.37	34.1	0	Complete
1 ANN10	0.5	0.036	0.0247	0.575	1.55	0.166	0.805	0.0301	0.4	0.497	0	Complete
1 ANN10	1	0.0195	0.0418	0.201	3.26	0.268	0.926	0.0313	0.764	41.5	0	Complete
2 P-only DeepONet10	0.05	0.0284	0.101	1.06	0.831	0.179	0.85	0.0367	0.458	0.622	0	Complete
2 P-only DeepONet10	0.1	0.0629	0.063	0.97	0.99	0.447	0.972	0.0492	1.24	1.51	0	Complete
2 P-only DeepONet10	0.2	0.102	0.277	3.65	0.992	0.558	0.65	0.0444	1.43	1.93	0	Complete
2 P-only DeepONet10	0.3	0.441	1.09	0.953	46.4	4.36	4.95	0.593	1.71	13	0	Complete
2 P-only DeepONet10	0.5	0.0261	0.152	1.6	0.969	0.289	0.89	0.0422	0.895	1.05	0	Complete
2 P-only DeepONet10	1	0.0223	0.148	0.939	3.65	0.733	0.491	0.0544	3.57	17.2	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0407	0.181	2.09	0.915	0.17	0.949	0.0419	0.391	0.494	0	Complete
3 ANN10 + P-only DeepONet10	0.1	8.62e-03	0.0401	0.15	0.275	0.276	0.0993	0.0162	0.778	1	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0793	0.232	3.02	0.951	0.365	0.67	0.0372	1.02	1.39	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.316	0.807	0.892	24.4	3.12	5.25	0.46	1.63	35.7	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.132	0.193	0.145	10.9	0.245	3.65	0.161	1.29	155	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0194	0.139	0.912	3.84	0.634	0.54	0.0512	3.54	17	0	Complete
4 T/P DeepONet10	0.05	0.0424	0.11	0.997	1.01	0.702	0.997	0.0274	1.29	2.48	0	Complete
4 T/P DeepONet10	0.1	0.04	0.127	0.968	1	0.844	0.993	0.0346	2.01	13	0	Complete
4 T/P DeepONet10	0.2	8.38e-03	0.0456	0.133	0.308	0.317	0.0687	0.0183	1.54	6.94	0	Complete
4 T/P DeepONet10	0.3	0.0223	0.094	0.18	0.573	0.709	0.129	0.0407	1.01	0.916	0	Complete
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.114 ms
4 T/P DeepONet10	1	0.421	0.681	11.5	1.12	0.713	1.1	0.0578	7.46	64.4	0	Complete
5 ANN10 + T/P DeepONet10	0.05	9.00e-03	0.219	0.992	0.986	1.52	0.995	0.0599	0.759	1.2	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0538	0.0417	0.965	0.912	0.184	0.98	0.0292	0.891	0.997	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.012	0.058	0.139	0.337	0.415	0.0629	0.0242	1.44	1.78	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0185	0.0817	0.162	0.389	0.6	0.0871	0.0351	1.25	1.09	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.107	0.322	0.856	4.48	2.24	1.35	0.161	4.25	8.54	0	Complete
5 ANN10 + T/P DeepONet10	1	0.318	1.19	0.68	1.79	8.83	0.61	0.551	0.984	56.4	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.634	0.138	9.11	0.991	0.261	12.4	0.435	7.69	26.4	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.285	0.0903	3.13	1.31	0.345	6.75	0.254	2.61	7.38	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0148	0.0538	0.163	0.361	0.391	0.0706	0.024	1.7	2.68	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.105	0.297	3.8	0.514	0.737	0.87	0.0563	1	0.923	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.212	0.348	0.321	13.9	0.799	6.35	0.276	1.23	2	0	Complete
6 T/P/ϕ DeepONet10	1	0.147	0.161	3.06	6.39	0.726	0.94	0.0634	2.32	3.05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	6.96e-03	0.221	0.989	0.97	1.53	0.993	0.0604	0.534	0.947	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.272	0.13	2.39	0.126	0.0193	7.65	0.272	1.92	9.92	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0123	0.0597	0.148	0.32	0.426	0.0626	0.0248	1.34	1.5	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0868	0.263	3.3	0.546	0.637	0.866	0.052	1.13	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.114	0.376	1.13	3.06	2.56	1.29	0.18	2.88	8.17	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.152	0.171	3.24	6.04	0.713	0.935	0.0618	3.26	4.68	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 32

T_0=1140.7 K P_0=3.354 atm ϕ_0=1.463 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] · Y_{H_2} · Y_H · Y_O · Y_{O_2} · Y_{OH} · Y_{H_2O} · Y_{HO_2} · $Y_{H_2O_2}$ · Y_{N_2}

g=1: stop 0.202 ms

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 32

Repaired | Cantera vs NN NRMSE | T0=1140.7K, P0=3.354atm, phi0=1.463

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0749	0.116	1.85	0.816	0.183	1.26	0.0413	0.445	0.588	0	Complete
1 ANN10	0.1	9.01e-03	0.0869	0.463	7.61	0.316	0.916	0.0206	1.71	11.7	0	Complete
1 ANN10	0.2	0.0497	0.164	0.596	0.423	0.366	2.19	0.101	0.156	0.165	0	Complete
1 ANN10	0.3	0.0159	0.0363	0.119	1.36	0.061	0.827	0.0297	0.138	3.17	0	Complete
1 ANN10	0.5	0.0157	0.0522	0.141	0.482	0.278	0.51	0.0284	0.748	0.957	0	Complete
1 ANN10	1	0.519	1.9	0.979	0.888	14.1	0.636	0.884	1	25.9	0	Complete
2 P-only DeepONet10	0.05	0.0539	0.191	2.34	0.885	0.244	0.765	0.0367	0.61	0.817	0	Complete
2 P-only DeepONet10	0.1	4.12e-03	0.0241	0.0913	0.205	0.159	0.0408	9.28e-03	0.393	0.454	0	Complete
2 P-only DeepONet10	0.2	3.64e-03	0.0249	0.0857	0.2	0.165	0.0379	9.53e-03	0.427	0.521	0	Complete
2 P-only DeepONet10	0.3	5.02e-03	0.0334	0.0989	0.242	0.227	0.0433	0.013	0.574	0.73	0	Complete
2 P-only DeepONet10	0.5	0.0121	0.0838	0.539	0.943	0.0703	0.979	0.0419	0.199	0.519	0	Complete
2 P-only DeepONet10	1	0.743	0.0888	9.57	55.2	0.985	5.68	0.435	2.28	1.8e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0943	0.259	3.5	0.869	0.214	0.703	0.0339	0.528	0.671	0	Complete
3 ANN10 + P-only DeepONet10	0.1	2.66e-03	0.0204	0.0726	0.168	0.133	0.0299	7.65e-03	0.315	0.306	0	Complete
3 ANN10 + P-only DeepONet10	0.2	2.26e-03	0.0197	0.0689	0.164	0.128	0.0337	7.37e-03	0.311	0.298	0	Complete
3 ANN10 + P-only DeepONet10	0.3	1.80e-03	0.0176	0.0661	0.169	0.11	0.0386	6.57e-03	0.265	0.239	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0111	0.0863	0.566	0.943	0.125	0.968	0.0419	0.325	0.446	0	Complete
3 ANN10 + P-only DeepONet10	1	0.567	1.01	2.13	32.6	5.53	5.77	0.658	2.71	893	0	Complete
4 T/P DeepONet10	0.05	0.625	0.175	9.42	4.99	0.292	10.8	0.397	14.5	146	0	Complete
4 T/P DeepONet10	0.1	0.187	0.112	3.31	0.464	0.414	3.09	0.112	2.21	5.07	0	Complete
4 T/P DeepONet10	0.2	0.0942	0.282	0.517	8.83	1.14	1.6	0.149	2.75	3.93	0	Complete
4 T/P DeepONet10	0.3	8.90e-03	0.0504	0.131	0.38	0.346	0.0934	0.0205	1.07	1.7	0	Complete
4 T/P DeepONet10	0.5	0.101	0.196	2.99	2.55	0.396	0.317	0.026	1.86	6.2	0	Complete
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.202 ms
5 ANN10 + T/P DeepONet10	0.05	0.0544	0.0517	0.938	0.967	0.303	0.994	0.0345	0.558	2.93	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.186	0.113	3.37	0.682	0.236	3.06	0.105	1.66	5.32	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0101	0.0544	0.131	0.323	0.383	0.0558	0.0222	1.05	1.33	0	Complete
5 ANN10 + T/P DeepONet10	0.3	8.05e-03	0.0473	0.132	0.339	0.324	0.0789	0.019	0.897	1.29	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0159	0.087	0.558	0.609	0.511	0.207	0.026	1.26	2.87	0	Complete
5 ANN10 + T/P DeepONet10	1	0.716	0.69	15.6	7.73	0.934	4.13	0.221	1.61	1.48e+03	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.482	0.0985	6.15	0.992	0.274	10.9	0.382	0.906	10.5	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.544	0.288	9.48	1.01	0.51	8.21	0.3	2.57	11.8	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0306	0.0931	0.555	0.504	0.655	0.158	0.0394	1.45	1.74	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.235	0.662	0.962	5.53	2.84	5.5	0.391	8.68	8.44	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.236	0.259	1.42	5.67	1.01	5.69	0.263	3.42	10.5	0	Complete
6 T/P/ϕ DeepONet10	1	0.83	0.987	19.2	16.8	0.731	2.79	0.171	0.538	5.87e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.606	0.56	2.58	0.918	0.231	20.1	0.708	0.601	1.8	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.055	0.0547	0.952	0.861	0.328	0.99	0.0357	0.933	3.41	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0717	0.215	2.74	0.453	0.459	0.714	0.04	1.2	1.49	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.166	0.54	0.968	3.27	2.51	3.59	0.298	1.3	4.2	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.265	0.556	0.208	2.65	2.09	7.13	0.393	1.76	5.88	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.6	0.112	9.83	6.42	2.79	4.8	0.369	0.849	882	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 39

T_0=1250.1 K P_0=2.297 atm ϕ_0=0.646 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 39

Repaired | Cantera vs NN NRMSE | T0=1250.1K, P0=2.297atm, phi0=0.646

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0169	0.127	6.43	0.805	0.142	0.901	0.0109	0.589	0.568	0	Complete
1 ANN10	0.1	0.15	0.476	2.07	0.897	0.329	5.49	0.241	0.78	0.837	0	Complete
1 ANN10	0.2	0.0857	0.454	0.774	0.918	0.173	3.28	0.149	0.915	0.953	0	Complete
1 ANN10	0.3	4.50e-03	0.269	0.194	0.428	0.055	0.61	0.0105	0.221	0.925	0	Complete
1 ANN10	0.5	0.0385	0.514	1.04	0.583	0.0379	0.823	0.0568	0.954	0.984	0	Complete
1 ANN10	1	0.0298	0.68	8.55	0.465	0.133	0.913	0.0289	0.621	73	0	Complete
2 P-only DeepONet10	0.05	0.0363	0.12	0.793	2.21	0.12	0.71	0.0292	0.674	0.383	0	Complete
2 P-only DeepONet10	0.1	3.93e-03	0.236	0.758	1.17	6.65e-03	0.321	6.61e-03	0.331	0.24	0	Complete
2 P-only DeepONet10	0.2	0.035	0.485	0.831	0.779	6.57e-03	0.301	0.0412	0.953	0.979	0	Complete
2 P-only DeepONet10	0.3	0.132	0.512	8.75	0.535	0.16	3	0.142	0.962	0.479	0	Complete
2 P-only DeepONet10	0.5	0.54	17	1.38	1.87	1.37	1.87	0.938	0.986	60.7	0	Complete
2 P-only DeepONet10	1	0.542	17.7	1.21	3.7	1.42	0.828	0.929	0.551	74.1	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.191	0.0779	4.02	0.821	0.381	6.17	0.266	2.79	1.97	0	Complete
3 ANN10 + P-only DeepONet10	0.1	5.60e-03	0.233	0.703	1.3	7.30e-03	0.33	6.99e-03	0.292	0.178	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0153	0.448	0.755	0.529	0.0508	0.286	0.0155	0.941	0.969	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0609	0.506	2.3	0.781	0.167	2.36	0.0812	0.955	0.737	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.107	0.452	1.61	0.515	0.312	3.76	0.136	0.943	31.7	0	Complete
3 ANN10 + P-only DeepONet10	1	0.065	0.581	5.62	0.927	0.0889	0.882	0.0453	0.902	84.8	0	Complete
4 T/P DeepONet10	0.05	0.0572	3.13	0.796	0.948	0.362	0.997	0.129	1.36	1.84	0	Complete
4 T/P DeepONet10	0.1	0.0406	0.212	0.694	0.675	0.0647	0.897	0.0488	0.769	0.794	0	Complete
4 T/P DeepONet10	0.2	4.09e-03	0.216	0.339	0.129	0.0224	0.0389	9.57e-03	0.138	0.121	0	Complete
4 T/P DeepONet10	0.3	4.42e-03	0.276	0.373	0.143	0.0287	0.0329	0.012	0.176	0.146	0	Complete
4 T/P DeepONet10	0.5	0.0582	0.646	3.83	1.49	0.129	0.927	0.0406	6.97	9.67	0	Complete
4 T/P DeepONet10	1	0.281	9.34	1.41	2.52	0.794	0.242	0.474	0.798	55.8	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.026	1.57	0.828	0.921	0.224	0.995	0.0514	1.93	2.36	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0362	0.13	0.568	0.562	0.0688	0.891	0.0437	0.833	0.881	0	Complete
5 ANN10 + T/P DeepONet10	0.2	1.88e-03	0.148	0.223	0.0828	0.0152	0.0166	6.17e-03	0.106	0.0986	0	Complete
5 ANN10 + T/P DeepONet10	0.3	2.92e-03	0.216	0.323	0.0988	0.0216	0.0201	9.00e-03	0.144	0.118	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0545	0.551	1.76	1.96	0.187	1.29	0.0393	9.56	12.7	0	Complete
5 ANN10 + T/P DeepONet10	1	0.467	14.7	3.22	4.35	1.24	0.229	0.759	0.73	113	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0372	1.79	0.741	0.959	0.236	0.997	0.0708	0.9	1.29	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0408	0.197	0.787	0.562	0.0679	0.955	0.0499	0.634	0.567	0	Complete
6 T/P/ϕ DeepONet10	0.2	8.71e-03	0.242	0.583	0.181	0.0243	0.0644	0.0126	0.18	0.121	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0205	0.45	0.892	0.171	0.0387	0.612	0.0395	0.309	0.436	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.484	0.683	16	18.5	0.974	6.44	0.322	95.3	174	0	Complete
6 T/P/ϕ DeepONet10	1	0.156	5.78	0.573	0.782	0.516	0.238	0.285	0.373	9.41	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0107	1.23	0.833	0.962	0.206	0.995	0.0195	0.343	0.356	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0351	0.121	0.697	0.309	0.0652	0.934	0.0446	0.686	0.692	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	5.87e-03	0.161	0.512	0.088	0.0165	0.053	8.32e-03	0.182	0.117	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0183	0.405	0.874	0.128	0.0305	0.551	0.036	0.313	0.423	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0663	2.05	2.81	0.887	0.204	0.257	0.104	0.834	0.756	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.483	14.8	2.49	4.84	1.18	0.343	0.778	0.67	89.5	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 39

T_0=1250.1 K P_0=2.297 atm ϕ_0=0.646 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] · Y_{H_2} · Y_H · Y_O · Y_{O_2} · Y_{OH} · Y_{H_2O} · Y_{HO_2} · $Y_{H_2O_2}$ · Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 39

Repaired | Cantera vs NN NRMSE | T0=1250.1K, P0=2.297atm, phi0=0.646

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0864	0.206	0.616	0.717	0.27	3.78	0.139	0.5	0.381	0	Complete
1 ANN10	0.1	0.174	0.522	0.434	0.831	0.38	6.24	0.273	0.429	0.37	0	Complete
1 ANN10	0.2	7.56e-03	0.208	0.672	1.46	0.0449	0.774	0.028	0.616	2.65	0	Complete
1 ANN10	0.3	0.0108	0.364	0.335	0.215	0.0186	0.0692	0.0193	0.106	0.0468	0	Complete
1 ANN10	0.5	0.0152	0.796	0.289	0.105	0.0795	0.197	0.032	0.0634	0.415	0	Complete
1 ANN10	1	0.33	11.7	2.49	0.676	1.04	1.07	0.612	15.5	106	0	Complete
2 P-only DeepONet10	0.05	0.188	0.381	9.28	0.362	0.245	4.61	0.232	1.4	1.17	0	Complete
2 P-only DeepONet10	0.1	1.24e-03	0.0311	0.031	0.0623	1.13e-03	5.79e-03	1.34e-03	0.0319	0.0516	0	Complete
2 P-only DeepONet10	0.2	1.25e-03	0.0261	0.0212	0.0464	1.18e-03	2.87e-03	1.38e-03	0.0447	0.0213	0	Complete
2 P-only DeepONet10	0.3	1.27e-03	0.0294	0.017	0.0437	1.60e-03	4.19e-03	1.51e-03	0.07	0.0995	0	Complete
2 P-only DeepONet10	0.5	0.0229	1.33	0.812	0.248	0.127	0.318	0.0507	0.488	0.682	0	Complete
2 P-only DeepONet10	1	0.754	4.54	30.6	18.7	0.347	6.09	0.641	0.465	690	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.252	0.116	9.27	0.624	0.367	6.66	0.318	2.48	2.25	0	Complete
3 ANN10 + P-only DeepONet10	0.1	1.21e-03	0.0413	0.0471	0.083	1.90e-03	0.017	1.31e-03	0.0373	0.072	0	Complete
3 ANN10 + P-only DeepONet10	0.2	7.96e-04	0.0198	0.0177	0.0339	1.11e-03	5.51e-03	8.95e-04	0.0297	0.0152	0	Complete
3 ANN10 + P-only DeepONet10	0.3	1.06e-03	0.0319	0.0183	0.0437	1.68e-03	4.28e-03	1.42e-03	0.063	0.0884	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0324	1.58	0.802	0.412	0.142	0.279	0.0653	0.6	0.514	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0259	0.521	5.96	0.0921	0.0385	0.315	9.63e-03	0.965	0.744	0	Complete
4 T/P DeepONet10	0.05	0.281	0.978	12.6	3.73	0.22	4.28	0.289	3.88	1.78	0	Complete
4 T/P DeepONet10	0.1	0.173	0.154	5.21	0.255	0.295	4.79	0.22	0.959	0.952	0	Complete
4 T/P DeepONet10	0.2	8.43e-04	0.0762	0.123	0.0368	7.44e-03	6.39e-03	3.11e-03	0.0601	0.0425	0	Complete
4 T/P DeepONet10	0.3	5.73e-04	0.0314	0.0664	0.0265	2.95e-03	7.58e-03	1.39e-03	0.0477	0.0652	0	Complete
4 T/P DeepONet10	0.5	0.0869	2.55	0.853	1.61	0.147	0.354	0.14	0.387	0.69	0	Complete
4 T/P DeepONet10	1	0.275	8.87	1.24	1.9	0.731	0.291	0.469	0.27	54.8	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.268	0.564	11.1	2.33	0.284	5.11	0.293	2.8	1.36	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.253	0.278	8.86	0.355	0.43	7.01	0.315	0.925	0.957	0	Complete
5 ANN10 + T/P DeepONet10	0.2	3.22e-04	0.0361	0.0662	0.0153	3.31e-03	4.07e-03	1.42e-03	0.0282	0.0196	0	Complete
5 ANN10 + T/P DeepONet10	0.3	6.23e-04	0.0212	0.0397	0.0222	1.61e-03	8.51e-03	1.10e-03	0.0408	0.0592	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0334	1.2	0.79	0.653	0.0776	0.0497	0.059	0.308	0.239	0	Complete
5 ANN10 + T/P DeepONet10	1	0.415	11	7.75	3.17	0.916	0.724	0.632	0.739	141	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.367	0.846	18.3	3.54	0.284	5.85	0.376	2.53	0.756	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.37	0.536	18.1	2.29	0.425	7.61	0.407	1.37	5.3	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0217	0.109	0.817	0.411	0.0223	0.334	0.0224	0.18	0.326	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0217	0.868	0.842	0.0331	0.057	0.184	0.0471	0.0754	0.216	0	Complete
6 T/P/ϕ DeepONet10	0.5	2.72e-03	0.221	0.328	0.117	0.0213	0.0177	9.10e-03	0.241	0.16	0	Complete
6 T/P/ϕ DeepONet10	1	0.188	6.47	0.878	1.35	0.538	0.146	0.333	0.528	22.5	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.314	0.491	13.9	1.7	0.325	6.24	0.35	3.02	1.52	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0331	0.202	0.61	0.216	0.0392	0.78	0.0447	0.579	0.624	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	4.58e-03	0.107	0.731	0.095	7.72e-03	0.0247	1.31e-03	0.0841	0.314	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0292	1.02	0.835	0.181	0.0612	0.249	0.0577	0.0378	0.238	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	1.84e-03	0.134	0.187	0.0673	0.0133	0.0136	5.67e-03	0.121	0.105	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.388	7.94	7.65	4.65	0.478	2.35	0.538	1.15	197	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 44

T_0=1031.1 K P_0=2.272 atm ϕ_0=0.770 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 44

Repaired | Cantera vs NN NRMSE | T0=1031.1K, P0=2.272atm, phi0=0.770

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.229	0.829	1.2	1.02	0.614	5.08	0.368	1.25	1.63	0	Complete
1 ANN10	0.1	0.231	0.688	2.26	0.999	0.481	5.74	0.363	1.11	1.23	0	Complete
1 ANN10	0.2	0.196	0.796	1.33	1.01	0.552	3.1	0.328	1.19	1.16	0	Complete
1 ANN10	0.3	0.748	2.69	1.29	13.8	1.55	0.967	1.04	5.96	1.76	0	Complete
1 ANN10	0.5	0.218	0.914	1.12	0.542	0.616	0.994	0.359	1.93	3.52	0	Complete
1 ANN10	1	0.597	2.57	0.753	0.852	1.77	0.964	0.993	0.999	0.994	0	Complete
2 P-only DeepONet10	0.05	0.697	1.93	24.4	1	1.46	1	0.932	55.4	2.51e+03	0	Complete
2 P-only DeepONet10	0.1	0.252	0.701	8.52	0.997	0.528	5.19	0.335	1.82	1.38	0	Complete
2 P-only DeepONet10	0.2	0.21	0.881	1.14	0.918	0.601	0.474	0.342	1.11	1.02	0	Complete
2 P-only DeepONet10	0.3	0.203	0.923	4.58	0.721	0.71	3.97	0.329	1.04	0.975	0	Complete
2 P-only DeepONet10	0.5	0.616	2.51	0.94	1.27	1.65	0.368	0.991	0.906	3.02	0	Complete
2 P-only DeepONet10	1	0.593	2.58	0.95	0.995	1.77	0.998	0.99	1	0.985	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.684	2	2.1	13.5	0.903	3.04	0.893	9.65	2.43	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.194	0.773	1.77	1.02	0.628	5.32	0.317	1.03	0.986	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.202	0.888	1.14	0.676	0.618	0.588	0.336	1.12	1.02	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.2	0.925	0.837	0.91	0.67	1.94	0.337	1.08	0.998	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.601	2.51	0.927	0.985	1.68	0.688	0.976	0.815	0.972	0	Complete
3 ANN10 + P-only DeepONet10	1	0.592	2.57	0.934	0.992	1.77	0.997	0.989	1	0.987	0	Complete
4 T/P DeepONet10	0.05	0.63	2.54	1.67	5.43	1.64	0.995	0.979	23.6	1.83	0	Complete
4 T/P DeepONet10	0.1	0.169	0.784	4.31	0.488	0.547	0.67	0.288	10.8	819	0	Complete
4 T/P DeepONet10	0.2	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
4 T/P DeepONet10	0.3	0.599	2.6	1	1	1.79	1	1	0.998	1	0	Complete
4 T/P DeepONet10	0.5	0.596	2.58	0.86	0.992	1.78	0.997	0.993	5.34	1.85	0	Complete
4 T/P DeepONet10	1	0.349	1.38	1.36	3.23	0.897	0.434	0.542	1.07	6.72	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.146	0.648	2	2.44	0.471	1.04	0.244	3.91	1.28	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.782	0.972	57.9	0.965	1	0.638	0.733	221	3.74e+03	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.203	0.948	1.99	5.09	0.767	2.6	0.339	12.5	3.02	0	Complete
5 ANN10 + T/P DeepONet10	1	0.48	1.77	3.32	4.9	1.13	0.464	0.706	1.09	15.4	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.599	2.6	1	1	1.79	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.599	2.6	1	1	1.79	1	1	0.998	1	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.514	0.175	24.5	14.2	0.52	6.58	0.336	38.8	27.8	0	Complete
6 T/P/ϕ DeepONet10	1	0.206	0.979	6.97	2.69	0.689	1.02	0.355	0.866	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.599	2.6	1	1	1.79	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.599	2.6	1	1	1.79	1	1	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.599	2.6	1	1	1.79	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.599	2.6	1	1	1.79	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.206	0.896	4.15	1.36	0.619	0.508	0.341	1.32	0.994	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.489	1.74	3.17	5.47	1.08	0.389	0.71	0.915	11.4	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 44

T_0=1031.1 K P_0=2.272 atm ϕ_0=0.770 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 44

Repaired | Cantera vs NN NRMSE | T0=1031.1K, P0=2.272atm, phi0=0.770

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.226	0.813	1.11	0.995	0.629	5.45	0.36	1.13	1.08	0	Complete
1 ANN10	0.1	0.25	0.575	1.31	0.997	0.461	7.2	0.373	1.1	1.23	0	Complete
1 ANN10	0.2	0.191	0.845	1.17	2.81	0.584	0.976	0.323	1.31	1.25	0	Complete
1 ANN10	0.3	0.181	0.788	1.63	0.512	0.536	0.524	0.304	1.08	1.14	0	Complete
1 ANN10	0.5	0.206	0.886	0.853	0.479	0.607	0.45	0.343	1.08	1.25	0	Complete
1 ANN10	1	0.347	1.43	1.55	0.791	0.956	1.05	0.571	9.09	9.42	0	Complete
2 P-only DeepONet10	0.05	0.323	0.873	8.78	0.553	0.641	6.28	0.437	1.29	1.02	0	Complete
2 P-only DeepONet10	0.1	0.194	0.847	0.795	0.425	0.584	0.336	0.325	0.984	1.01	0	Complete
2 P-only DeepONet10	0.2	0.207	0.899	0.779	0.436	0.618	0.344	0.346	0.984	1	0	Complete
2 P-only DeepONet10	0.3	0.183	0.795	0.749	0.406	0.547	0.308	0.306	0.916	0.988	0	Complete
2 P-only DeepONet10	0.5	0.205	0.886	1.02	0.647	0.611	0.411	0.342	0.879	0.987	0	Complete
2 P-only DeepONet10	1	0.813	0.926	34.7	23.4	0.734	6.38	0.611	0.935	106	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.317	0.857	9.37	0.703	0.632	6.06	0.422	1.54	1.05	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.189	0.823	0.857	0.38	0.567	0.337	0.316	1.01	1.03	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.203	0.884	0.791	0.425	0.608	0.343	0.34	1	1.01	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.176	0.766	0.767	0.39	0.527	0.301	0.294	0.949	0.999	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.201	0.909	14.9	1.02	0.593	0.963	0.336	1.16	1.03	0	Complete
3 ANN10 + P-only DeepONet10	1	0.492	1.61	7.18	4.23	1.03	0.712	0.689	0.931	22.2	0	Complete
4 T/P DeepONet10	0.05	0.347	1.51	1.02	0.95	1.04	0.992	0.58	5.09	3.18	0	Complete
4 T/P DeepONet10	0.1	0.269	0.881	10.8	0.537	0.707	5.76	0.353	1.06	1.01	0	Complete
4 T/P DeepONet10	0.2	0.208	0.905	0.844	0.429	0.623	0.358	0.347	0.947	1	0	Complete
4 T/P DeepONet10	0.3	0.202	0.881	0.829	0.425	0.606	0.344	0.338	0.908	0.989	0	Complete
4 T/P DeepONet10	0.5	0.598	2.6	0.999	1	1.79	1	1	1	1	0	Complete
4 T/P DeepONet10	1	0.325	1.3	0.856	2.04	0.84	0.441	0.519	0.807	5.97	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.158	0.619	1.04	0.742	0.413	0.997	0.252	1.5	2.25	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.338	0.857	13.6	0.713	0.751	8.27	0.404	1.11	1.02	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.206	0.898	0.852	0.423	0.618	0.356	0.345	0.953	0.996	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.205	0.895	0.84	0.427	0.616	0.349	0.344	0.932	0.989	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.598	2.6	0.999	1	1.79	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.42	1.46	6.57	3.42	0.963	0.442	0.608	0.886	13	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.071	0.232	0.973	0.893	0.16	0.991	0.103	1.18	1.37	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.537	0.286	25.1	0.394	0.448	10.9	0.512	1.21	2.08	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.197	0.872	4.07	0.947	0.599	0.444	0.331	1.02	1.02	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.2	0.862	1.02	0.448	0.596	0.355	0.332	0.949	1.01	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.232	0.974	0.909	0.729	0.653	0.61	0.383	0.959	0.995	0	Complete
6 T/P/ϕ DeepONet10	1	0.285	0.91	3.07	10.2	0.888	5.6	0.361	1.26	17	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.185	0.729	0.989	0.789	0.485	0.993	0.296	1.34	1.25	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0489	0.149	0.885	0.412	0.104	0.871	0.0712	0.694	1.1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.207	0.887	1.01	0.456	0.611	0.341	0.343	1.01	1.01	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.199	0.865	1.04	0.537	0.599	0.406	0.332	0.969	0.999	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.228	0.961	0.792	0.694	0.647	0.604	0.377	0.964	0.995	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.2	0.95	4.72	1.78	0.661	0.683	0.35	1.16	1.26	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 45

T_0=1181.9 K P_0=4.842 atm ϕ_0=1.099 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 45

Repaired | Cantera vs NN NRMSE | T0=1181.9K, P0=4.842atm, phi0=1.099

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0251	0.397	2.47	1.36	0.258	0.936	0.056	1.19	5.72	0	Complete
1 ANN10	0.1	8.22e-03	0.379	1.98	1.07	0.214	0.942	0.0629	1.61	5.76	0	Complete
1 ANN10	0.2	0.0284	0.306	0.716	0.965	0.357	0.921	0.0736	1.79	6.68	0	Complete
1 ANN10	0.3	0.0553	0.379	0.289	0.405	0.558	0.929	0.0964	2.68	21	0	Complete
1 ANN10	0.5	0.0752	0.316	0.576	0.366	0.616	0.899	0.1	0.635	0.797	0	Complete
1 ANN10	1	0.0523	0.464	0.595	0.325	0.83	0.928	0.103	1.05	39.1	0	Complete
2 P-only DeepONet10	0.05	0.0487	0.304	0.43	0.677	0.28	0.993	0.081	0.905	0.939	0	Complete
2 P-only DeepONet10	0.1	0.25	1.59	0.904	5.59	2.08	2.48	0.364	2	7.4	0	Complete
2 P-only DeepONet10	0.2	0.362	2.27	0.952	13.2	3.1	1.99	0.475	9.65	84.2	0	Complete
2 P-only DeepONet10	0.3	0.343	2.03	1.2	12.9	2.26	2.4	0.443	29.7	159	0	Complete
2 P-only DeepONet10	0.5	0.23	0.957	8.85	0.59	0.724	1.09	0.044	1.03	215	0	Complete
2 P-only DeepONet10	1	0.263	0.372	7.52	1.82	0.742	1.07	0.134	1	611	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0372	0.33	0.839	0.59	0.266	0.988	0.0778	0.851	0.898	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.148	0.982	0.896	1.99	1.19	2.01	0.236	2.46	12.9	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.245	1.68	0.968	6.73	2.51	1.74	0.355	6.96	69.1	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.132	1.18	0.974	3.41	2.28	0.433	0.217	6.31	20.4	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.105	0.917	6.14	0.581	0.688	0.0449	0.049	0.877	107	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0215	0.862	4.06	0.961	0.735	0.986	0.114	0.881	15.1	0	Complete
4 T/P DeepONet10	0.05	0.032	0.376	1	1	1.37	1	0.0291	1.07	0.919	0	Complete
4 T/P DeepONet10	0.1	0.137	1.06	1.85	0.835	2.96	1.05	0.199	4.01	5.37	0	Complete
4 T/P DeepONet10	0.2	0.258	1.68	0.881	9.1	2.31	1.6	0.351	5.19	3.06	0	Complete
4 T/P DeepONet10	0.3	0.404	2.48	0.964	14.4	3.33	2.37	0.525	7.08	13	0	Complete
4 T/P DeepONet10	0.5	0.163	0.919	7.44	0.269	0.738	0.29	0.0355	11.4	31.2	0	Complete
4 T/P DeepONet10	1	0.525	4.78	0.969	3.33	12.8	0.658	0.853	0.969	104	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0605	0.811	1	0.994	2.52	0.998	0.117	1.02	1.03	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0687	0.168	0.958	0.974	0.537	0.996	0.0649	1.19	0.682	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0964	0.75	0.744	3.32	1.11	0.69	0.148	2.06	2.06	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.163	1.16	0.881	5.37	1.68	1.1	0.238	1.17	1.98	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0452	0.372	0.964	0.703	0.729	0.724	0.075	0.97	0.741	0	Complete
5 ANN10 + T/P DeepONet10	1	0.208	0.893	8.08	1.07	0.77	0.884	0.0382	6.47	30.4	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0598	0.183	1	1	0.734	1	0.0495	1.05	0.958	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0835	0.76	0.851	1.55	2.22	0.889	0.134	2.86	5.55	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.34	1.68	0.132	7.06	1.71	3.91	0.448	2.86	6.66	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.412	1.7	1.36	9.67	1.74	4.05	0.484	2.47	19.6	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.0178	0.849	3.7	0.73	0.836	0.95	0.117	1.28	1.65	0	Complete
6 T/P/ϕ DeepONet10	1	0.234	0.911	8.7	2.74	0.819	0.713	0.0447	4.24	17.7	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0663	0.707	0.999	0.998	2.19	0.999	0.109	1.03	0.983	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0594	0.181	0.981	0.969	0.723	0.997	0.0499	0.892	0.616	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0202	0.258	0.545	0.397	0.553	0.258	0.0443	0.793	0.632	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.183	1.14	0.8	4.31	1.31	2.1	0.268	0.837	0.858	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.711	1.46	7.29	3.58	1.45	9.13	0.724	8.03	168	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.182	0.891	7.56	1.76	0.791	0.419	0.0277	4.56	22.8	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 45

T_0=1181.9 K P_0=4.842 atm ϕ_0=1.099 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 45

Repaired | Cantera vs NN NRMSE | T0=1181.9K, P0=4.842atm, phi0=1.099

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.126	0.444	0.762	0.918	0.589	3.49	0.193	0.575	0.617	0	Complete
1 ANN10	0.1	0.012	0.14	0.396	3.55	0.133	0.824	0.0268	0.855	4.48	0	Complete
1 ANN10	0.2	0.0226	0.152	0.18	1.72	0.242	0.761	0.051	0.301	0.38	0	Complete
1 ANN10	0.3	0.019	0.14	0.362	1.56	0.261	0.544	0.0333	0.81	2.03	0	Complete
1 ANN10	0.5	0.052	0.274	0.647	1.13	0.575	0.648	0.0754	1.31	6.43	0	Complete
1 ANN10	1	0.048	0.556	1.23	0.889	0.674	0.871	0.108	1.05	5.21	0	Complete
2 P-only DeepONet10	0.05	1.52e-03	0.053	0.178	0.0602	0.0721	0.0373	7.55e-03	0.118	0.216	0	Complete
2 P-only DeepONet10	0.1	6.65e-04	0.0116	0.0223	0.0334	0.0255	8.99e-03	1.52e-03	0.0568	0.125	0	Complete
2 P-only DeepONet10	0.2	2.74e-03	0.0133	0.0962	0.042	0.025	0.0366	1.08e-03	0.1	0.185	0	Complete
2 P-only DeepONet10	0.3	6.18e-03	0.32	1.23	0.599	0.5	0.129	0.0307	0.499	0.385	0	Complete
2 P-only DeepONet10	0.5	0.0698	0.522	0.604	0.995	0.646	0.983	0.119	0.814	4.54	0	Complete
2 P-only DeepONet10	1	0.503	0.494	9.14	11.2	1.17	2.43	0.348	2.65	847	0	Complete
3 ANN10 + P-only DeepONet10	0.05	1.16e-03	0.0397	0.174	0.0282	0.044	0.0314	4.88e-03	0.0644	0.167	0	Complete
3 ANN10 + P-only DeepONet10	0.1	1.07e-03	4.43e-03	0.0198	0.0176	0.0104	0.0171	1.23e-03	0.0455	0.143	0	Complete
3 ANN10 + P-only DeepONet10	0.2	8.58e-04	0.0202	0.0746	0.0594	0.0409	0.0305	1.29e-03	0.0464	0.0664	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0743	0.787	0.821	1.67	1.62	0.142	0.136	3.48	19	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0664	0.525	0.698	0.987	0.628	0.967	0.117	0.739	4.23	0	Complete
3 ANN10 + P-only DeepONet10	1	0.533	3.09	1.52	9.52	6.42	2.32	0.687	2.95	504	0	Complete
4 T/P DeepONet10	0.05	0.149	1.63	0.967	1.75	4.68	0.956	0.271	7.06	152	0	Complete
4 T/P DeepONet10	0.1	0.108	0.177	1.22	0.582	0.414	2.04	0.121	2.51	3.5	0	Complete
4 T/P DeepONet10	0.2	0.0588	0.544	0.538	2.14	1	0.141	0.0955	1.86	1.59	0	Complete
4 T/P DeepONet10	0.3	0.0125	0.143	0.126	0.123	0.398	0.0365	0.0237	0.87	0.669	0	Complete
4 T/P DeepONet10	0.5	0.239	1.82	0.851	2.82	3.57	1.64	0.38	0.761	4.24	0	Complete
4 T/P DeepONet10	1	0.26	0.945	9.35	2.26	0.883	1.04	0.0645	6.16	15.8	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0847	0.19	0.996	0.965	0.266	0.996	0.0871	0.917	0.963	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0403	0.0968	0.65	0.336	0.36	0.829	0.0407	0.921	0.879	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0336	0.372	0.484	1.32	0.717	0.0416	0.0607	1.28	1.22	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0101	0.124	0.126	0.122	0.343	0.0338	0.0202	0.855	0.706	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.149	1.48	0.944	0.89	3.51	0.354	0.269	0.906	1.47	0	Complete
5 ANN10 + T/P DeepONet10	1	0.203	0.956	8.27	1.59	0.835	0.548	0.0353	6.12	17.1	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.205	0.392	2.65	1.09	0.887	3.31	0.212	1.63	12.9	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.514	0.269	8.64	0.29	0.382	6.61	0.446	6.52	9.65	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0959	0.71	0.693	1.76	0.993	1.11	0.157	1.09	1.82	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.327	2.58	1	4.06	5.51	1.53	0.516	0.96	2.2	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.334	1.64	0.527	5.43	2.18	3.52	0.438	0.837	5.5	0	Complete
6 T/P/ϕ DeepONet10	1	0.164	0.829	6.88	1.99	0.797	0.297	0.0286	3.8	6.05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0869	0.218	0.992	0.846	0.333	0.996	0.0922	0.942	0.994	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0693	0.161	0.852	0.118	0.293	0.966	0.076	0.939	0.826	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0391	0.386	0.672	0.84	0.55	0.533	0.0762	0.909	1.04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.2	1.86	0.998	1.98	4.38	0.404	0.343	0.836	1.67	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.369	2.24	0.173	1.39	4.18	3.68	0.547	0.803	4.3	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.188	0.853	7.45	1.83	0.791	0.566	0.0383	6.43	12.9	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 53

T_0=1257.8 K P_0=2.103 atm ϕ_0=1.799 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 53

Repaired | Cantera vs NN NRMSE | T0=1257.8K, P0=2.103atm, phi0=1.799

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0362	0.0222	0.466	0.494	0.0361	0.895	0.0292	0.139	0.551	0	Complete
1 ANN10	0.1	0.0249	0.105	0.84	0.642	0.0369	3.25	0.0957	0.118	0.566	0	Complete
1 ANN10	0.2	0.0819	0.19	0.818	3.04	1.08	4.2	0.18	0.169	0.626	0	Complete
1 ANN10	0.3	0.028	0.0151	0.411	3.67	0.0417	0.925	0.0208	8.14	193	0	Complete
1 ANN10	0.5	0.0278	2.82e-03	0.299	1.12	0.0472	0.807	0.0236	0.182	0.654	0	Complete
1 ANN10	1	0.164	0.0701	0.976	3.61	0.569	5.49	0.176	40.1	655	0	Complete
2 P-only DeepONet10	0.05	0.0428	0.133	0.891	5.36	0.241	3.21	0.116	0.351	0.36	0	Complete
2 P-only DeepONet10	0.1	0.0551	0.0218	0.754	0.845	0.0818	0.868	0.03	0.247	0.641	0	Complete
2 P-only DeepONet10	0.2	8.21e-03	6.75e-03	0.117	0.262	0.0918	0.0954	5.37e-03	0.23	0.218	0	Complete
2 P-only DeepONet10	0.3	5.43e-03	7.74e-03	0.0681	0.282	0.119	0.0687	5.82e-03	0.338	1.1	0	Complete
2 P-only DeepONet10	0.5	0.349	0.428	0.255	48.8	2.1	8.57	0.463	0.795	110	0	Complete
2 P-only DeepONet10	1	0.0332	0.034	0.413	2.24	0.514	0.448	0.0272	1.08	2.69	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0561	0.133	0.913	6.07	0.212	3.65	0.124	0.255	0.397	0	Complete
3 ANN10 + P-only DeepONet10	0.1	2.57e-03	2.74e-03	0.0432	0.0484	0.0322	0.0285	1.59e-03	0.131	0.184	0	Complete
3 ANN10 + P-only DeepONet10	0.2	4.41e-03	5.04e-03	0.0555	0.191	0.0758	0.0651	4.06e-03	0.169	0.223	0	Complete
3 ANN10 + P-only DeepONet10	0.3	1.67e-03	2.61e-03	0.0171	0.153	0.0379	0.0297	2.08e-03	0.107	0.458	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.13	0.226	0.824	18.5	0.469	5.43	0.226	0.529	156	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0168	0.0442	0.466	0.692	0.484	0.313	0.022	1.19	11	0	Complete
4 T/P DeepONet10	0.05	0.475	0.16	6.01	2.15	0.175	8.85	0.273	4.24	20.3	0	Complete
4 T/P DeepONet10	0.1	0.0311	0.0186	0.439	0.467	0.196	0.463	0.0176	0.488	0.557	0	Complete
4 T/P DeepONet10	0.2	3.31e-03	0.0221	0.0967	0.271	0.333	0.0606	0.0131	0.762	0.565	0	Complete
4 T/P DeepONet10	0.3	0.464	0.514	0.139	87.1	2.2	7.96	0.538	64.1	111	0	Complete
4 T/P DeepONet10	0.5	0.441	0.395	0.507	87.3	0.479	8.58	0.468	93.7	476	0	Complete
4 T/P DeepONet10	1	0.0774	0.0701	0.864	3.23	1.39	0.502	0.0716	1.42	25.3	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.362	0.0696	2.68	0.445	0.132	11.6	0.35	1.38	9.81	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0172	0.0109	0.236	0.198	0.13	0.287	0.0107	0.341	0.331	0	Complete
5 ANN10 + T/P DeepONet10	0.2	2.43e-03	0.0158	0.071	0.205	0.234	0.0676	9.51e-03	0.585	0.451	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0104	0.024	0.246	0.287	0.289	0.0983	0.012	0.705	0.605	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.168	0.263	0.996	14.4	3.31	4.37	0.249	14.3	73	0	Complete
5 ANN10 + T/P DeepONet10	1	0.13	0.222	0.396	2.47	4.31	0.145	0.192	1.07	75.1	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.514	0.174	6.46	0.454	0.21	9.93	0.299	6.08	45.1	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.149	0.0297	1.67	1.47	0.293	3.28	0.106	1.41	6.57	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0207	0.0421	0.527	0.305	0.407	0.18	0.0168	0.847	0.578	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.305	0.107	1.95	10.1	0.558	8.93	0.304	9.11	42.5	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.78	0.552	12.5	2.97	0.535	7.05	0.229	195	5.67e+03	0	Complete
6 T/P/ϕ DeepONet10	1	0.0611	0.0711	0.462	1.04	1.54	0.137	0.0724	1.14	49.2	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.492	0.435	0.37	0.865	0.164	23.9	0.713	0.876	14.2	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0109	0.0127	0.15	0.173	0.181	0.196	9.52e-03	0.429	0.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0125	0.0308	0.36	0.225	0.292	0.192	0.0131	0.683	0.543	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0366	0.0392	0.67	0.539	0.342	0.288	0.0164	0.806	0.738	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0369	0.0303	0.451	0.841	0.445	0.684	0.03	0.981	0.798	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.0347	0.0522	0.282	0.696	1.03	0.0748	0.0455	1.1	30.8	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 53

T_0=1257.8 K P_0=2.103 atm ϕ_0=1.799 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 53

Repaired | Cantera vs NN NRMSE | T0=1257.8K, P0=2.103atm, phi0=1.799

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0467	0.033	0.591	0.618	0.0369	1.55	0.0439	0.0615	0.261	0	Complete
1 ANN10	0.1	0.0146	0.0273	0.321	5.33	0.11	0.737	0.0126	1.09	19.9	0	Complete
1 ANN10	0.2	0.0905	0.172	0.754	0.557	0.435	5.8	0.192	0.0893	0.552	0	Complete
1 ANN10	0.3	0.0128	0.0178	0.097	0.691	0.0555	0.76	0.0231	0.303	15.5	0	Complete
1 ANN10	0.5	0.0167	0.0228	0.0469	0.0804	0.321	0.353	0.0249	0.0648	0.612	0	Complete
1 ANN10	1	0.153	0.0545	1.04	3.76	0.55	4.82	0.154	41.2	573	0	Complete
2 P-only DeepONet10	0.05	0.269	0.273	5.25	0.448	0.0297	1.19	0.0339	0.558	5.93	0	Complete
2 P-only DeepONet10	0.1	5.15e-03	5.83e-03	0.0958	0.113	0.0452	0.0489	2.61e-03	0.193	0.174	0	Complete
2 P-only DeepONet10	0.2	1.69e-03	7.24e-03	0.0481	0.0712	0.101	0.0237	4.07e-03	0.265	0.444	0	Complete
2 P-only DeepONet10	0.3	2.49e-03	8.02e-03	0.0392	0.116	0.117	0.0626	5.20e-03	0.303	0.203	0	Complete
2 P-only DeepONet10	0.5	0.534	0.285	2.61	61.8	1.21	9.92	0.496	42.1	1.66e+03	0	Complete
2 P-only DeepONet10	1	0.84	0.048	8.18	88.9	0.234	8.04	0.457	1.08	4.09e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.185	0.233	4.07	0.74	0.0407	0.156	7.79e-03	0.0871	2.12	0	Complete
3 ANN10 + P-only DeepONet10	0.1	2.45e-03	5.37e-03	0.0668	0.151	0.0331	0.0347	2.16e-03	0.121	0.164	0	Complete
3 ANN10 + P-only DeepONet10	0.2	8.50e-04	3.96e-03	0.0317	0.0658	0.0481	0.0324	2.20e-03	0.14	0.339	0	Complete
3 ANN10 + P-only DeepONet10	0.3	8.45e-04	4.03e-03	0.0191	0.0553	0.0559	0.0408	2.63e-03	0.16	0.137	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.128	0.23	0.799	13.1	0.789	5.58	0.233	3.28	280	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0694	0.054	0.169	6.75	0.0995	2.11	0.083	0.766	965	0	Complete
4 T/P DeepONet10	0.05	0.529	0.231	7.15	0.967	0.0917	9.27	0.274	0.346	45.9	0	Complete
4 T/P DeepONet10	0.1	0.147	0.0762	2.2	2.06	0.119	1.84	0.0649	0.52	1.3	0	Complete
4 T/P DeepONet10	0.2	5.95e-03	0.0106	0.0994	0.193	0.155	0.0454	6.50e-03	0.342	0.303	0	Complete
4 T/P DeepONet10	0.3	6.50e-03	0.0151	0.0887	0.273	0.231	0.107	0.0101	0.579	0.445	0	Complete
4 T/P DeepONet10	0.5	0.457	0.291	1.9	59.8	0.792	9.14	0.448	37.7	337	0	Complete
4 T/P DeepONet10	1	0.545	1.32	1	0.997	23.3	0.998	1	1	0.752	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.4	0.128	5.01	2.02	0.0523	7.75	0.237	1.34	9.62	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.372	0.262	6.09	0.209	0.0492	4.22	0.125	0.133	0.625	0	Complete
5 ANN10 + T/P DeepONet10	0.2	2.89e-03	5.95e-03	0.0513	0.126	0.0853	0.0202	3.57e-03	0.17	0.199	0	Complete
5 ANN10 + T/P DeepONet10	0.3	2.39e-03	7.59e-03	0.0344	0.143	0.115	0.045	4.98e-03	0.279	0.257	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0222	0.0214	0.396	0.573	0.115	0.225	5.52e-03	0.257	0.466	0	Complete
5 ANN10 + T/P DeepONet10	1	0.363	0.315	4.67	8.89	5.33	2.89	0.291	22.2	397	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.604	0.321	8.72	3.5	0.0341	8.36	0.258	3.3	4.85	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.223	0.0931	3.11	4.06	0.231	3.03	0.11	0.536	0.961	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.028	0.0205	0.447	0.513	0.182	0.277	0.0127	0.392	0.371	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.197	0.242	4.29	3.03	0.281	0.603	0.0158	1.39	0.557	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.152	0.192	3.37	3.28	0.177	0.74	0.0158	21.4	26.4	0	Complete
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.124 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.438	0.123	5.3	3.11	0.0159	8.73	0.269	1.04	4.79	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.395	0.174	5.4	0.765	0.0585	6.81	0.204	0.135	0.639	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0209	0.0255	0.447	0.276	0.103	0.0342	4.14e-03	0.205	0.198	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.175	0.226	3.91	1.98	0.148	0.664	0.0175	0.912	0.456	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.135	0.171	3	3.76	0.0968	0.749	0.0146	18.3	17.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.408	0.334	6.08	7.72	4.26	3.47	0.249	18.7	461	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 57

T_0=1412.0 K P_0=4.083 atm ϕ_0=0.677 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 57

Repaired | Cantera vs NN NRMSE | T0=1412.0K, P0=4.083atm, phi0=0.677

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0511	0.555	9.53	1.08	0.219	0.944	0.025	0.605	1.14	0	Complete
1 ANN10	0.1	0.0428	0.666	8.62	0.853	0.176	0.956	0.0319	1.51	4.49	0	Complete
1 ANN10	0.2	0.417	12.1	1.12	0.894	0.715	4.53	0.776	0.867	1.03	0	Complete
1 ANN10	0.3	5.57e-03	0.681	0.863	0.504	0.0621	0.367	0.0135	0.46	0.675	0	Complete
1 ANN10	0.5	0.0137	0.82	3.98	0.859	0.105	0.792	0.0484	0.972	0.992	0	Complete
1 ANN10	1	6.36e-03	0.826	5.57	0.576	0.13	0.932	0.0444	0.929	7.06	0	Complete
2 P-only DeepONet10	0.05	0.0533	0.107	0.962	2.62	0.245	1.01	0.042	0.518	0.0693	0	Complete
2 P-only DeepONet10	0.1	0.0101	0.104	0.933	0.986	0.0348	0.33	0.0267	0.132	1.47	0	Complete
2 P-only DeepONet10	0.2	0.066	0.517	2.58	0.279	4.14e-03	0.95	0.0908	4.06	32.8	0	Complete
2 P-only DeepONet10	0.3	0.0465	0.433	0.975	0.275	0.0316	0.729	0.0637	1.76	0.65	0	Complete
2 P-only DeepONet10	0.5	3.73e-03	1.08	0.979	0.194	0.147	0.504	0.0176	0.107	0.82	0	Complete
2 P-only DeepONet10	1	0.485	19.7	1.32	1.88	1.73	0.112	0.909	0.768	55.5	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0442	0.14	0.951	2.72	0.23	0.753	0.0261	0.562	0.126	0	Complete
3 ANN10 + P-only DeepONet10	0.1	5.19e-03	0.0455	0.906	1.27	0.0366	0.352	0.0232	0.0753	1.05	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0693	0.304	2.95	0.565	0.0234	1.2	0.0957	1.82	22	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0394	0.365	0.963	0.0158	0.0181	0.665	0.0572	1.53	0.723	0	Complete
3 ANN10 + P-only DeepONet10	0.5	2.65e-03	0.992	0.961	0.073	0.128	0.459	0.0159	0.128	0.847	0	Complete
3 ANN10 + P-only DeepONet10	1	0.452	19.2	1.56	1.31	1.74	0.298	0.867	0.99	108	0	Complete
4 T/P DeepONet10	0.05	0.107	0.306	0.93	0.933	0.255	2.47	0.144	2.27	2.26	0	Complete
4 T/P DeepONet10	0.1	3.80e-03	0.0682	0.0852	0.0645	2.64e-03	0.0271	4.83e-03	0.0327	0.023	0	Complete
4 T/P DeepONet10	0.2	5.81e-03	0.0854	0.125	0.107	4.06e-03	0.0478	6.94e-03	0.0564	0.0335	0	Complete
4 T/P DeepONet10	0.3	3.52e-03	0.0497	0.0986	0.0762	2.30e-03	0.0203	3.79e-03	0.0464	0.0639	0	Complete
4 T/P DeepONet10	0.5	6.59e-03	0.0276	0.29	0.0277	0.0102	0.128	7.79e-03	0.795	1.37	0	Complete
4 T/P DeepONet10	1	0.408	17.7	0.768	1.49	1.65	0.685	0.783	0.99	67.9	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0345	2.61	0.971	0.892	0.357	0.995	0.0814	3.09	3.4	0	Complete
5 ANN10 + T/P DeepONet10	0.1	4.76e-03	0.0786	0.0856	0.0623	1.90e-03	0.045	6.40e-03	0.0294	0.0218	0	Complete
5 ANN10 + T/P DeepONet10	0.2	5.14e-03	0.0823	0.0367	0.119	4.01e-03	0.0508	6.44e-03	0.0779	0.0806	0	Complete
5 ANN10 + T/P DeepONet10	0.3	3.92e-03	0.0661	0.0641	0.0957	1.58e-03	0.0208	4.49e-03	0.0478	0.0672	0	Complete
5 ANN10 + T/P DeepONet10	0.5	3.13e-03	0.664	0.321	0.193	0.107	0.395	8.63e-03	3.09	5.45	0	Complete
5 ANN10 + T/P DeepONet10	1	0.072	0.922	6.7	0.195	0.0627	0.545	0.0468	1.02	15.4	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0966	0.247	0.37	0.699	0.263	2.5	0.138	2.16	2.42	0	Complete
6 T/P/ϕ DeepONet10	0.1	8.28e-03	0.0376	0.0669	0.036	0.03	0.288	0.0133	0.265	0.371	0	Complete
6 T/P/ϕ DeepONet10	0.2	5.90e-03	0.115	0.0431	0.119	6.02e-03	0.0648	8.24e-03	0.0259	0.0246	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0728	1.14	0.774	1.4	0.089	1.16	0.105	0.395	0.529	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.0105	0.179	0.134	0.187	3.98e-03	0.108	0.0142	0.151	0.0645	0	Complete
6 T/P/ϕ DeepONet10	1	0.29	12.3	0.989	1.45	1.08	0.167	0.554	0.966	33.1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0841	0.0563	0.157	0.407	0.264	2.44	0.124	1.96	2.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	3.84e-03	0.0728	0.087	0.0721	0.0256	0.213	7.18e-03	0.213	0.325	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	5.76e-03	0.129	0.151	0.141	6.02e-03	0.082	8.97e-03	0.0717	0.0214	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0437	0.78	0.899	0.739	0.0572	0.807	0.07	0.396	0.568	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0121	0.247	0.0265	0.208	1.05e-03	0.127	0.0177	0.201	0.114	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.0362	0.835	3.74	0.493	0.0775	0.295	0.0137	0.37	4.09	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 57

T_0=1412.0 K P_0=4.083 atm ϕ_0=0.677 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 57

Repaired | Cantera vs NN NRMSE | T0=1412.0K, P0=4.083atm, phi0=0.677

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0362	0.142	0.896	3.72	0.189	0.365	0.0144	0.536	0.336	0	Complete
1 ANN10	0.1	0.0278	0.533	0.748	3.87	0.0693	0.905	0.0251	0.635	3.59	0	Complete
1 ANN10	0.2	5.60e-03	0.069	0.553	2.4	0.0307	0.893	0.0478	0.428	2.45	0	Complete
1 ANN10	0.3	0.0115	0.479	0.312	0.141	0.0282	0.0705	0.023	0.106	0.204	0	Complete
1 ANN10	0.5	8.73e-03	0.426	0.215	0.0904	0.0833	0.488	0.0114	0.184	0.651	0	Complete
1 ANN10	1	0.0207	0.788	2.71	0.394	0.0709	0.807	0.0563	0.898	1.68	0	Complete
2 P-only DeepONet10	0.05	0.079	0.408	11.4	0.676	0.227	0.645	0.0324	1.49	1.94	0	Complete
2 P-only DeepONet10	0.1	0.0558	0.539	4.78	0.739	0.0323	0.864	0.0535	0.174	0.396	0	Complete
2 P-only DeepONet10	0.2	7.20e-04	0.0236	0.0595	0.036	2.65e-03	0.0309	1.12e-03	0.0375	0.0786	0	Complete
2 P-only DeepONet10	0.3	6.83e-03	0.435	0.648	0.116	0.0618	0.362	2.71e-03	1.14	2.71	0	Complete
2 P-only DeepONet10	0.5	0.0241	1.75	0.957	0.276	0.167	0.304	0.0589	0.458	0.462	0	Complete
2 P-only DeepONet10	1	0.693	3.36	22.7	18.3	0.63	3.94	0.523	1.29	672	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.037	0.16	0.911	1.53	0.2	0.974	0.035	0.683	0.714	0	Complete
3 ANN10 + P-only DeepONet10	0.1	1.40e-03	0.19	0.455	0.45	2.91e-03	0.125	1.31e-03	0.0908	0.275	0	Complete
3 ANN10 + P-only DeepONet10	0.2	2.63e-04	0.0272	0.0186	0.0291	2.67e-03	0.0175	2.96e-04	0.0258	0.0273	0	Complete
3 ANN10 + P-only DeepONet10	0.3	7.80e-04	0.406	0.446	0.231	0.0418	0.214	4.88e-03	0.506	0.831	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0255	1.72	0.944	0.323	0.158	0.263	0.0598	0.472	0.367	0	Complete
3 ANN10 + P-only DeepONet10	1	0.448	12.6	2.78	4.54	0.816	1.88	0.701	1.17	338	0	Complete
4 T/P DeepONet10	0.05	0.0631	0.54	0.97	0.761	0.0728	0.992	0.0818	0.906	0.794	0	Complete
4 T/P DeepONet10	0.1	2.91e-03	0.0793	0.0983	0.0269	8.64e-03	0.0564	4.88e-03	0.0952	0.0646	0	Complete
4 T/P DeepONet10	0.2	1.54e-03	0.0641	0.0711	0.0472	5.45e-03	9.61e-03	2.64e-03	0.0599	0.0237	0	Complete
4 T/P DeepONet10	0.3	1.46e-03	0.0612	0.0944	0.0681	1.54e-03	5.09e-03	2.34e-03	0.055	0.0407	0	Complete
4 T/P DeepONet10	0.5	0.0329	1.24	1.17	0.485	0.0744	0.156	0.0585	0.274	0.437	0	Complete
4 T/P DeepONet10	1	0.304	12.1	2.69	2.1	1.1	0.665	0.569	4.97	47.7	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.059	0.478	0.939	0.562	0.0688	0.989	0.0786	0.937	0.92	0	Complete
5 ANN10 + T/P DeepONet10	0.1	1.95e-03	0.0258	0.0483	9.58e-03	4.39e-03	0.0491	3.11e-03	0.052	0.0407	0	Complete
5 ANN10 + T/P DeepONet10	0.2	1.04e-03	0.0315	0.0314	0.0334	1.60e-03	6.40e-03	1.49e-03	0.0339	0.0164	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.34e-03	0.065	0.106	0.0727	1.29e-03	3.41e-03	2.29e-03	0.0556	0.0419	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0345	1.06	2.64	0.346	0.0652	0.255	0.0529	0.323	0.755	0	Complete
5 ANN10 + T/P DeepONet10	1	0.417	16.2	2.67	0.916	1.53	0.568	0.763	1.82	81.8	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0652	0.636	0.943	0.784	0.0647	0.987	0.0858	0.966	0.973	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0559	0.554	0.955	0.546	0.052	0.87	0.0753	1.86	2.55	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0159	0.544	0.88	0.239	6.53e-03	0.295	0.0335	0.103	0.85	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.021	1.26	0.977	0.266	0.0956	0.0429	0.0494	0.428	0.317	0	Complete
6 T/P/ϕ DeepONet10	0.5	1.67e-03	0.15	0.26	0.0876	7.57e-03	0.0177	4.52e-03	0.0399	0.159	0	Complete
6 T/P/ϕ DeepONet10	1	0.355	8	0.707	3.8	0.242	3.46	0.548	0.961	75.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0595	0.543	0.915	0.505	0.0601	0.987	0.0813	0.955	0.973	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0418	0.362	0.668	0.185	0.0426	0.763	0.06	0.718	0.74	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.015	0.519	0.814	0.303	2.78e-03	0.241	0.0302	0.0777	0.65	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0239	1.26	0.965	0.401	0.0852	6.64e-03	0.0511	0.356	0.311	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	1.51e-03	0.165	0.287	0.09	8.75e-03	0.0206	4.74e-03	0.0363	0.165	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.216	8.68	2.08	1.19	0.806	0.867	0.415	5.29	28.4	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 64

T_0=1045.3 K P_0=4.301 atm ϕ_0=0.595 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 64

Repaired | Cantera vs NN NRMSE | T0=1045.3K, P0=4.301atm, phi0=0.595

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.305	0.863	2.59	1.28	0.522	8.08	0.571	1.03	1.01	0	Complete
1 ANN10	0.1	0.284	0.791	1.02	1.04	0.509	11.4	0.524	1.02	1.01	0	Complete
1 ANN10	0.2	0.503	1.21	1	1	0.559	1	1	35.9	194	0	Complete
1 ANN10	0.3	0.559	1.27	3.99	3.58	0.627	2.67	0.986	4.59	7.84	0	Complete
1 ANN10	0.5	0.375	0.944	1.18	0.741	0.502	0.629	0.691	1.09	0.999	0	Complete
1 ANN10	1	0.542	1.37	0.964	0.985	0.733	1	1	1	0.988	0	Complete
2 P-only DeepONet10	0.05	0.542	1.37	1	1	0.733	1	1	0.996	1	0	Complete
2 P-only DeepONet10	0.1	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
2 P-only DeepONet10	0.2	0.542	1.37	1	1	0.733	1	1	0.995	1	0	Complete
2 P-only DeepONet10	0.3	0.542	1.37	1	1	0.733	1	1	0.973	0.998	0	Complete
2 P-only DeepONet10	0.5	0.542	1.37	1	0.995	0.733	1	1	0.941	0.995	0	Complete
2 P-only DeepONet10	1	0.542	1.37	1	0.999	0.73	1	1	9.5	0.952	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.245	0.662	1	5.74	0.381	2.28	0.467	0.979	0.999	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.542	1.37	1	1	0.733	1	1	0.982	1	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.542	1.37	0.999	1	0.733	1	1	0.953	0.996	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.542	1.37	1	0.995	0.733	1	1	0.931	0.993	0	Complete
3 ANN10 + P-only DeepONet10	1	0.531	1.34	0.876	0.964	0.719	0.999	0.98	0.998	0.987	0	Complete
4 T/P DeepONet10	0.05	0.317	0.919	25.2	7.01	0.637	15.1	0.565	4.38	1.43	0	Complete
4 T/P DeepONet10	0.1	0.346	0.949	6.56	1.71	0.539	3.79	0.654	1.21	0.992	0	Complete
4 T/P DeepONet10	0.2	0.542	1.37	1	1	0.733	1	1	0.996	1	0	Complete
4 T/P DeepONet10	0.3	0.542	1.37	1	1	0.733	1	1	0.998	1	0	Complete
4 T/P DeepONet10	0.5	0.543	1.37	2.3	1.17	0.728	1.02	1	17.5	2.2	0	Complete
4 T/P DeepONet10	1	0.542	1.37	1	1	0.73	1	1	9	0.956	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.353	0.904	44.7	0.973	0.654	20.6	0.566	3.33	1.32	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.331	0.919	4.93	1.11	0.528	4.59	0.625	1.09	0.99	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.542	1.37	1	1	0.733	1	1	0.994	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.542	1.37	1	1	0.733	1	1	0.987	0.999	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.367	0.963	11.2	2.25	0.522	1.44	0.69	0.924	0.995	0	Complete
5 ANN10 + T/P DeepONet10	1	0.541	1.37	1	1	0.729	1	0.998	9.1	0.969	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.298	0.901	4.63	3.8	0.614	13.4	0.559	3.23	0.988	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.542	1.37	1	1	0.733	1	1	0.998	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.542	1.37	1	1	0.733	1	1	0.995	1	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.377	0.991	20.7	5.14	0.545	2.18	0.706	5.54	1.02	0	Complete
6 T/P/ϕ DeepONet10	1	0.322	0.803	0.987	10.3	0.429	0.632	0.587	1.09	0.962	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.291	0.872	6.99	2.03	0.582	12.7	0.544	2.43	0.996	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.542	1.37	1	1	0.733	1	1	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.542	1.37	1	1	0.733	1	1	0.997	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.542	1.37	1	1	0.733	1	1	0.989	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.361	0.949	15.7	3.07	0.515	1.52	0.679	0.969	0.997	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.541	1.36	1	0.983	0.728	1	0.998	9.28	0.995	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 64

T_0=1045.3 K P_0=4.301 atm ϕ_0=0.595 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 64

Repaired | Cantera vs NN NRMSE | T0=1045.3K, P0=4.301atm, phi0=0.595

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.27	0.742	1.16	1.22	0.441	6.51	0.503	1.1	1.04	0	Complete
1 ANN10	0.1	0.274	0.634	1.2	1.16	0.374	9.66	0.489	1.11	1.15	0	Complete
1 ANN10	0.2	0.303	0.762	1.04	1.4	0.405	1.09	0.559	1.83	3.92	0	Complete
1 ANN10	0.3	0.147	0.34	2.49	1.1	0.17	2.28	0.272	4.78	10	0	Complete
1 ANN10	0.5	0.357	0.904	1.43	0.786	0.484	0.696	0.66	0.875	0.996	0	Complete
1 ANN10	1	0.505	1.28	0.997	0.998	0.688	1	0.935	1	0.983	0	Complete
2 P-only DeepONet10	0.05	0.554	1.2	38.4	1	0.631	1	0.975	22.4	105	0	Complete
2 P-only DeepONet10	0.1	0.582	1.34	2.54	15.6	0.653	0.793	0.979	3.7	0.952	0	Complete
2 P-only DeepONet10	0.2	0.323	0.817	1.45	0.75	0.437	0.59	0.597	1.04	1.16	0	Complete
2 P-only DeepONet10	0.3	0.542	1.37	1	1	0.733	1	1	0.999	1	0	Complete
2 P-only DeepONet10	0.5	0.367	0.921	1.08	1.22	0.492	0.595	0.676	0.904	1	0	Complete
2 P-only DeepONet10	1	0.729	0.805	63.3	102	0.712	11.3	0.655	0.926	4.54	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.641	1.24	7.51	23.6	0.518	4.86	0.998	5.33	1.07	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.629	1.33	3.15	24.9	0.604	1.34	0.997	1.92	0.958	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.302	0.764	1.48	0.725	0.408	0.548	0.558	1.08	1.22	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.542	1.37	1	1	0.733	1	1	0.998	1	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.542	1.37	0.988	1	0.733	1	0.999	0.994	1	0	Complete
3 ANN10 + P-only DeepONet10	1	0.494	0.901	26	23.6	0.433	5.28	0.739	0.866	2.8	0	Complete
4 T/P DeepONet10	0.05	0.358	0.844	47.6	13.1	0.618	17.2	0.541	3.14	0.999	0	Complete
4 T/P DeepONet10	0.1	0.483	0.951	82.1	0.747	0.74	28.4	0.648	5.16	0.964	0	Complete
4 T/P DeepONet10	0.2	0.542	1.37	1	1	0.733	1	1	1	1	0	Complete
4 T/P DeepONet10	0.3	0.542	1.37	1	1	0.733	1	1	1	1	0	Complete
4 T/P DeepONet10	0.5	0.542	1.37	1	1	0.733	1	1	1	1	0	Complete
4 T/P DeepONet10	1	0.35	0.887	33.2	10.9	0.477	1.92	0.639	1.42	0.978	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.376	0.855	47.2	2.47	0.648	22.2	0.564	2.53	1.01	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.373	0.917	43.8	0.772	0.686	23.1	0.584	1.94	0.982	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.542	1.37	1	1	0.733	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.542	1.37	1	1	0.733	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.542	1.37	1	1	0.733	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.414	0.879	39.5	7.85	0.462	1.77	0.682	0.969	1.18	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.393	0.86	59.6	4.71	0.636	21.2	0.57	4.93	0.986	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.542	1.37	1	1	0.733	1	1	0.997	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.542	1.37	1	1	0.733	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.542	1.37	1	1	0.733	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.542	1.37	0.999	1	0.733	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	1	0.359	0.987	13.3	5.3	0.544	1.73	0.697	0.912	0.995	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.442	0.854	46.2	0.848	0.716	29	0.627	2.55	0.998	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.363	0.838	86.5	0.836	0.531	15.7	0.539	3.46	0.983	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.542	1.37	1	1	0.733	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.542	1.37	1	1	0.733	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.542	1.37	0.999	1	0.733	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.622	0.807	44.1	81.8	0.57	7.63	0.654	0.959	10.5	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 68

T_0=1086.3 K P_0=3.702 atm ϕ_0=1.168 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 68

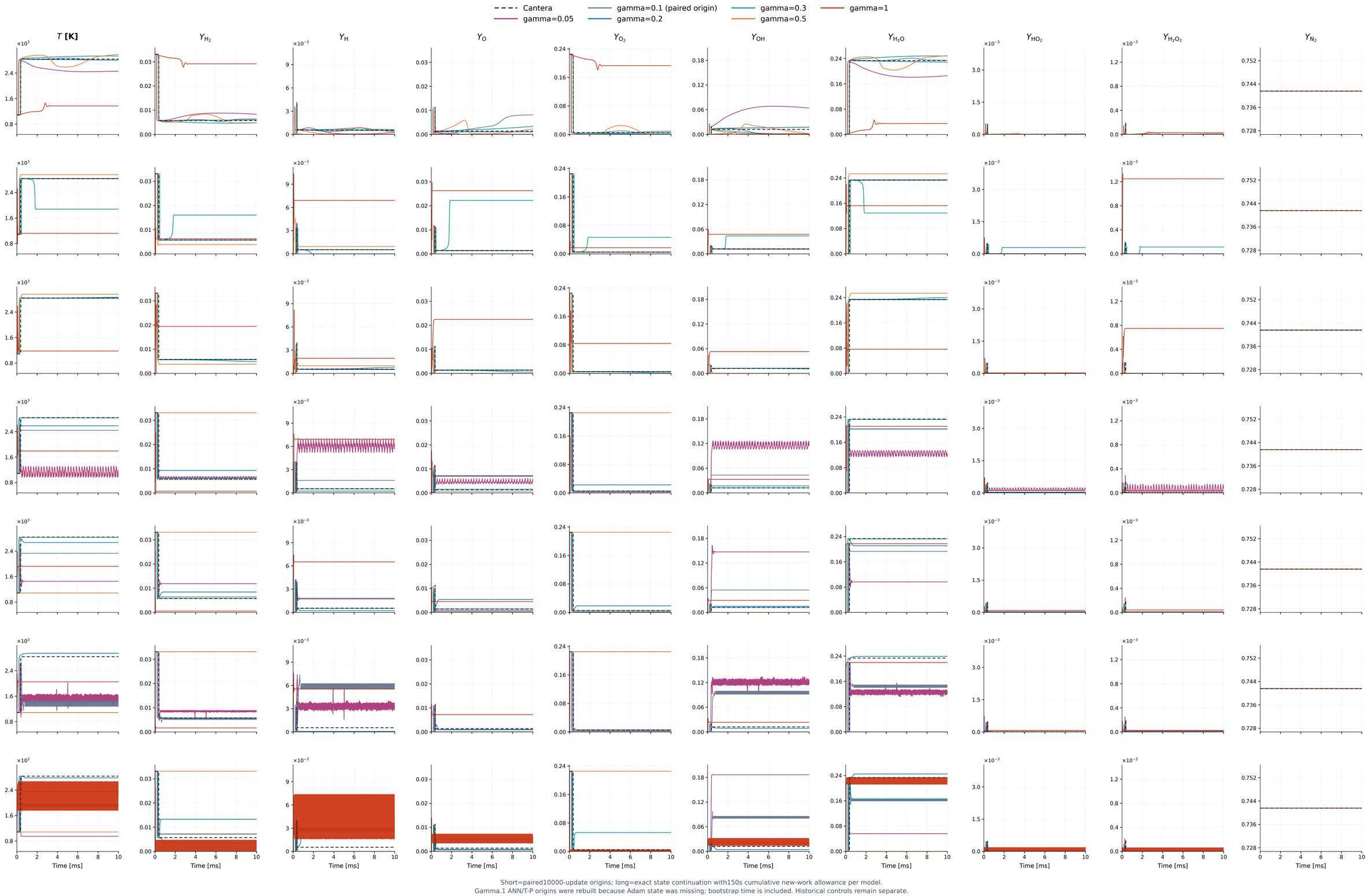
Repaired | Cantera vs NN NRMSE | T0=1086.3K, P0=3.702atm, phi0=1.168

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0962	0.543	2.09	0.919	0.778	0.992	0.166	1.29	1.14	0	Complete
1 ANN10	0.1	0.092	0.499	1.5	1.33	0.738	0.979	0.157	1.47	1.33	0	Complete
1 ANN10	0.2	0.0966	0.521	0.998	0.474	0.759	0.969	0.169	1.32	1.14	0	Complete
1 ANN10	0.3	0.11	0.557	0.52	0.492	0.825	0.983	0.184	1.81	3.83	0	Complete
1 ANN10	0.5	0.105	0.527	0.55	0.752	0.829	0.34	0.169	3.26	8.21	0	Complete
1 ANN10	1	0.115	0.581	1.37	0.803	0.893	0.346	0.181	5.79	15.8	0	Complete
2 P-only DeepONet10	0.05	0.123	0.452	0.931	1.59	0.573	2.51	0.186	1.36	1.45	0	Complete
2 P-only DeepONet10	0.1	0.615	3.15	1	1	4.94	1	1	0.986	0.989	0	Complete
2 P-only DeepONet10	0.2	0.0794	0.459	1.43	0.952	0.654	0.283	0.137	1.16	1.27	0	Complete
2 P-only DeepONet10	0.3	0.615	3.15	1	1	4.94	1	1	0.985	1	0	Complete
2 P-only DeepONet10	0.5	0.241	0.905	9.43	0.54	0.918	0.633	0.178	1	8.11	0	Complete
2 P-only DeepONet10	1	0.615	3.15	1	1	4.92	1	0.999	23	6.78	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0732	0.402	0.959	0.758	0.55	1	0.132	1.29	1.43	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0875	0.309	0.756	0.914	0.368	1.32	0.126	1.17	1.26	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0813	0.473	1.8	0.965	0.661	0.247	0.138	1.18	1.05	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0377	0.329	2.54	0.981	0.28	0.396	0.0727	0.87	1.64	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0964	0.573	2.17	1.27	0.79	1.01	0.175	0.899	2.25	0	Complete
3 ANN10 + P-only DeepONet10	1	0.107	0.552	0.944	1.19	0.857	0.191	0.173	1.81	1.44	0	Complete
4 T/P DeepONet10	0.05	0.373	0.413	4.58	2.39	0.522	5.9	0.356	2.28	2.54	0	Complete
4 T/P DeepONet10	0.1	0.124	0.51	1.4	0.786	0.778	1.36	0.174	2.17	1.35	0	Complete
4 T/P DeepONet10	0.2	0.4	1.39	0.953	20.1	1.2	2.54	0.51	4.46	1.12	0	Complete
4 T/P DeepONet10	0.3	0.448	1.52	0.993	23.3	1.29	2.93	0.562	6.85	2.46	0	Complete
4 T/P DeepONet10	0.5	0.244	0.779	8.73	0.913	0.851	0.682	0.166	5.43	4.69	0	Complete
4 T/P DeepONet10	1	0.436	2.19	0.841	3.69	3.27	0.57	0.696	1	10.3	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.399	0.584	2.04	0.238	0.308	9	0.479	1.35	1.96	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0776	0.325	1.01	0.993	0.528	1.02	0.102	1.28	1.09	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.171	0.698	0.823	6.75	0.715	1.44	0.25	1.8	1.06	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.102	0.526	0.227	0.347	0.827	0.19	0.166	1.06	0.998	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.206	0.793	7.54	0.552	0.903	0.497	0.178	4.94	3.97	0	Complete
5 ANN10 + T/P DeepONet10	1	0.275	0.873	9.56	1.97	0.944	1.09	0.183	3.34	3.98	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.644	0.155	11.1	1	0.256	9.4	0.493	6.66	3.05	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.132	0.389	0.981	1.04	0.557	1.8	0.167	1.73	1.92	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.615	3.15	1	1	4.94	1	1	0.991	0.999	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.385	0.915	1.51	10.3	0.912	5	0.449	1.54	2.18	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.214	0.71	0.637	8.9	0.865	3.39	0.288	4.4	2.35	0	Complete
6 T/P/ϕ DeepONet10	1	0.321	0.899	10.7	4.37	0.959	1.12	0.188	2.43	4.56	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.639	1.04	1.92	1	0.396	14.7	0.789	1.3	1.85	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.219	0.369	2.06	0.265	0.472	4.04	0.245	1.75	1.6	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0638	0.333	0.224	0.245	0.521	0.121	0.105	1.18	1.08	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.208	0.788	0.8	5.44	0.866	2.87	0.308	1.32	1.18	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.129	0.717	5.8	0.733	0.793	0.929	0.174	1.71	1.48	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.245	0.842	8.46	2.92	0.963	0.712	0.186	2.32	4.14	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 68

T_0=1086.3 K P_0=3.702 atm ϕ_0=1.168 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 68

Repaired | Cantera vs NN NRMSE | T0=1086.3K, P0=3.702atm, phi0=1.168

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.148	0.539	0.725	0.967	0.731	3.95	0.238	1.28	1.11	0	Complete
1 ANN10	0.1	0.081	0.422	0.344	2.73	0.658	0.76	0.133	1.32	1.23	0	Complete
1 ANN10	0.2	0.0197	0.113	0.193	0.303	0.172	0.331	0.0354	0.728	0.844	0	Complete
1 ANN10	0.3	0.0276	0.106	0.2	0.724	0.0777	0.728	0.0508	0.216	0.268	0	Complete
1 ANN10	0.5	0.0966	0.455	0.52	1.16	0.704	0.571	0.152	1.41	1.51	0	Complete
1 ANN10	1	0.532	2.77	0.975	0.938	4.32	0.868	0.876	1	2.92	0	Complete
2 P-only DeepONet10	0.05	0.0502	0.265	0.177	0.254	0.418	0.139	0.0826	1.12	1.19	0	Complete
2 P-only DeepONet10	0.1	0.0384	0.207	0.191	0.242	0.323	0.0872	0.0642	1.06	1.15	0	Complete
2 P-only DeepONet10	0.2	0.0586	0.307	0.208	0.243	0.479	0.11	0.0965	1.23	1.19	0	Complete
2 P-only DeepONet10	0.3	0.316	1.14	0.93	13.9	0.981	2.39	0.423	11.1	12.2	0	Complete
2 P-only DeepONet10	0.5	0.0865	0.455	0.799	1.02	0.608	1	0.151	1.12	1.02	0	Complete
2 P-only DeepONet10	1	0.603	0.587	11.7	18.3	0.919	2.99	0.369	1.39	147	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0742	0.386	0.227	0.249	0.604	0.144	0.122	1.26	1.23	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0579	0.305	0.204	0.247	0.476	0.114	0.0955	1.21	1.19	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.069	0.359	0.212	0.246	0.561	0.125	0.113	1.25	1.17	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.059	0.313	0.305	0.406	0.483	0.118	0.0979	1.14	1.08	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0847	0.447	0.843	1.01	0.606	1	0.148	1.17	1.06	0	Complete
3 ANN10 + P-only DeepONet10	1	0.584	1.61	2.65	15.4	1.87	3.42	0.678	1.53	87.2	0	Complete
4 T/P DeepONet10	0.05	0.601	0.196	9.95	2.29	0.307	8.43	0.464	6.56	8.25	0	Complete
4 T/P DeepONet10	0.1	0.148	0.313	1.95	0.488	0.485	2.56	0.16	1.62	1.1	0	Complete
4 T/P DeepONet10	0.2	0.12	0.585	0.624	4.19	0.768	0.43	0.188	1.78	1.24	0	Complete
4 T/P DeepONet10	0.3	0.0758	0.393	0.22	0.287	0.615	0.137	0.124	1.14	1.07	0	Complete
4 T/P DeepONet10	0.5	0.615	3.15	1	1	4.94	1	1	1	1	0	Complete
4 T/P DeepONet10	1	0.371	0.898	11.7	4.12	0.948	1.77	0.196	4.76	4.55	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.487	0.74	2.2	0.491	0.372	11	0.589	1.55	1.74	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.184	0.303	2.35	0.75	0.463	3.47	0.192	1.67	1.26	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0981	0.511	0.634	2.86	0.707	0.265	0.161	1.48	1.19	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.083	0.431	0.245	0.28	0.673	0.146	0.136	1.21	1.12	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.615	3.15	1	1	4.94	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.327	0.921	11	2.36	0.956	1.48	0.188	3.63	4.77	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.462	0.414	5.18	1.02	0.43	8.86	0.464	1.05	1.96	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.494	0.166	9.21	0.38	0.253	6.53	0.369	2.76	1.8	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.09	0.417	0.881	0.434	0.658	0.304	0.134	1.22	1.08	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.615	3.15	1	1	4.94	1	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.615	3.15	1	1	4.94	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	1	0.285	0.834	9.23	4.32	0.96	1	0.188	2.43	3.08	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.663	0.914	3.64	1.01	0.496	14.3	0.764	1.16	1.24	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.321	0.272	4.17	0.546	0.342	5.76	0.31	1.69	1.33	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0815	0.476	2.25	0.425	0.646	0.635	0.139	1.22	1.08	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.176	0.897	1	2.58	1.13	0.715	0.296	1.06	1.02	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.615	3.15	1	1	4.94	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.232	0.755	7.23	2.65	0.949	1.03	0.188	3.33	3.47	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 69

T_0=935.0 K P_0=2.045 atm ϕ_0=0.523 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 69

Repaired | Cantera vs NN NRMSE | T0=935.0K, P0=2.045atm, phi0=0.523

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	1.11e-06	1.29e-06	0.999	1.38e+03	1.51e-06	1	1	0.988	0.999	0	Complete
1 ANN10	0.1	9.88e-07	1.28e-06	1	185	1.53e-06	1	1	0.967	0.997	0	Complete
1 ANN10	0.2	9.71e-07	1.29e-06	1	1	1.56e-06	1	1	0.995	1	0	Complete
1 ANN10	0.3	9.72e-07	1.24e-06	0.969	0.753	1.46e-06	0.615	1	0.895	0.985	0	Complete
1 ANN10	0.5	1.1	0.812	8.10e+06	2.69e+08	0.436	1.21e+08	1.64e+06	283	473	0	Complete
1 ANN10	1	3.63e-03	4.29e-03	1.41e+06	4.88e+05	3.35e-03	1.11e+07	5.86e+03	1.42e+03	899	0	Complete
2 P-only DeepONet10	0.05	9.52e-07	1.27e-06	0.999	4.7	1.54e-06	1.29	0.973	0.985	0.999	0	Complete
2 P-only DeepONet10	0.1	2.83e-03	8.45e-03	9.87e+06	5.45e+07	6.11e-03	7.94e+07	1.15e+04	139	52.9	0	Complete
2 P-only DeepONet10	0.2	9.70e-07	1.25e-06	1	0.998	1.49e-06	1	1	0.924	0.984	0	Complete
2 P-only DeepONet10	0.3	0.0436	0.0558	1.26e+06	1.34e+05	0.0606	6.04e+04	57.7	5.31e+03	1.54e+05	0	Complete
2 P-only DeepONet10	0.5	1.33e-04	1.60e-04	3.98e+05	9.16e+03	6.63e-05	9.46e+03	121	33.7	17	0	Complete
2 P-only DeepONet10	1	0.0923	0.076	2.17e+07	7.26e+05	0.0382	8.45e+06	1.43e+05	968	170	0	Complete
3 ANN10 + P-only DeepONet10	0.05	9.58e-07	1.28e-06	1	5.32	1.55e-06	0.733	0.982	0.986	0.999	0	Complete
3 ANN10 + P-only DeepONet10	0.1	9.70e-07	1.29e-06	1	0.492	1.55e-06	0.998	0.999	0.988	0.999	0	Complete
3 ANN10 + P-only DeepONet10	0.2	9.71e-07	1.28e-06	1	0.996	1.54e-06	1	1	0.968	0.996	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.132	0.0201	1.81e+07	1.24e+09	0.0635	2.04e+08	2.76e+04	1.76e+03	2.24e+04	0	Complete
3 ANN10 + P-only DeepONet10	0.5	9.66e-06	3.56e-05	2.20e+04	1.24e+03	3.92e-05	3.8e+03	21.6	25.7	25.3	0	Complete
3 ANN10 + P-only DeepONet10	1	0.125	0.0915	1.03e+07	7.97e+05	0.0476	6.59e+06	1.79e+05	1.1e+03	248	0	Complete
4 T/P DeepONet10	0.05	7.36e-07	1.10e-06	0.996	20.2	1.45e-06	1.29	0.699	0.992	1	0	Complete
4 T/P DeepONet10	0.1	9.45e-07	1.27e-06	1	2.28	1.55e-06	0.952	0.963	0.997	1	0	Complete
4 T/P DeepONet10	0.2	9.33e-07	1.26e-06	4	2.27	1.55e-06	29	0.944	0.996	1	0	Complete
4 T/P DeepONet10	0.3	1.00e-04	2.42e-04	205	1.83e+05	3.51e-04	5.75e+04	22.6	202	402	0	Complete
4 T/P DeepONet10	0.5	3.20e-06	8.58e-06	2.07e+03	2.81e+04	1.62e-05	2.39e+04	7.36	15.4	2.01	0	Complete
4 T/P DeepONet10	1	0.584	0.6	7.56e+07	1.31e+09	0.378	2.46e+09	1.11e+06	1.01	906	0	Complete
5 ANN10 + T/P DeepONet10	0.05	1.43e-04	1.08e-04	0.992	151	7.14e-05	9.54	179	8.64	62.3	0	Complete
5 ANN10 + T/P DeepONet10	0.1	3.86e-04	5.04e-04	1	13	5.70e-04	0.999	8.88	94.2	1.33e+03	0	Complete
5 ANN10 + T/P DeepONet10	0.2	1.80e-06	5.60e-06	1.39	61.2	9.14e-06	11.5	0.822	7.3	5.63	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.74e-04	3.32e-04	56	1.85e+05	4.46e-04	9.03e+04	23.4	205	655	0	Complete
5 ANN10 + T/P DeepONet10	0.5	2.11e-06	1.19e-05	1.38e+03	2.44e+04	2.19e-05	2.05e+04	6.31	19.9	4.85	0	Complete
5 ANN10 + T/P DeepONet10	1	0.189	0.397	1.41e+08	1.82e+09	0.284	2.17e+09	6.79e+05	6.13	2.17e+03	0	Complete
6 T/P/ϕ DeepONet10	0.05	7.93e-07	1.15e-06	0.892	27.9	1.48e-06	24.8	0.753	0.983	1	0	Complete
6 T/P/ϕ DeepONet10	0.1	2.78e-05	2.32e-05	87.3	109	1.84e-05	399	35.7	7.04	8.79	0	Complete
6 T/P/ϕ DeepONet10	0.2	9.63e-07	1.28e-06	2.79	0.512	1.56e-06	15.3	0.986	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	1.52e-04	3.19e-04	3.34e+05	1.77e+05	1.59e-04	1.70e+06	456	19.5	1.17	0	Complete
6 T/P/ϕ DeepONet10	0.5	5.50e-05	1.14e-03	1.83e+06	2.36e+05	4.24e-04	1.78e+06	1.48e+03	7.68	0.936	0	Complete
6 T/P/ϕ DeepONet10	1	0.857	0.735	1.06e+07	1.01e+09	0.444	2.56e+09	1.41e+06	9.54	31.2	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	6.13e-07	9.56e-07	0.944	108	1.35e-06	39.3	0.605	0.967	0.999	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	9.40e-06	8.95e-06	18.9	16.9	8.37e-06	52.1	11.3	4.03	5.05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	9.68e-07	1.29e-06	1.01	0.899	1.56e-06	1.73	0.995	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	1.98e-04	3.01e-04	2.51e+05	1.02e+05	1.48e-04	1.22e+06	469	9.71	0.972	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	1.60e-04	4.32e-04	5.45e+05	1.23e+05	1.79e-04	6.97e+05	629	9.1	0.926	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.113	0.349	8.01e+07	2.10e+09	0.283	3.07e+09	5.82e+05	9.96	1.61e+03	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 69

T_0=935.0 K P_0=2.045 atm ϕ_0=0.523 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

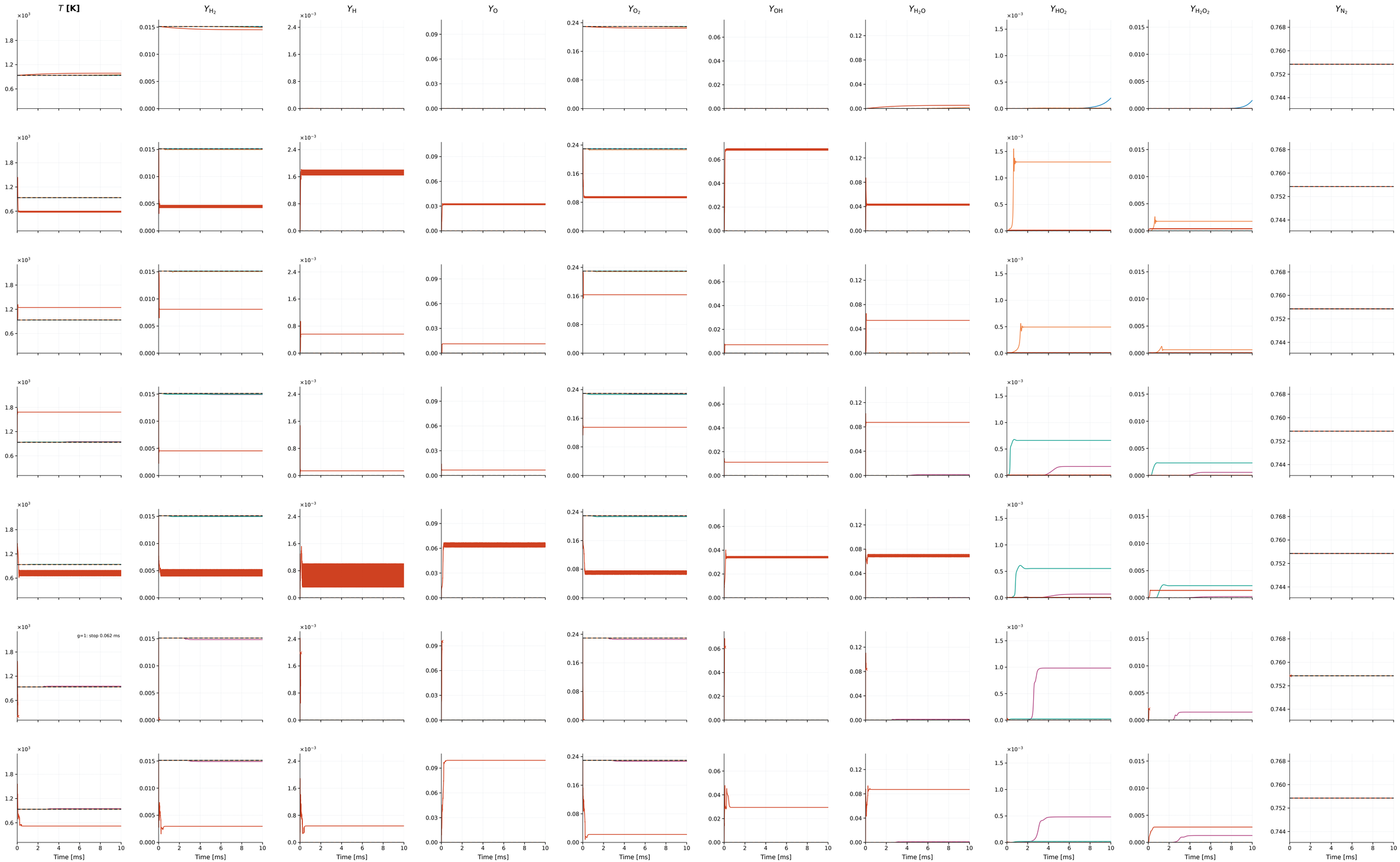
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 69

Repaired | Cantera vs NN NRMSE | T0=935.0K, P0=2.045atm, phi0=0.523

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	1.08e-06	1.28e-06	0.999	1e+03	1.50e-06	0.937	0.999	0.975	0.998	0	Complete
1 ANN10	0.1	3.71e-06	2.67e-05	1	9.83e+04	3.96e-05	0.999	1	15.7	58.2	0	Complete
1 ANN10	0.2	8.59e-04	1.12e-03	1	3.37e+04	1.26e-03	0.824	1	188	3.01e+03	0	Complete
1 ANN10	0.3	9.80e-07	3.17e-06	94.3	4.6	4.99e-06	77.2	1	3.04	6.56	0	Complete
1 ANN10	0.5	8.12e-03	5.54e-03	5.08e+05	7.23e+05	2.93e-03	2.74e+06	1.11e+04	35.2	173	0	Complete
1 ANN10	1	0.0478	0.0326	8.93e+04	3.25e+03	0.0171	1.89e+04	6.54e+04	0.735	287	0	Complete
2 P-only DeepONet10	0.05	9.71e-07	1.29e-06	1	0.971	1.56e-06	0.997	1	0.995	1	0	Complete
2 P-only DeepONet10	0.1	9.70e-07	1.29e-06	0.985	0.924	1.56e-06	0.99	0.999	0.995	1	0	Complete
2 P-only DeepONet10	0.2	9.71e-07	1.29e-06	1	0.987	1.57e-06	0.991	1	1	1	0	Complete
2 P-only DeepONet10	0.3	9.74e-07	1.29e-06	0.803	33.2	1.55e-06	58.4	0.999	0.988	0.999	0	Complete
2 P-only DeepONet10	0.5	4.80e-03	9.28e-03	7.97e+04	5.11e+06	0.0126	8.06e+06	606	5.6e+03	1.95e+04	0	Complete
2 P-only DeepONet10	1	0.37	0.702	5.52e+08	3.39e+09	0.588	1.91e+10	6.46e+05	49.4	4.43e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	9.71e-07	1.29e-06	1	0.913	1.56e-06	0.996	1	0.996	1	0	Complete
3 ANN10 + P-only DeepONet10	0.1	9.71e-07	1.29e-06	1	0.646	1.56e-06	0.997	1	0.996	1	0	Complete
3 ANN10 + P-only DeepONet10	0.2	9.71e-07	1.29e-06	1	0.989	1.57e-06	0.995	1	0.999	1	0	Complete
3 ANN10 + P-only DeepONet10	0.3	9.72e-07	1.28e-06	0.799	16.8	1.55e-06	33.5	0.999	0.987	0.999	0	Complete
3 ANN10 + P-only DeepONet10	0.5	4.75e-03	5.49e-03	2.62e+05	1.13e+06	5.61e-03	1.73e+06	4.34e+03	2.05e+03	7.03e+03	0	Complete
3 ANN10 + P-only DeepONet10	1	0.331	0.464	1.80e+08	1.19e+09	0.287	1.99e+09	8.03e+05	48.8	1.15e+03	0	Complete
4 T/P DeepONet10	0.05	0.013	9.84e-03	2.59e+03	3.22e+03	6.42e-03	5.14e+03	1.60e+04	558	4.86e+03	0	Complete
4 T/P DeepONet10	0.1	7.58e-07	9.61e-07	0.876	0.967	1.13e-06	0.854	0.79	0.691	0.752	0	Complete
4 T/P DeepONet10	0.2	1.01e-06	1.35e-06	194	153	1.62e-06	552	1.11	1.05	1.01	0	Complete
4 T/P DeepONet10	0.3	7.50e-03	0.0103	1.64e+04	4.38e+05	0.0121	7.76e+04	402	2.89e+03	2.56e+04	0	Complete
4 T/P DeepONet10	0.5	6.00e-05	1.04e-04	7.96e+04	1.32e+04	6.45e-05	2.51e+05	136	16.1	26.7	0	Complete
4 T/P DeepONet10	1	0.8	0.7	4.56e+07	6.98e+08	0.413	3.13e+09	1.30e+06	45.2	513	0	Complete
5 ANN10 + T/P DeepONet10	0.05	3.35e-03	2.64e-03	470	570	1.85e-03	908	3.9e+03	224	1.75e+03	0	Complete
5 ANN10 + T/P DeepONet10	0.1	6.52e-05	6.05e-05	1.34e+03	38	5.41e-05	78.1	69.5	17.9	56.7	0	Complete
5 ANN10 + T/P DeepONet10	0.2	9.71e-07	1.29e-06	5.71	4.43	1.56e-06	11.4	1	0.989	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	7.07e-03	9.53e-03	1.65e+04	2.70e+05	0.011	7.79e+04	388	2.36e+03	2.41e+04	0	Complete
5 ANN10 + T/P DeepONet10	0.5	7.83e-05	1.22e-04	8.49e+04	1.82e+04	7.56e-05	3.41e+05	165	16.7	31.1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.243	0.7	2.49e+08	6.70e+09	0.689	9.55e+09	1.03e+06	25.2	1.54e+04	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0173	0.0157	1.02e+03	1.15e+05	0.0134	2.46e+04	1.83e+04	3.73e+03	1.43e+04	0	Complete
6 T/P/ϕ DeepONet10	0.1	9.68e-07	1.28e-06	0.56	0.956	1.55e-06	0.684	0.994	0.989	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	8.65e-06	4.36e-06	1.83e+04	290	1.04e-06	3.27e+03	2.26	0.968	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	2.06e-04	2.50e-04	4.42e+04	6.36e+04	2.35e-04	1.14e+05	253	98	200	0	Complete
6 T/P/ϕ DeepONet10	0.5	4.58e-04	5.52e-04	2.70e+05	3.68e+05	3.01e-04	3.10e+06	911	6.73	0.758	0	Complete
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.062 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0136	0.0118	572	2.76e+04	9.58e-03	6.67e+03	1.40e+04	1.76e+03	1.18e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	9.67e-07	1.28e-06	0.724	0.959	1.55e-06	0.777	0.994	0.982	0.999	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	9.71e-07	1.29e-06	1.79	0.956	1.56e-06	1.47	1	0.997	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	1.75e-04	2.04e-04	3.27e+04	2.71e+04	1.91e-04	4.66e+04	207	78.9	169	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	6.10e-04	6.40e-04	2.54e+05	3.35e+05	3.45e-04	3.00e+06	1.1e+03	5.41	0.745	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.442	0.801	1.60e+08	1.04e+10	0.897	8.35e+09	1.29e+06	4	3.17e+04	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 73

T_0=1106.0 K P_0=1.911 atm ϕ_0=1.915 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 73

Repaired | Cantera vs NN NRMSE | T0=1106.0K, P0=1.911atm, phi0=1.915

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0366	0.0242	0.713	0.692	0.158	0.891	0.0204	0.339	2.02	0	Complete
1 ANN10	0.1	0.0381	0.0486	1.14	0.997	0.157	0.82	0.0187	0.373	6.14	0	Complete
1 ANN10	0.2	0.059	0.109	0.74	3.46	0.351	5.47	0.124	0.491	0.586	0	Complete
1 ANN10	0.3	0.0121	0.0257	0.182	3.11	0.24	0.899	0.0223	2.97	62.4	0	Complete
1 ANN10	0.5	0.0155	8.72e-03	0.219	1.02	0.0787	0.735	0.0138	0.181	0.259	0	Complete
1 ANN10	1	0.158	0.0655	1.15	4.75	0.374	9.03	0.169	19.1	238	0	Complete
2 P-only DeepONet10	0.05	0.0364	0.0922	0.908	8.35	0.336	4.07	0.0897	0.867	1.2	0	Complete
2 P-only DeepONet10	0.1	0.0376	0.025	0.757	0.734	0.154	0.77	0.019	0.37	0.648	0	Complete
2 P-only DeepONet10	0.2	6.10e-03	5.14e-03	0.117	0.256	0.0418	0.127	4.12e-03	0.179	0.243	0	Complete
2 P-only DeepONet10	0.3	6.52e-03	0.0135	0.133	0.415	0.12	0.143	9.84e-03	0.359	0.485	0	Complete
2 P-only DeepONet10	0.5	0.382	0.421	0.474	90.1	1.72	15.5	0.51	1	22.3	0	Complete
2 P-only DeepONet10	1	0.0879	0.0546	1.36	9.76	0.52	1.93	0.0619	1.3	1.67	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0457	0.0277	0.926	0.734	0.135	0.945	0.0206	0.31	0.602	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.03	0.0143	0.592	0.58	7.95e-03	0.633	0.0124	0.0794	0.188	0	Complete
3 ANN10 + P-only DeepONet10	0.2	4.40e-03	0.0147	0.104	0.324	0.134	0.0917	0.0105	0.315	0.351	0	Complete
3 ANN10 + P-only DeepONet10	0.3	7.16e-03	0.0271	0.166	0.511	0.253	0.0831	0.0195	0.6	0.732	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0122	0.0418	0.236	1.34	0.385	0.364	0.0314	0.871	0.981	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0501	0.0462	0.784	6.19	0.444	1	0.0427	1.19	1.64	0	Complete
4 T/P DeepONet10	0.05	0.048	0.0282	0.98	0.835	0.105	0.996	0.0198	0.221	0.769	0	Complete
4 T/P DeepONet10	0.1	0.0466	0.0258	0.956	0.772	0.0161	0.986	0.0171	0.459	6.48	0	Complete
4 T/P DeepONet10	0.2	8.22e-03	0.029	0.169	0.523	0.274	0.0808	0.0212	0.768	1.02	0	Complete
4 T/P DeepONet10	0.3	0.568	0.406	1.68	194	1.12	14.3	0.556	32.1	5.72	0	Complete
4 T/P DeepONet10	0.5	0.525	0.345	1.58	193	0.622	14.8	0.499	52.8	270	0	Complete
4 T/P DeepONet10	1	0.101	0.0769	1.78	6.08	0.788	1.22	0.0742	1.4	7.49	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0461	0.0277	0.962	0.734	0.054	0.987	0.0164	0.111	2.44	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0444	0.0276	0.93	0.673	0.0709	0.961	0.0159	0.422	2.4	0	Complete
5 ANN10 + T/P DeepONet10	0.2	7.19e-03	0.0269	0.161	0.504	0.252	0.0689	0.0195	0.683	0.892	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0108	0.0379	0.222	0.627	0.36	0.0917	0.0278	1.02	1.3	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.264	0.283	1.07	60.9	1.12	13.3	0.358	10.1	42	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0524	0.0657	0.842	2.97	0.674	0.599	0.0574	1.34	6.68	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.045	0.0326	0.98	0.822	0.11	0.995	0.0144	0.288	0.524	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0497	0.0333	0.96	1.13	0.242	1.05	0.0279	0.634	0.879	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0147	0.0462	0.255	0.73	0.446	0.111	0.0345	1.22	1.35	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.543	0.292	10.7	4.62	0.512	11.1	0.204	17.7	435	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.728	0.561	17	3.17	0.634	5.75	0.122	16.9	1.18e+03	0	Complete
6 T/P/ϕ DeepONet10	1	0.0414	0.084	0.635	1.27	0.837	0.406	0.0668	1	6.85	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.0398	0.0409	0.969	0.769	0.188	0.99	8.33e-03	0.218	2.14	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0248	0.0245	0.472	0.561	0.21	0.58	0.0195	0.528	0.672	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0107	0.0373	0.216	0.635	0.354	0.0889	0.0273	0.944	1.13	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0331	0.0683	1.15	0.776	0.444	0.858	0.0379	1.21	1.32	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0564	0.112	0.638	2.88	0.651	4.39	0.118	1.46	2.91	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.0312	0.0715	0.424	0.88	0.717	0.219	0.0569	1	6.1	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 73

T_0=1106.0 K P_0=1.911 atm ϕ_0=1.915 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

g=1: stop 0.070 ms

g=1: stop 0.278 ms

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 73

Repaired | Cantera vs NN NRMSE | T0=1106.0K, P0=1.911atm, phi0=1.915

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0232	0.0199	0.509	0.464	4.13e-03	1.04	0.0178	0.0161	0.0185	0	Complete
1 ANN10	0.1	0.0129	0.0132	0.31	1.03	0.0634	0.443	7.29e-03	0.15	0.297	0	Complete
1 ANN10	0.2	0.0497	0.0905	0.715	0.84	0.129	5.65	0.11	0.0217	0.11	0	Complete
1 ANN10	0.3	0.0118	6.73e-03	0.123	1.15	0.0359	0.808	0.0139	0.159	3.78	0	Complete
1 ANN10	0.5	0.0102	0.0143	0.0619	0.321	0.116	0.444	0.0159	0.204	0.281	0	Complete
1 ANN10	1	0.129	0.262	0.34	0.858	2.8	0.185	0.234	0.946	2.11	0	Complete
2 P-only DeepONet10	0.05	0.0227	0.0586	0.913	4.09	0.314	0.745	0.018	0.0514	0.0561	0	Complete
2 P-only DeepONet10	0.1	7.79e-03	0.0215	0.171	0.447	0.199	0.112	0.0155	0.475	0.572	0	Complete
2 P-only DeepONet10	0.2	0.0109	0.0357	0.199	0.605	0.343	0.0838	0.0265	0.847	0.982	0	Complete
2 P-only DeepONet10	0.3	0.0141	0.0443	0.23	0.692	0.431	0.109	0.0333	1.08	1.12	0	Complete
2 P-only DeepONet10	0.5	0.0487	0.0804	1.58	0.865	0.452	0.799	0.0381	1.02	1.02	0	Complete
2 P-only DeepONet10	1	0.786	0.539	18.4	3.76	1.81	1.14	0.134	0.993	1.32e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.173	0.163	4.57	0.354	0.0207	0.558	9.36e-03	0.0786	1.31	0	Complete
3 ANN10 + P-only DeepONet10	0.1	4.63e-03	0.017	0.113	0.338	0.157	0.0561	0.0121	0.367	0.43	0	Complete
3 ANN10 + P-only DeepONet10	0.2	6.13e-03	0.0239	0.146	0.443	0.223	0.0557	0.0171	0.532	0.641	0	Complete
3 ANN10 + P-only DeepONet10	0.3	9.13e-03	0.0327	0.186	0.567	0.31	0.0779	0.0239	0.772	0.904	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0927	0.151	0.833	17.8	0.485	6.63	0.169	1.12	74.9	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0642	0.0584	0.337	10.7	0.436	3.14	0.0784	0.954	344	0	Complete
4 T/P DeepONet10	0.05	0.0483	0.0246	0.972	0.776	0.0278	0.994	0.0191	0.0938	6.44	0	Complete
4 T/P DeepONet10	0.1	0.0695	0.043	1.39	1.76	0.245	1.32	0.0327	0.618	0.89	0	Complete
4 T/P DeepONet10	0.2	0.0129	0.0402	0.235	0.683	0.385	0.151	0.0299	1.11	1.2	0	Complete
4 T/P DeepONet10	0.3	9.82e-03	0.0347	0.197	0.584	0.331	0.0811	0.0255	0.968	1.1	0	Complete
4 T/P DeepONet10	0.5	0.582	1.13	1	1	12.3	1	1	1	1	0	Complete
4 T/P DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.070 ms
5 ANN10 + T/P DeepONet10	0.05	0.0447	0.027	0.933	0.729	0.0372	0.989	0.0153	0.0569	3.01	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.297	0.179	6.27	1.19	0.129	5.3	0.0946	0.292	0.421	0	Complete
5 ANN10 + T/P DeepONet10	0.2	9.31e-03	0.0318	0.192	0.585	0.3	0.118	0.0232	0.813	0.983	0	Complete
5 ANN10 + T/P DeepONet10	0.3	7.82e-03	0.0291	0.173	0.525	0.274	0.0688	0.0211	0.762	0.906	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0348	0.0571	0.73	2.05	0.523	0.276	0.041	1.36	1.62	0	Complete
5 ANN10 + T/P DeepONet10	1	0.352	0.209	6.3	12.2	1.98	5	0.219	11.5	164	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.545	0.278	10.6	0.453	0.0158	11.8	0.207	1.03	642	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.831	0.77	21.2	1.13	0.171	1.41	0.0282	0.45	2.89	0	Complete
6 T/P/ϕ DeepONet10	0.2	8.29e-03	0.0223	0.191	0.445	0.204	0.111	0.0158	0.55	0.651	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.169	0.155	4.42	6.27	0.178	0.198	0.015	4.4	4.26	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.166	0.17	4.52	4.98	0.457	0.486	0.0355	6.5	14.5	0	Complete
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.278 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.041	0.0234	0.832	0.722	0.0502	0.985	0.0158	0.0921	6.69	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.361	0.13	6.28	2.63	0.0935	10.2	0.183	0.221	2.08	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	7.04e-03	0.0194	0.175	0.382	0.175	0.0701	0.0135	0.448	0.558	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.151	0.148	4.08	4.46	0.145	0.385	0.0112	0.465	0.501	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.146	0.145	3.94	6.42	0.303	0.525	0.0234	5.32	8.07	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.397	0.211	7.86	11.8	0.869	6.34	0.166	9.24	176	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 74

T_0=1222.9 K P_0=4.971 atm ϕ_0=1.766 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 74

Repaired | Cantera vs NN NRMSE | T0=1222.9K, P0=4.971atm, phi0=1.766

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	8.50e-03	0.0316	0.358	0.887	0.195	0.977	0.0245	0.633	0.919	0	Complete
1 ANN10	0.1	0.0331	0.0696	1.38	0.728	0.0408	0.804	0.0201	0.211	1.99	0	Complete
1 ANN10	0.2	0.0301	0.07	1.34	0.96	0.0569	0.934	0.0229	0.335	0.666	0	Complete
1 ANN10	0.3	0.0188	0.0236	0.536	7.33	0.0741	0.875	9.15e-03	0.883	2.7	0	Complete
1 ANN10	0.5	0.0257	0.0111	0.505	2.37	0.0427	0.809	0.0167	0.228	0.587	0	Complete
1 ANN10	1	0.0188	0.0392	0.415	7.61	0.627	0.966	0.0267	1.11	7.94	0	Complete
2 P-only DeepONet10	0.05	0.0222	0.0382	0.33	1.54	0.181	1.7	0.0406	0.705	0.878	0	Complete
2 P-only DeepONet10	0.1	0.0501	0.0284	1.01	1.17	0.177	1.1	0.0287	0.692	1.41	0	Complete
2 P-only DeepONet10	0.2	0.107	0.17	3.76	0.916	0.0997	0.969	0.0246	0.795	9.46	0	Complete
2 P-only DeepONet10	0.3	0.406	0.639	0.764	83.4	6.4	10.5	0.589	1.05	39.6	0	Complete
2 P-only DeepONet10	0.5	0.639	0.695	4.41	13	14.8	8	0.765	1.02	235	0	Complete
2 P-only DeepONet10	1	0.836	0.439	18.4	2.06	4.86	4.88	0.301	1.02	2.09e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	7.85e-03	0.0244	0.189	0.457	0.264	0.775	0.0205	0.778	0.946	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.045	0.0562	1.06	2.06	0.212	1.67	0.0447	0.662	3.9	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.081	0.138	2.97	0.931	0.197	0.952	0.0249	0.596	0.965	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.304	0.464	0.724	45.2	3.96	11.2	0.464	2.21	79.4	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.614	0.97	1.45	6.34	17.9	9.16	0.898	1.03	147	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0153	0.0678	0.642	9.49	0.624	1.29	0.0458	2.27	29.7	0	Complete
4 T/P DeepONet10	0.05	0.0423	0.0504	1.01	1.16	0.618	1	0.0214	1.03	0.966	0	Complete
4 T/P DeepONet10	0.1	0.04	0.0532	0.97	1.09	0.72	1	0.0224	1.15	9.51	0	Complete
4 T/P DeepONet10	0.2	0.373	0.543	0.717	94.5	4.15	10.6	0.517	6.57	6.54	0	Complete
4 T/P DeepONet10	0.3	0.0102	0.036	0.218	0.827	0.667	0.134	0.0221	0.975	0.972	0	Complete
4 T/P DeepONet10	0.5	—	—	—	—	—	—	—	—	—	—	Stop 0.092 ms
4 T/P DeepONet10	1	0.192	0.217	3.19	15.6	4.37	1.58	0.183	1.72	29.8	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.33	0.0694	3.48	0.735	0.471	14.1	0.32	1.14	7.73	0	Complete
5 ANN10 + T/P DeepONet10	0.1	8.40e-03	0.0292	0.178	0.524	0.521	0.199	0.0177	1.07	0.981	0	Complete
5 ANN10 + T/P DeepONet10	0.2	8.55e-03	0.031	0.127	0.515	0.582	0.137	0.0197	1.06	0.979	0	Complete
5 ANN10 + T/P DeepONet10	0.3	9.64e-03	0.0326	0.226	0.582	0.584	0.203	0.0202	1.02	0.977	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0206	0.0343	0.382	0.858	0.622	0.472	0.023	1.01	0.983	0	Complete
5 ANN10 + T/P DeepONet10	1	0.17	0.43	0.425	2.92	8.45	0.549	0.314	1.03	36.1	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.0365	0.061	1.01	1.13	0.782	1	0.018	1.01	0.984	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.155	0.0323	2.06	2	0.572	5.48	0.131	1.5	5.07	0	Complete
6 T/P/ϕ DeepONet10	0.2	9.55e-03	0.0352	0.164	0.553	0.643	0.112	0.022	0.99	0.959	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0883	0.14	3.03	0.684	0.675	0.742	0.0289	1.08	0.98	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.0126	0.0437	0.217	0.881	0.727	0.647	0.0296	0.996	0.975	0	Complete
6 T/P/ϕ DeepONet10	1	0.121	0.28	0.122	0.567	5.72	0.346	0.211	1.09	33	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.028	0.0755	0.998	1.01	1.05	0.999	0.0194	0.99	0.993	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.152	0.0322	1.86	0.606	0.517	5.99	0.138	1.4	6.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	8.21e-03	0.0318	0.135	0.488	0.581	0.124	0.02	1.05	0.971	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0675	0.114	2.39	0.412	0.591	0.718	0.0266	1.06	0.979	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0135	0.0365	0.138	0.581	0.626	0.645	0.0272	1.01	0.976	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.114	0.286	0.215	0.536	5.75	0.463	0.21	1.07	33.2	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 74

T_0=1222.9 K P_0=4.971 atm ϕ_0=1.766 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 74

Repaired | Cantera vs NN NRMSE | T0=1222.9K, P0=4.971atm, phi0=1.766

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.073	0.109	2.45	0.832	0.0231	1.03	0.0234	0.119	0.191	0	Complete
1 ANN10	0.1	0.011	0.0564	0.647	17.3	0.272	0.778	0.0175	1.29	6.78	0	Complete
1 ANN10	0.2	0.0929	0.186	0.875	1.12	1.15	6.3	0.188	0.232	0.445	0	Complete
1 ANN10	0.3	7.34e-03	1.88e-03	0.0988	1.11	0.0328	0.423	7.39e-03	0.179	0.398	0	Complete
1 ANN10	0.5	0.0808	0.216	0.788	0.791	2.86	2.95	0.176	0.447	32.2	0	Complete
1 ANN10	1	0.19	0.0313	2.43	6.68	0.517	6.66	0.162	40.6	150	0	Complete
2 P-only DeepONet10	0.05	4.74e-03	0.0152	0.269	0.128	0.0138	0.25	6.10e-03	0.0992	0.0984	0	Complete
2 P-only DeepONet10	0.1	2.32e-03	2.93e-03	0.0333	0.163	0.0277	0.105	2.89e-03	0.0435	0.066	0	Complete
2 P-only DeepONet10	0.2	8.49e-04	3.57e-03	0.0428	0.177	0.0291	0.0657	2.04e-03	0.0889	0.139	0	Complete
2 P-only DeepONet10	0.3	1.86e-03	3.22e-03	0.0369	0.219	0.0295	0.0771	2.45e-03	0.0842	0.242	0	Complete
2 P-only DeepONet10	0.5	0.554	0.509	3.43	17.9	7.97	15	0.665	3.67	634	0	Complete
2 P-only DeepONet10	1	0.55	0.295	4.37	130	2.1	10.1	0.477	3.79	2.22e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	4.53e-03	0.0131	0.24	0.0247	0.016	0.222	5.09e-03	0.0332	0.155	0	Complete
3 ANN10 + P-only DeepONet10	0.1	1.03e-03	4.24e-03	0.0514	0.147	0.0193	0.108	2.92e-03	0.0761	0.171	0	Complete
3 ANN10 + P-only DeepONet10	0.2	7.32e-04	4.25e-03	0.0588	0.203	0.0227	0.0683	2.18e-03	0.11	0.211	0	Complete
3 ANN10 + P-only DeepONet10	0.3	1.12e-03	3.04e-03	0.0371	0.179	0.0224	0.06	1.95e-03	0.098	0.274	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.248	0.179	1.09	29	0.337	10.8	0.294	2.28	544	0	Complete
3 ANN10 + P-only DeepONet10	1	0.539	0.773	0.771	91.3	9.98	9.89	0.725	4.5	1.37e+03	0	Complete
4 T/P DeepONet10	0.05	0.611	0.283	11.7	0.49	0.467	13.5	0.306	2.51	155	0	Complete
4 T/P DeepONet10	0.1	0.224	0.158	5.12	0.891	0.531	3.75	0.086	2.36	9.95	0	Complete
4 T/P DeepONet10	0.2	9.39e-03	0.0357	0.173	0.558	0.643	0.106	0.0222	1.02	0.962	0	Complete
4 T/P DeepONet10	0.3	0.0114	0.0339	0.118	0.559	0.659	0.086	0.0229	0.994	0.968	0	Complete
4 T/P DeepONet10	0.5	0.0158	0.0398	0.354	1.19	0.705	0.158	0.0241	0.995	1.09	0	Complete
4 T/P DeepONet10	1	0.111	0.0433	1.86	8.36	0.613	2.56	0.0707	2.22	4.53	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.312	0.0244	4.26	0.851	0.363	11.3	0.256	1	3.93	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.192	0.158	4.72	0.856	0.372	2.57	0.058	1.47	5.02	0	Complete
5 ANN10 + T/P DeepONet10	0.2	8.84e-03	0.0289	0.117	0.48	0.535	0.123	0.019	1.09	0.974	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0106	0.0316	0.12	0.413	0.605	0.114	0.0212	1.02	0.976	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0124	0.0323	0.189	0.557	0.612	0.113	0.0215	1.03	0.99	0	Complete
5 ANN10 + T/P DeepONet10	1	0.0726	0.0314	1.01	7.51	0.59	1.92	0.0559	2.58	5.08	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.545	0.0691	7.89	1.04	0.526	17.9	0.406	1.11	12.4	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.352	0.0722	5.58	2.06	0.532	10.6	0.246	2.31	4.97	0	Complete
6 T/P/ϕ DeepONet10	0.2	7.66e-03	0.0368	0.203	0.561	0.642	0.121	0.0214	1.02	0.968	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.359	0.569	0.511	6	6.72	14.1	0.583	1.1	24.5	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.682	0.734	18.9	1.25	0.803	2.51	0.0693	52.9	1.05e+03	0	Complete
6 T/P/ϕ DeepONet10	1	0.171	0.0664	3.07	15.6	0.675	3.16	0.0936	6.04	16.4	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.567	0.408	1.73	0.867	0.369	32.5	0.738	1.03	5.87	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.211	0.0799	3.87	0.377	0.405	5.49	0.126	1.39	2.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	8.59e-03	0.031	0.136	0.451	0.564	0.0944	0.0198	1.1	0.973	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.121	0.261	0.767	4.42	3.11	4.77	0.231	1.04	3.56	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0153	0.0347	0.155	0.514	0.647	0.356	0.0257	1	0.971	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.179	0.0474	2.9	17.1	0.639	3.9	0.112	9.8	30	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 76

T_0=1523.6 K P_0=3.626 atm ϕ_0=1.349 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 76

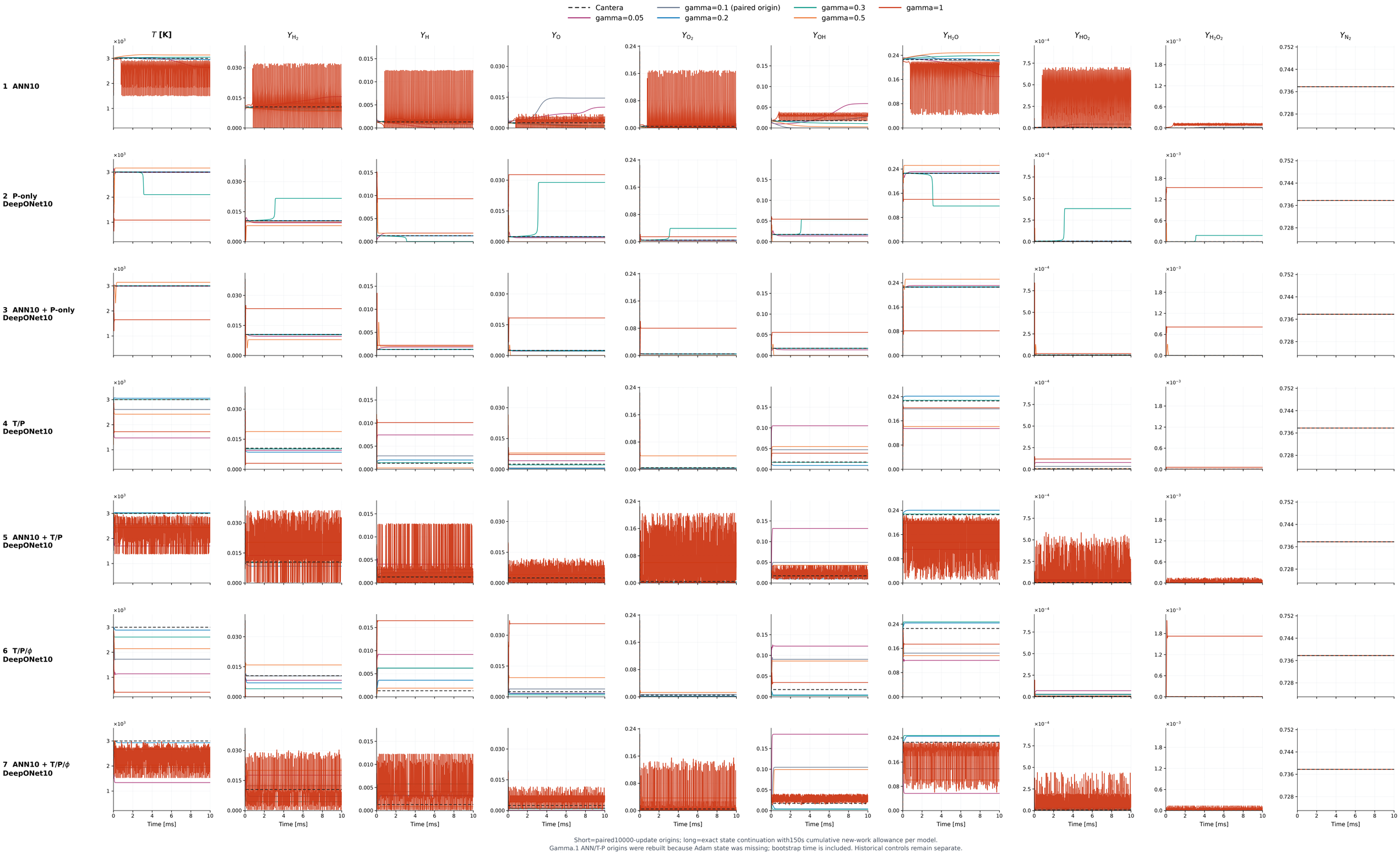
Repaired | Cantera vs NN NRMSE | T0=1523.6K, P0=3.626atm, phi0=1.349

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0183	0.242	1.07	0.664	0.184	0.974	0.0868	0.819	1.92	0	Complete
1 ANN10	0.1	7.28e-03	0.157	0.733	1.66	0.164	0.967	0.0661	3.08	10.3	0	Complete
1 ANN10	0.2	0.0229	0.227	0.706	0.103	0.589	0.925	0.0964	1.24	3.61	0	Complete
1 ANN10	0.3	0.0572	0.0754	0.333	1.33	0.682	0.943	0.0834	7.83	67.4	0	Complete
1 ANN10	0.5	0.0462	0.0909	0.199	0.44	0.721	0.447	0.0621	5	153	0	Complete
1 ANN10	1	0.0602	0.29	0.747	0.455	3.06	0.264	0.107	25.2	188	0	Complete
2 P-only DeepONet10	0.05	0.0242	0.18	0.556	0.419	0.163	0.868	0.0819	0.221	0.174	0	Complete
2 P-only DeepONet10	0.1	0.0606	0.124	0.537	0.631	0.585	0.637	0.075	1.43	3.56	0	Complete
2 P-only DeepONet10	0.2	3.07e-03	0.269	1.04	0.971	0.717	0.556	0.0814	0.915	0.481	0	Complete
2 P-only DeepONet10	0.3	0.0177	0.309	1.38	0.969	0.716	0.537	0.0797	0.956	0.925	0	Complete
2 P-only DeepONet10	0.5	0.336	0.773	6.26	0.458	0.457	0.513	0.0247	1.45	77.9	0	Complete
2 P-only DeepONet10	1	0.622	0.597	8.39	1.29	2.36	2.18	0.279	1.04	1.92e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.0138	0.243	0.912	0.573	0.262	0.938	0.0928	0.367	0.322	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0447	0.119	0.469	0.535	0.487	0.537	0.062	1.43	4.32	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0119	0.295	1.26	0.961	0.713	0.55	0.0806	0.948	0.969	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0101	0.275	1.16	0.925	0.689	0.508	0.0758	0.925	0.908	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.23	0.363	3.48	0.591	0.72	1.1	0.0748	1.04	202	0	Complete
3 ANN10 + P-only DeepONet10	1	0.131	0.301	2.51	0.254	0.221	0.122	0.0163	1.21	518	0	Complete
4 T/P DeepONet10	0.05	0.102	0.0507	0.995	0.992	0.707	0.999	0.0716	0.984	0.901	0	Complete
4 T/P DeepONet10	0.1	0.101	0.138	1.02	0.564	0.519	1.13	0.102	4.47	2.83	0	Complete
4 T/P DeepONet10	0.2	0.0278	0.0729	0.0142	0.579	0.425	0.237	0.0406	0.561	0.404	0	Complete
4 T/P DeepONet10	0.3	0.388	1.29	0.921	13.6	6.93	2.29	0.587	14.6	27.4	0	Complete
4 T/P DeepONet10	0.5	0.15	0.613	0.808	4.44	3.15	1.51	0.276	2.5	1.69	0	Complete
4 T/P DeepONet10	1	0.433	0.961	7.87	0.111	0.906	0.73	0.0406	21.1	84	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0437	0.314	0.994	0.991	4.07	0.998	0.0396	0.776	0.169	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0593	0.113	0.713	0.206	0.165	0.838	0.067	0.92	1.56	0	Complete
5 ANN10 + T/P DeepONet10	0.2	0.0249	0.122	0.212	0.71	0.53	0.329	0.0534	0.619	0.533	0	Complete
5 ANN10 + T/P DeepONet10	0.3	0.0127	0.0557	0.0843	0.352	0.249	0.154	0.0252	0.184	0.036	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.153	0.853	0.827	1.01	9.45	0.634	0.332	7.72	8.62	0	Complete
5 ANN10 + T/P DeepONet10	1	0.276	0.569	4.81	0.749	0.695	0.599	0.0365	11.2	53.7	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.114	7.48e-03	0.995	0.992	0.0794	0.999	0.0944	0.982	0.922	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0794	0.0678	0.632	0.485	0.728	0.848	0.0779	2.08	2.75	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0146	0.133	0.389	0.515	0.378	0.407	0.0524	0.413	0.475	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0599	0.489	2.56	0.754	0.712	0.942	0.11	0.434	0.66	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.201	0.518	0.607	5.92	0.809	3.12	0.313	1.83	1.34	0	Complete
6 T/P/ϕ DeepONet10	1	0.33	0.577	5.32	2.35	0.875	0.706	0.0659	8.01	47.4	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.083	0.227	0.992	0.991	2.95	0.999	0.0681	0.828	0.636	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0672	0.0325	0.52	0.132	0.106	0.894	0.0693	0.916	1.18	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0129	0.149	0.47	0.535	0.402	0.442	0.0562	0.38	0.504	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.05	0.467	2.37	0.79	0.715	0.941	0.111	0.6	0.725	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.653	0.518	3.77	2.09	2.56	7.49	0.69	6.1	153	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.231	0.456	3.92	1.54	0.722	0.409	0.0298	8.09	28	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 76

T_0=1523.6 K P_0=3.626 atm ϕ_0=1.349 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 76

Repaired | Cantera vs NN NRMSE | T0=1523.6K, P0=3.626atm, phi0=1.349

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0591	0.291	0.77	1.77	0.505	1.3	0.135	0.654	1.42	0	Complete
1 ANN10	0.1	0.0122	0.142	0.362	3.91	0.852	0.951	0.027	4.68	22.8	0	Complete
1 ANN10	0.2	6.68e-03	0.0455	0.167	0.381	0.289	0.189	0.0122	0.305	0.126	0	Complete
1 ANN10	0.3	0.0126	0.128	0.353	0.458	0.522	0.299	0.0473	0.504	0.252	0	Complete
1 ANN10	0.5	0.0432	0.171	0.279	0.506	0.711	0.733	0.0887	0.744	0.888	0	Complete
1 ANN10	1	0.189	0.602	2.5	0.607	6.25	0.733	0.229	66.9	126	0	Complete
2 P-only DeepONet10	0.05	7.33e-03	0.0992	0.434	0.238	0.259	0.21	0.0277	0.0963	0.0923	0	Complete
2 P-only DeepONet10	0.1	5.37e-03	0.0164	0.026	0.106	0.11	0.0454	8.18e-03	0.116	0.156	0	Complete
2 P-only DeepONet10	0.2	8.80e-04	7.35e-03	0.0334	0.0171	0.0355	3.22e-03	1.48e-03	0.102	0.212	0	Complete
2 P-only DeepONet10	0.3	0.248	0.878	0.811	8.7	4.36	1.73	0.397	47.7	186	0	Complete
2 P-only DeepONet10	0.5	0.0593	0.241	0.621	0.994	0.707	0.989	0.117	1.39	28	0	Complete
2 P-only DeepONet10	1	0.638	0.0835	6.14	11.9	1.51	2.11	0.38	4.23	1.93e+03	0	Complete
3 ANN10 + P-only DeepONet10	0.05	4.76e-03	0.086	0.391	0.151	0.164	0.213	0.0244	0.0509	0.0779	0	Complete
3 ANN10 + P-only DeepONet10	0.1	3.07e-03	0.0152	0.025	0.072	0.0806	0.0401	6.65e-03	0.0432	0.109	0	Complete
3 ANN10 + P-only DeepONet10	0.2	2.77e-04	4.80e-03	0.0189	5.90e-03	0.0192	8.80e-03	1.41e-03	0.0614	0.17	0	Complete
3 ANN10 + P-only DeepONet10	0.3	2.26e-03	0.0167	0.0339	0.108	0.1	0.0144	5.81e-03	0.153	0.0713	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.0499	0.253	0.658	0.982	0.697	0.975	0.115	1.71	36.5	0	Complete
3 ANN10 + P-only DeepONet10	1	0.449	1.22	0.779	6.25	11.5	2.18	0.637	5.02	1.01e+03	0	Complete
4 T/P DeepONet10	0.05	0.508	0.108	4.68	0.658	0.495	4.99	0.403	10.9	10.5	0	Complete
4 T/P DeepONet10	0.1	0.131	0.0419	1.22	0.739	0.438	1.69	0.116	4.17	5.45	0	Complete
4 T/P DeepONet10	0.2	0.0194	0.188	0.532	0.847	0.64	0.468	0.07	0.728	0.608	0	Complete
4 T/P DeepONet10	0.3	2.30e-03	0.0306	0.0979	0.154	0.119	0.0514	9.85e-03	0.21	0.151	0	Complete
4 T/P DeepONet10	0.5	0.192	0.784	0.79	2.15	5.28	2.12	0.374	0.284	6.38	0	Complete
4 T/P DeepONet10	1	0.425	0.715	6.73	1.88	0.806	1.21	0.101	17	68.4	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.432	0.308	2.13	0.525	0.672	6.5	0.506	4.81	11.7	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.136	0.0424	1.25	0.829	0.536	1.83	0.123	3.83	6.28	0	Complete
5 ANN10 + T/P DeepONet10	0.2	8.99e-03	0.197	0.67	0.814	0.617	0.416	0.0646	0.665	0.566	0	Complete
5 ANN10 + T/P DeepONet10	0.3	2.30e-03	0.0323	0.099	0.167	0.125	0.0547	0.0106	0.182	0.155	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.186	0.926	0.889	0.617	8.37	1.44	0.4	0.551	6.19	0	Complete
5 ANN10 + T/P DeepONet10	1	0.252	0.704	3.25	1.46	7.65	0.835	0.268	27.3	63.9	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.619	0.216	6	0.301	0.671	5.98	0.467	9.77	11	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.424	0.0142	3.72	0.523	0.356	4.15	0.36	3.77	2.76	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0383	0.334	1.74	0.378	0.421	0.748	0.0799	0.142	0.52	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.13	0.615	3.75	0.6	0.658	0.811	0.0956	2.83	1.02	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.284	0.51	0.445	2.7	1.23	3.92	0.397	0.576	8.34	0	Complete
6 T/P/ϕ DeepONet10	1	0.863	0.995	11.6	13.1	0.729	0.974	0.229	1.07	2.15e+03	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.552	0.676	1.41	0.942	0.723	9.5	0.746	0.602	6.86	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.349	0.173	2.1	0.527	0.457	4.93	0.384	3.17	5.48	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0226	0.32	1.52	0.503	0.514	0.769	0.0864	0.156	0.569	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.11	0.577	3.4	0.666	0.689	0.824	0.0986	1.52	0.885	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.316	0.896	0.358	0.668	4.86	4.59	0.545	0.52	9.61	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.26	0.534	3.86	1.29	4.14	0.824	0.171	14.5	65.1	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 78

T_0=962.3 K P_0=2.443 atm ϕ_0=1.140 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

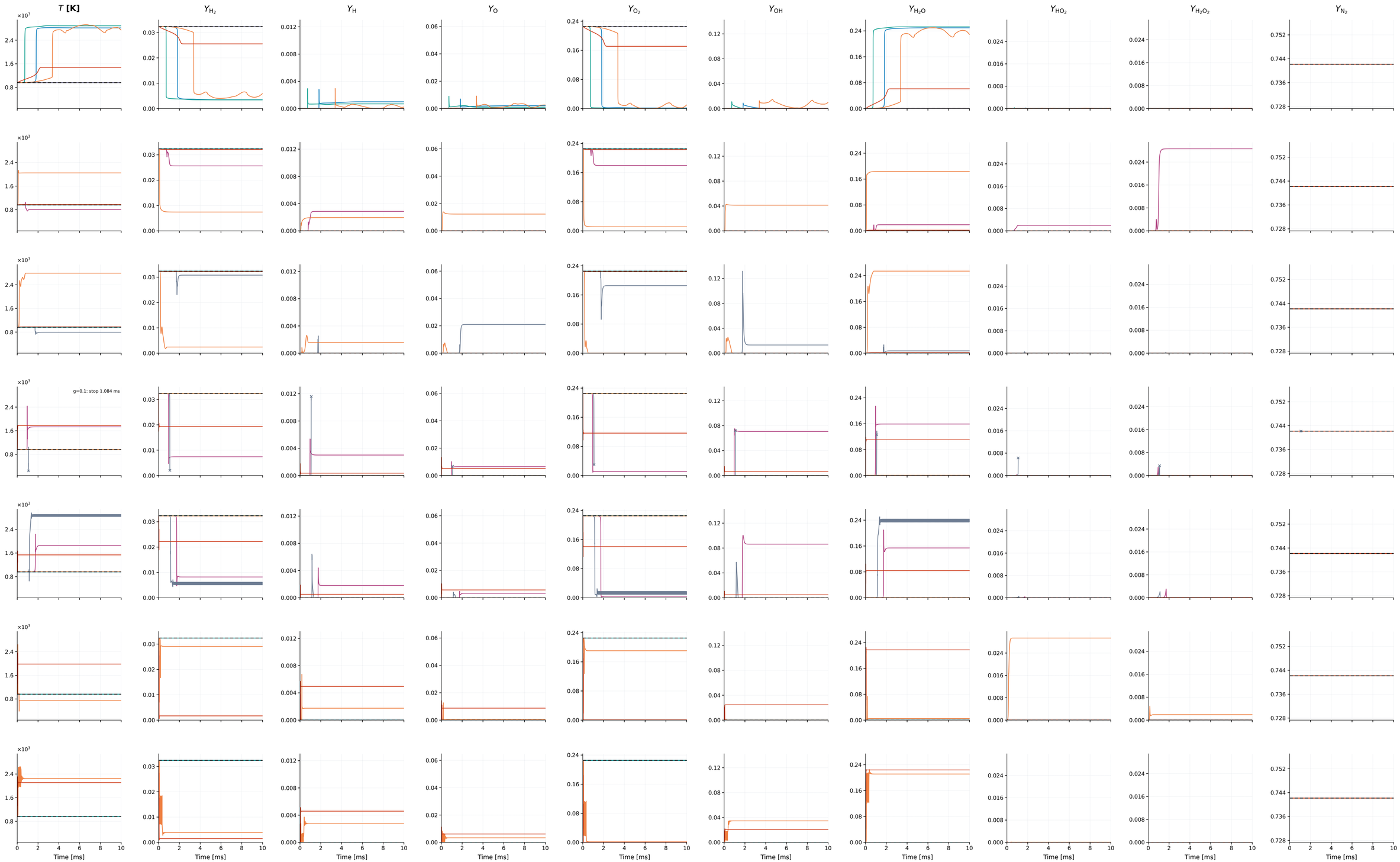
3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10



Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 78

Repaired | Cantera vs NN NRMSE | T0=962.3K, P0=2.443atm, phi0=1.140

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	4.55e-05	2.59e-05	1	15.2	5.42e-05	1	1	0.998	1	0	Complete
1 ANN10	0.1	4.72e-05	2.63e-05	3.24	4.29e+03	6.52e-05	1	1	1.07	2.06	0	Complete
1 ANN10	0.2	1.74	0.805	4.52e+06	6.09e+06	0.896	9.12e+06	6.00e+04	2.46	0.69	0	Complete
1 ANN10	0.3	1.92	0.858	3.43e+06	4.21e+06	0.961	8.21e+06	6.46e+04	8.08	7.31	0	Complete
1 ANN10	0.5	1.55	0.686	1.64e+06	6.52e+06	0.792	4.45e+07	5.15e+04	7.99	6.16	0	Complete
1 ANN10	1	0.484	0.192	1.08e+04	3.97e+05	0.219	1.39e+06	1.47e+04	1	5.81	0	Complete
2 P-only DeepONet10	0.05	0.159	0.197	1.42e+07	1.86e+03	0.192	1.25e+04	4.8e+03	401	5.82e+03	0	Complete
2 P-only DeepONet10	0.1	3.54e-05	1.95e-05	1	0.876	4.00e-05	0.999	0.869	0.848	0.718	0	Complete
2 P-only DeepONet10	0.2	6.58e-05	3.44e-05	0.999	1.19	6.49e-05	0.976	1.37	0.702	1.6	0	Complete
2 P-only DeepONet10	0.3	4.88e-05	2.22e-05	1.33	93.1	4.15e-05	208	1.4	0.816	0.933	0	Complete
2 P-only DeepONet10	0.5	1.12	0.765	1.01e+07	4.29e+07	0.945	2.85e+08	4.87e+04	0.942	8.93	0	Complete
2 P-only DeepONet10	1	0.0188	7.89e-03	1.99e+04	1.52e+05	9.18e-03	4.29e+05	590	0.965	1.52	0	Complete
3 ANN10 + P-only DeepONet10	0.05	4.54e-05	2.58e-05	1	0.806	5.42e-05	0.982	0.999	0.997	1	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.16	0.0496	8.47e+05	6.63e+07	0.166	1.05e+08	1.77e+03	16.5	4.06	0	Complete
3 ANN10 + P-only DeepONet10	0.2	2.92e-05	1.53e-05	1	0.292	2.92e-05	1.01	0.7	0.473	0.489	0	Complete
3 ANN10 + P-only DeepONet10	0.3	5.22e-04	1.98e-04	15.8	647	2.15e-04	4.22e+03	15.8	0.978	1	0	Complete
3 ANN10 + P-only DeepONet10	0.5	1.87	0.911	8.12e+06	3.52e+06	0.987	2.92e+07	6.65e+04	1.31	2.08	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0147	6.09e-03	1.00e+04	1.20e+05	7.13e-03	3.59e+05	458	0.976	1.21	0	Complete
4 T/P DeepONet10	0.05	0.759	0.734	1.50e+07	2.13e+07	0.904	4.66e+08	4.03e+04	7.75	31.2	0	Complete
4 T/P DeepONet10	0.1	—	—	—	—	—	—	—	—	—	—	Stop 1.084 ms
4 T/P DeepONet10	0.2	4.54e-05	2.59e-05	1	1	5.42e-05	1	1	1	1	0	Complete
4 T/P DeepONet10	0.3	4.20e-05	2.43e-05	1.76	42.3	5.23e-05	399	0.897	1	1	0	Complete
4 T/P DeepONet10	0.5	5.37e-03	2.39e-03	1.69e+04	471	2.64e-03	1.38e+05	175	0.999	1	0	Complete
4 T/P DeepONet10	1	0.848	0.403	1.79e+06	1.83e+07	0.483	4.25e+07	2.94e+04	0.956	9.3	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.841	0.679	8.72e+06	1.05e+07	0.892	5.44e+08	3.73e+04	7.28	49.5	0	Complete
5 ANN10 + T/P DeepONet10	0.1	1.86	0.777	1.64e+06	1.13e+06	0.886	3.31e+07	5.98e+04	8.67	37.4	0	Complete
5 ANN10 + T/P DeepONet10	0.2	4.54e-05	2.59e-05	1	1	5.42e-05	1	1	1	1	0	Complete
5 ANN10 + T/P DeepONet10	0.3	3.81e-05	2.25e-05	13.2	12.3	4.99e-05	557	0.803	1	1	0	Complete
5 ANN10 + T/P DeepONet10	0.5	4.79e-03	2.08e-03	1.30e+04	440	2.30e-03	8.63e+04	154	0.999	1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.597	0.315	2.71e+06	2.01e+07	0.376	3.34e+07	2.23e+04	1	17.9	0	Complete
6 T/P/ϕ DeepONet10	0.05	4.54e-05	2.59e-05	1	0.739	5.42e-05	0.988	1	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	0.1	4.54e-05	2.59e-05	1	0.884	5.42e-05	0.997	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	4.54e-05	2.59e-05	1	1	5.42e-05	1	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	4.05e-05	2.37e-05	2.2	25	5.15e-05	496	0.859	1	1	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.229	0.11	9.13e+06	2.30e+06	0.161	6.98e+06	3.3e+03	6.17e+03	408	0	Complete
6 T/P/ϕ DeepONet10	1	1.06	0.947	2.58e+07	3.06e+07	0.995	1.70e+08	5.78e+04	7.58	3.71	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	4.54e-05	2.58e-05	1	0.705	5.42e-05	0.988	0.999	0.998	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	4.54e-05	2.58e-05	1	0.819	5.42e-05	0.995	0.999	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	4.54e-05	2.59e-05	1	1	5.42e-05	1	1	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	4.49e-05	2.56e-05	0.803	2.77	5.40e-05	31.7	0.984	1	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	1.34	0.871	1.41e+07	1.19e+07	0.985	2.36e+08	5.60e+04	20.4	4.42	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	1.19	0.952	2.39e+07	2.15e+07	0.995	1.44e+08	5.96e+04	7.39	6.3	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 78

T_0=962.3 K P_0=2.443 atm ϕ_0=1.140 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 78

Repaired | Cantera vs NN NRMSE | T0=962.3K, P0=2.443atm, phi0=1.140

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	4.53e-05	2.57e-05	0.997	0.257	5.37e-05	0.98	0.999	0.98	0.992	0	Complete
1 ANN10	0.1	3.84e-05	1.37e-05	1	2.81e+03	2.20e-05	1	1	0.357	0.603	0	Complete
1 ANN10	0.2	4.11e-05	1.08e-05	2.06	1	2.09e-05	1	1.5	0.893	0.979	0	Complete
1 ANN10	0.3	1.81	0.817	3.52e+06	5.75e+06	0.916	1.24e+07	6.13e+04	3.61	0.695	0	Complete
1 ANN10	0.5	1.16	0.51	1.11e+06	5.26e+06	0.587	2.08e+07	3.85e+04	1.76	0.909	0	Complete
1 ANN10	1	0.188	0.074	6.69e+04	7.71e+05	0.0852	1.87e+04	5.7e+03	0.999	1.26	0	Complete
2 P-only DeepONet10	0.05	4.54e-05	2.59e-05	1	0.997	5.42e-05	0.999	1	0.999	1	0	Complete
2 P-only DeepONet10	0.1	0.149	0.0627	0.981	1.37e+04	0.0849	4.35e+03	3.98e+03	74	1.27e+03	0	Complete
2 P-only DeepONet10	0.2	2.86e-04	1.49e-04	6.48	3.22	2.78e-04	4.62	5.91	2.67	7.12	0	Complete
2 P-only DeepONet10	0.3	1.65e-04	8.07e-05	0.863	7.95	1.42e-04	30	3.96	1.69	2.59	0	Complete
2 P-only DeepONet10	0.5	4.49e-05	2.56e-05	0.962	49.6	5.37e-05	2.73	0.983	0.989	1	0	Complete
2 P-only DeepONet10	1	0.0169	7.59e-03	5.49e+04	1.06e+05	8.52e-03	1.28e+05	556	2.39	2.93	0	Complete
3 ANN10 + P-only DeepONet10	0.05	4.54e-05	2.59e-05	1	0.995	5.42e-05	0.999	1	0.999	1	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0634	0.0283	0.996	4.74e+03	0.0424	419	1.52e+03	40.2	986	0	Complete
3 ANN10 + P-only DeepONet10	0.2	2.29e-04	1.23e-04	11.8	3.74	2.38e-04	6.46	4.55	2.57	6.31	0	Complete
3 ANN10 + P-only DeepONet10	0.3	1.33e-04	6.29e-05	0.984	5.86	1.06e-04	21.7	3.32	1.19	1.75	0	Complete
3 ANN10 + P-only DeepONet10	0.5	4.45e-05	2.54e-05	0.931	37.9	5.36e-05	4.46	0.97	0.991	1	0	Complete
3 ANN10 + P-only DeepONet10	1	0.0215	9.35e-03	5.44e+04	9.89e+04	0.0105	9.28e+04	692	2.69	3.44	0	Complete
4 T/P DeepONet10	0.05	4.39e-05	2.47e-05	0.91	0.916	5.12e-05	0.937	0.968	0.909	0.96	0	Complete
4 T/P DeepONet10	0.1	5.45e-05	2.40e-05	491	0.957	5.40e-05	33.3	0.99	0.998	1	0	Complete
4 T/P DeepONet10	0.2	1.43	0.688	2.15e+06	2.79e+07	0.838	1.07e+08	5.00e+04	8.37	3.91	0	Complete
4 T/P DeepONet10	0.3	0.0353	0.0232	7.24e+04	5.64e+04	0.0495	1.03e+03	514	495	1.64e+03	0	Complete
4 T/P DeepONet10	0.5	3.76e-05	2.01e-05	79.8	429	4.31e-05	2.6e+03	1.04	0.949	1	0	Complete
4 T/P DeepONet10	1	0.987	0.93	2.70e+07	3.34e+07	0.989	2.12e+08	5.56e+04	20.2	2.94	0	Complete
5 ANN10 + T/P DeepONet10	0.05	4.35e-05	2.44e-05	0.888	0.9	5.05e-05	0.921	0.958	0.892	0.949	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0957	0.044	17.8	1.8e+03	0.0695	370	2.29e+03	311	1.51e+03	0	Complete
5 ANN10 + T/P DeepONet10	0.2	1.46	0.675	1.72e+06	1.77e+07	0.808	8.62e+07	4.97e+04	5.53	3.08	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.02	0.524	5.58e+05	4.32e+07	0.704	1.75e+08	3.68e+04	35.8	99	0	Complete
5 ANN10 + T/P DeepONet10	0.5	3.80e-05	2.01e-05	65.6	206	4.35e-05	1.25e+03	1.04	0.957	1	0	Complete
5 ANN10 + T/P DeepONet10	1	0.793	0.97	3.33e+07	2.50e+07	0.987	2.39e+08	5.49e+04	23.7	5.91	0	Complete
6 T/P/ϕ DeepONet10	0.05	4.51e-05	2.56e-05	0.964	0.968	5.35e-05	0.977	0.993	0.972	0.994	0	Complete
6 T/P/ϕ DeepONet10	0.1	4.54e-05	2.58e-05	0.982	0.983	5.41e-05	0.99	0.999	0.993	1	0	Complete
6 T/P/ϕ DeepONet10	0.2	4.53e-05	2.57e-05	9.09	1.48	5.41e-05	15.1	0.993	0.999	1	0	Complete
6 T/P/ϕ DeepONet10	0.3	2.34e-03	1.04e-03	348	3.22e+04	1.50e-03	5.23e+04	70.9	14.1	8.27	0	Complete
6 T/P/ϕ DeepONet10	0.5	1.06e-03	4.36e-04	935	2.29e+03	4.92e-04	4.11e+04	33	0.957	1	0	Complete
6 T/P/ϕ DeepONet10	1	—	—	—	—	—	—	—	—	—	—	Stop 0.530 ms
7 ANN10 + T/P/ϕ DeepONet10	0.05	4.52e-05	2.57e-05	0.957	0.972	5.37e-05	0.976	0.994	0.979	0.997	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	4.54e-05	2.58e-05	0.948	0.978	5.41e-05	0.977	0.998	0.994	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	1.41	0.738	8.16e+06	4.90e+06	0.809	5.24e+07	5.22e+04	9.45	22.7	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	2.34e-03	1.02e-03	486	1.80e+04	1.46e-03	3.04e+04	69	13.7	10.9	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	7.25e-04	2.92e-04	558	1.18e+03	3.23e-04	2.14e+04	22.4	0.961	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	1.19	0.921	2.24e+07	2.17e+07	0.977	1.62e+08	5.78e+04	13.9	4.7	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma short | Test IC 88

T_0=1502.1 K P_0=4.980 atm ϕ_0=0.859 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Short | Box-Cox gamma sweep | Test IC 88

Repaired | Cantera vs NN NRMSE | T0=1502.1K, P0=4.980atm, phi0=0.859

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.36	2.28	2.46	0.864	0.698	6.44	0.599	1.84	4.54	0	Complete
1 ANN10	0.1	0.0536	0.764	5.02	0.608	0.392	0.914	0.0409	1.27	4.48	0	Complete
1 ANN10	0.2	0.358	4.58	0.838	0.836	1.25	3.54	0.655	0.927	1.48	0	Complete
1 ANN10	0.3	0.0263	0.705	0.913	0.324	0.141	0.237	0.0689	0.366	0.798	0	Complete
1 ANN10	0.5	0.0368	0.955	1.72	0.8	0.0889	0.579	0.0974	0.975	0.996	0	Complete
1 ANN10	1	0.0461	0.941	1.58	0.353	0.0641	0.884	0.119	0.964	21.3	0	Complete
2 P-only DeepONet10	0.05	0.0909	0.601	0.822	1.01	0.318	1.55	0.141	3.16	2.78	0	Complete
2 P-only DeepONet10	0.1	0.101	0.92	0.803	0.75	0.101	1.19	0.157	5.76	17.3	0	Complete
2 P-only DeepONet10	0.2	0.188	2.3	0.797	3.28	0.294	0.974	0.279	15.8	148	0	Complete
2 P-only DeepONet10	0.3	0.247	0.711	7.05	0.742	0.356	1.94	0.179	12.2	295	0	Complete
2 P-only DeepONet10	0.5	0.25	0.789	5.57	0.309	0.812	3.29	0.224	1.01	320	0	Complete
2 P-only DeepONet10	1	0.356	5.71	2.78	1.57	2.62	1.04	0.612	1.1	312	0	Complete
3 ANN10 + P-only DeepONet10	0.05	0.057	0.319	0.674	0.96	0.33	1.12	0.0868	0.238	1.38	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.234	0.143	2.9	0.928	0.832	4.28	0.3	1.4	7.8	0	Complete
3 ANN10 + P-only DeepONet10	0.2	0.0442	0.716	0.485	0.226	0.143	0.412	0.0891	6.43	53	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.071	0.411	0.978	0.218	0.12	0.659	0.0997	2.04	0.657	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.126	0.93	2.62	1.25	0.923	2.11	0.0823	0.563	195	0	Complete
3 ANN10 + P-only DeepONet10	1	0.451	8.94	1.01	0.418	4	0.409	0.875	1.1	92.6	0	Complete
4 T/P DeepONet10	0.05	0.291	6.35	1.29	0.923	3.09	1.07	0.596	1.52	1.65	0	Complete
4 T/P DeepONet10	0.1	0.0871	0.306	1.15	0.406	0.106	0.998	0.107	2.11	2.76	0	Complete
4 T/P DeepONet10	0.2	9.09e-03	0.0582	0.36	0.0928	0.019	0.0261	5.99e-03	0.235	0.249	0	Complete
4 T/P DeepONet10	0.3	5.75e-03	0.0506	0.254	0.0628	0.0137	0.0208	4.09e-03	0.164	0.136	0	Complete
4 T/P DeepONet10	0.5	0.0114	0.929	0.185	0.98	0.667	0.492	0.0623	9.59	9.74	0	Complete
4 T/P DeepONet10	1	0.47	9.62	0.994	0.796	4.39	0.814	0.913	0.94	67.5	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.0643	1.17	0.978	0.952	0.756	1.02	0.0952	0.723	0.907	0	Complete
5 ANN10 + T/P DeepONet10	0.1	0.0113	0.0567	0.186	0.0697	0.0111	0.105	0.0136	0.111	0.106	0	Complete
5 ANN10 + T/P DeepONet10	0.2	5.18e-03	0.0202	0.232	0.0469	0.0211	5.80e-03	3.20e-03	0.212	0.292	0	Complete
5 ANN10 + T/P DeepONet10	0.3	4.64e-03	0.0254	0.131	0.0289	0.0183	3.49e-03	4.24e-03	0.0967	0.15	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0197	0.924	0.154	0.846	0.669	0.497	0.0663	9.35	10.9	0	Complete
5 ANN10 + T/P DeepONet10	1	0.136	2.51	1.35	0.408	1.16	0.27	0.256	2.02	12.4	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.185	1.99	3.7	0.716	1.05	1.76	0.244	2.3	1.69	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.024	0.107	0.479	0.124	0.0122	0.217	0.0275	0.295	0.232	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0599	0.659	0.59	0.683	0.0271	0.732	0.101	0.446	1.51	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0281	0.877	2.2	0.77	0.0846	0.763	0.094	0.154	0.293	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.103	1.12	0.651	1.82	0.0868	0.866	0.154	0.486	0.715	0	Complete
6 T/P/ϕ DeepONet10	1	0.257	4.53	2.2	0.899	2.12	0.579	0.466	6.29	8.21	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.213	0.194	3.23	0.112	0.397	2.87	0.252	1.91	1.52	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0144	0.076	0.19	0.0787	0.012	0.145	0.0184	0.122	0.129	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0151	0.231	0.482	0.221	0.0477	0.318	0.0336	0.222	0.823	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0237	0.846	2.23	0.742	0.0909	0.735	0.0884	0.0951	0.333	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0626	0.961	0.822	0.794	0.125	0.572	0.117	0.175	0.651	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.253	4.13	2.13	0.62	1.84	0.859	0.452	5.82	18.3	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.

Pure-MSE gamma long | Test IC 88

T_0=1502.1 K P_0=4.980 atm ϕ_0=0.859 | Repaired self-fed rollout, $\Delta t = 2\,\mu$s

Cantera
gamma=0.05
gamma=0.1 (paired origin)
gamma=0.2
gamma=0.3
gamma=0.5
gamma=1

1 ANN10

2 P-only DeepONet10

3 ANN10 + P-only DeepONet10

4 T/P DeepONet10

5 ANN10 + T/P DeepONet10

6 T/P/ϕ DeepONet10

7 ANN10 + T/P/ϕ DeepONet10

T [K] | Y_{H_2} | Y_H | Y_O | Y_{O_2} | Y_{OH} | Y_{H_2O} | Y_{HO_2} | $Y_{H_2O_2}$ | Y_{N_2}

Time [ms]

Short=paired10000-update origins; long=exact state continuation with150s cumulative new-work allowance per model.
Gamma.1 ANN/T-P origins were rebuilt because Adam state was missing; bootstrap time is included. Historical controls remain separate.
Both stages share y scales. Pure MSE; scalar train-loss plateau or time-limit termination. Actual updates and origins are exported.

Long | Box-Cox gamma sweep | Test IC 88

Repaired | Cantera vs NN NRMSE | T0=1502.1K, P0=4.980atm, phi0=0.859

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0453	0.526	0.909	1.99	0.174	0.372	0.0568	0.303	1.08	0	Complete
1 ANN10	0.1	0.103	1.34	0.842	2.21	0.16	0.662	0.154	0.232	1.12	0	Complete
1 ANN10	0.2	4.81e-03	0.3	0.548	1.17	0.035	0.334	3.09e-03	0.714	3.02	0	Complete
1 ANN10	0.3	0.0213	0.184	0.527	0.923	0.163	0.288	0.0447	0.826	1.99	0	Complete
1 ANN10	0.5	0.027	0.304	0.759	0.56	0.276	0.0427	0.0431	1.23	4.15	0	Complete
1 ANN10	1	0.378	6.18	0.305	0.545	2.45	1.43	0.712	13.8	55.6	0	Complete
2 P-only DeepONet10	0.05	0.0132	0.245	1.65	0.294	0.163	0.373	0.0164	0.645	0.794	0	Complete
2 P-only DeepONet10	0.1	0.0233	0.465	0.466	0.645	0.0778	0.0535	0.0417	0.188	1.11	0	Complete
2 P-only DeepONet10	0.2	1.48e-03	0.0246	8.62e-03	6.94e-03	8.10e-03	8.50e-03	2.82e-03	0.0252	0.0875	0	Complete
2 P-only DeepONet10	0.3	0.103	0.664	3.39	0.317	0.307	1.24	0.0703	0.458	2.72	0	Complete
2 P-only DeepONet10	0.5	0.193	0.963	4.61	2.02	0.887	2.16	0.121	0.469	451	0	Complete
2 P-only DeepONet10	1	0.552	0.225	11.5	8.23	0.593	2.22	0.372	1.7	775	0	Complete
3 ANN10 + P-only DeepONet10	0.05	8.08e-03	0.181	1.25	0.286	0.145	0.324	0.0143	0.294	0.366	0	Complete
3 ANN10 + P-only DeepONet10	0.1	0.0195	0.474	0.561	0.677	0.0836	0.0128	0.0377	0.21	1.32	0	Complete
3 ANN10 + P-only DeepONet10	0.2	1.71e-03	0.0478	0.0461	0.0313	0.0141	4.04e-03	3.99e-03	0.0331	0.11	0	Complete
3 ANN10 + P-only DeepONet10	0.3	0.0725	0.806	2.58	0.591	0.428	1.32	0.0461	0.409	4.1	0	Complete
3 ANN10 + P-only DeepONet10	0.5	0.147	0.922	3.26	1.4	0.888	2.12	0.096	1.36	231	0	Complete
3 ANN10 + P-only DeepONet10	1	0.417	5.71	0.625	2.78	1.94	1.26	0.669	1.46	311	0	Complete
4 T/P DeepONet10	0.05	0.408	0.498	6.95	1.09	0.44	4.16	0.431	4.38	2.51	0	Complete
4 T/P DeepONet10	0.1	7.67e-03	0.0353	0.0839	0.0272	6.15e-03	0.0842	0.0101	0.096	0.0849	0	Complete
4 T/P DeepONet10	0.2	2.74e-03	0.0167	0.0782	0.0172	6.89e-03	0.011	2.71e-03	0.0513	0.0594	0	Complete
4 T/P DeepONet10	0.3	3.11e-03	0.0187	0.0642	0.0272	9.10e-03	7.89e-03	3.19e-03	0.0303	0.0235	0	Complete
4 T/P DeepONet10	0.5	0.218	3.67	0.974	1.05	1.22	0.843	0.406	0.462	2.72	0	Complete
4 T/P DeepONet10	1	0.187	2.32	2.89	1.39	1.29	2.19	0.25	27.7	61.1	0	Complete
5 ANN10 + T/P DeepONet10	0.05	0.458	1.55	2.96	0.358	0.921	7.18	0.644	2.27	4.16	0	Complete
5 ANN10 + T/P DeepONet10	0.1	6.88e-03	0.0182	0.0651	0.0322	0.0136	0.116	0.01	0.109	0.123	0	Complete
5 ANN10 + T/P DeepONet10	0.2	7.37e-04	0.0326	0.0885	0.047	7.75e-03	5.24e-03	1.72e-03	0.0732	0.0838	0	Complete
5 ANN10 + T/P DeepONet10	0.3	1.30e-03	0.0115	0.0589	0.0111	5.12e-03	1.97e-03	1.06e-03	0.0297	0.0248	0	Complete
5 ANN10 + T/P DeepONet10	0.5	0.0777	1.89	0.957	0.222	0.743	0.0225	0.176	0.658	0.343	0	Complete
5 ANN10 + T/P DeepONet10	1	0.16	0.992	3.38	1.69	0.994	2.3	0.106	23.7	65	0	Complete
6 T/P/ϕ DeepONet10	0.05	0.152	0.466	1.13	0.215	0.338	2.48	0.217	2.68	3.23	0	Complete
6 T/P/ϕ DeepONet10	0.1	0.0158	0.0846	0.173	0.064	6.36e-03	0.145	0.0206	0.145	0.122	0	Complete
6 T/P/ϕ DeepONet10	0.2	0.0378	0.696	0.798	0.459	0.0824	0.408	0.0804	0.0262	0.966	0	Complete
6 T/P/ϕ DeepONet10	0.3	0.0464	1.4	0.993	0.502	0.537	0.187	0.113	0.596	0.461	0	Complete
6 T/P/ϕ DeepONet10	0.5	0.298	3.38	0.159	2.87	0.656	1.91	0.457	0.751	3.01	0	Complete
6 T/P/ϕ DeepONet10	1	0.281	5.42	1.33	1.12	2.45	0.866	0.546	7.2	29.8	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.602	2.39	3.37	0.97	0.981	9.25	0.861	0.975	11.8	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0227	0.0901	0.211	0.0447	0.0229	0.325	0.0327	0.315	0.34	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0298	0.602	0.774	0.422	0.0682	0.328	0.0665	0.0131	0.873	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0827	1.95	0.992	0.744	0.734	0.102	0.174	0.443	0.249	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.0298	0.827	0.688	0.221	0.308	0.14	0.071	0.424	0.53	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	1	0.286	4.88	1.96	1.27	2.16	1.61	0.525	9.14	47.4	0	Complete

NRMSE=RMSE(Cantera,NN)/RMS(Cantera), exact user util.py default; equal time samples, raw ratio without ×100.
Incomplete profiles have no full-horizon score. Summary cohorts are explicitly matched across short and long; empty sets are N/A.
Short origins:10000 updates. Long:variable actual updates; train-MSE plateau and150s active-time stops are distinguished. Gamma.1 ANN/T-P short origins are rebuilt.